

NiCE IA & QC v12.5 Troubleshooting Fundamentals

NiCE Interaction Analytics & Quality Central v12.5

Global Education Services

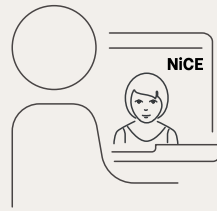


Course Overview

NiCE Interaction Analytics & Quality
Central Troubleshooting Fundamentals

Create a
NiCE..
world 

Training Expectations



During your session

Do these...

Setup Out of Office



Be on Time



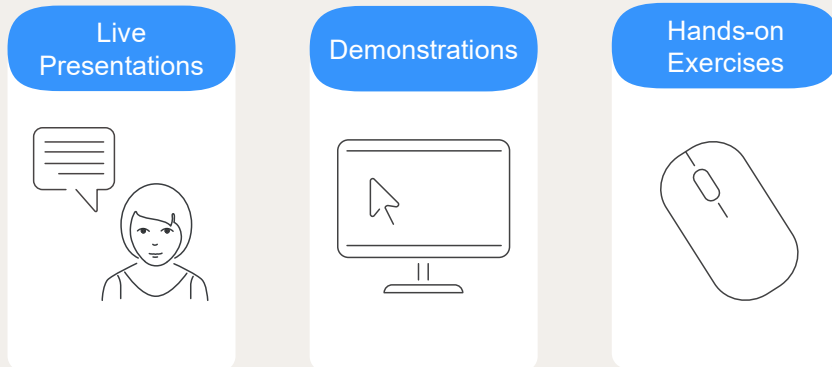
Ask Questions



NiCE

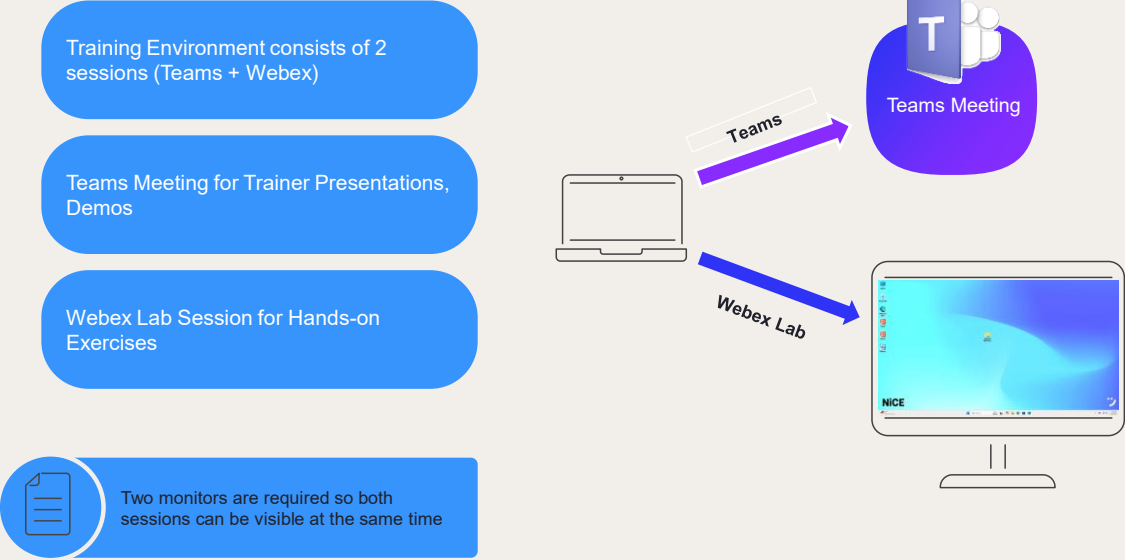
Even though this is a virtual training environment, it should be treated like a classroom environment. Setup your email out of office and update your voicemail that you will be unavailable during training. Be punctual when returning from the breaks and lunches. Active participation is appreciated and contributes to the learning process.

Instructor-led Training



This training is delivered over the web as 'Virtual Classroom Training' which combines live presentations and demonstrations. Every day during this training course, we will open a Teams session which will remain open throughout the day. This is your platform to communicate with the trainer, ask questions and comment. It will also be used for the live presentations and demos.

NICE Training Environment



This is a diagram of the virtual environment. A Teams session will be used for the audio conference. A second Webex connection will be made to computer.



Navigating the Teams Session

01:34

Take control

Pop out

Chat

People

Raise

React

View

Rooms

Apps

More

Camera

Mic

Share

Leave

Pop-out for Full Screen

Participants

React

Mute / Unmute

NiCE Product Intro and Application Design

NiCE Training

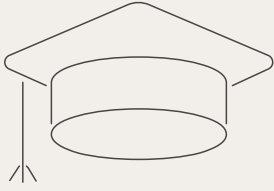
Create a NiCE world

Participant 1

Participant 2



Certification



Need 80%
or better



Practical



Theoretical



To receive a certification, you must have a scores above 80% on the practical and theoretical exams.



Course Materials



Course Book



Exercise Book

Let's take a look!



agenda

Day 1

- Data Flow & Configuration
- SQL Jobs
- Ingest Flow

Day 2

- Discovery Flow
- Search Grid
- Playback Flow

Day 3

- QC Flow Review

Day 4

- Theoretical Exam
- Practical Exam





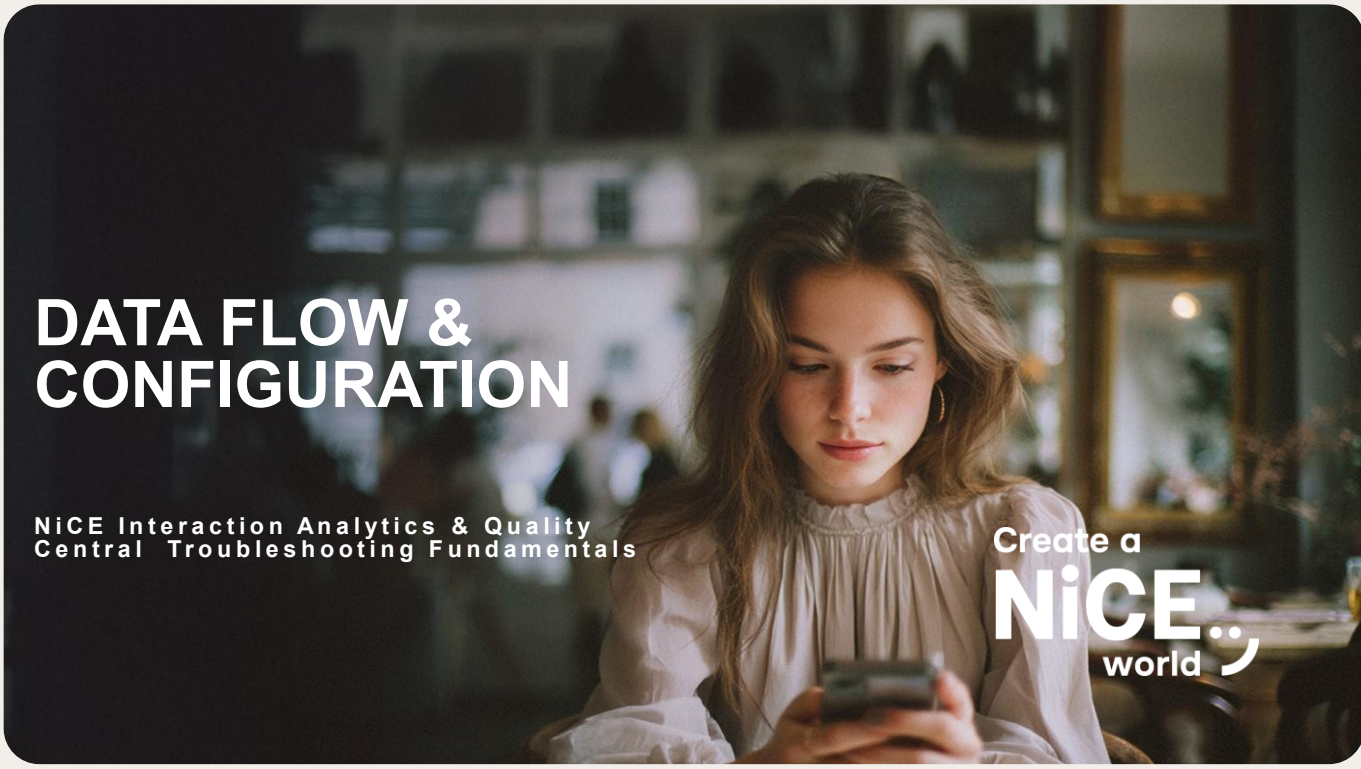
Before we begin...

- What is your experience with NiCE Solutions?
- Where are you located?
- What do you want to learn about in this course?





Thank You



DATA FLOW & CONFIGURATION

NiCE Interaction Analytics & Quality
Central Troubleshooting Fundamentals

Create a
NiCE..
world ☺

objectives

Describe

- Logs and Services of Interaction Analytics
- Interaction Analytics Databases
- Dataflow in Interaction Analytics
- Management Console



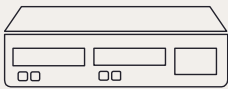
Training lab – Site Map



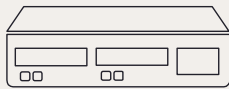
NIA - Application



NXDB1 - DB



Grid1 - Discovery (ASR)



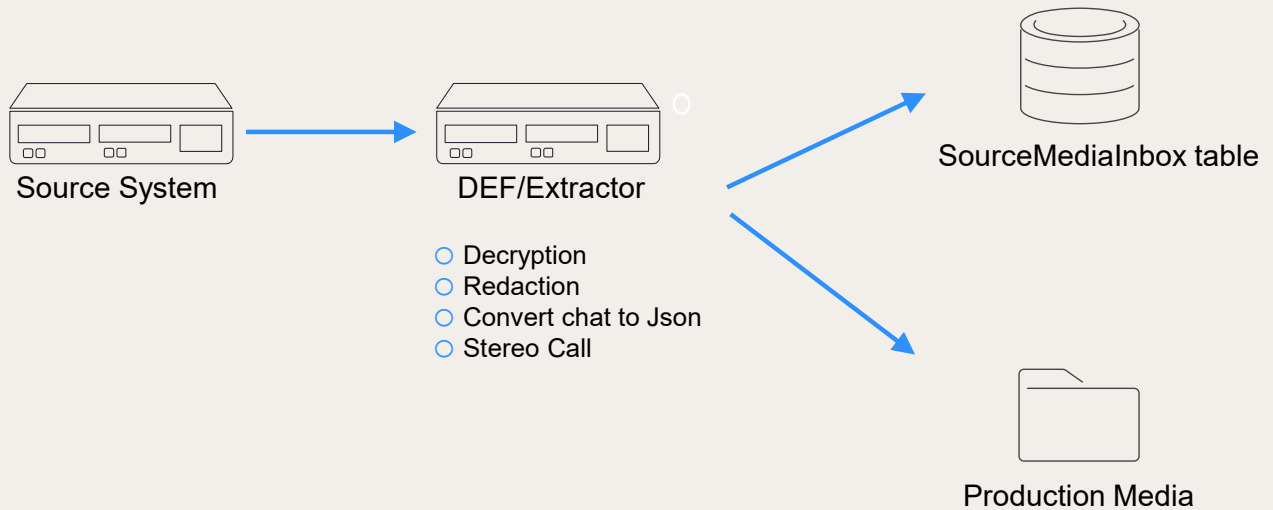
Grid2 - Phonetics SG



Grid3 - Phonetics SG



Preparing Data for Interaction Analytics



The DEF\extractor extracts the media files and metadata from the source system. DEF processes the audio and prepares it for NIA before pushing it to NIA to be ingested to the system.

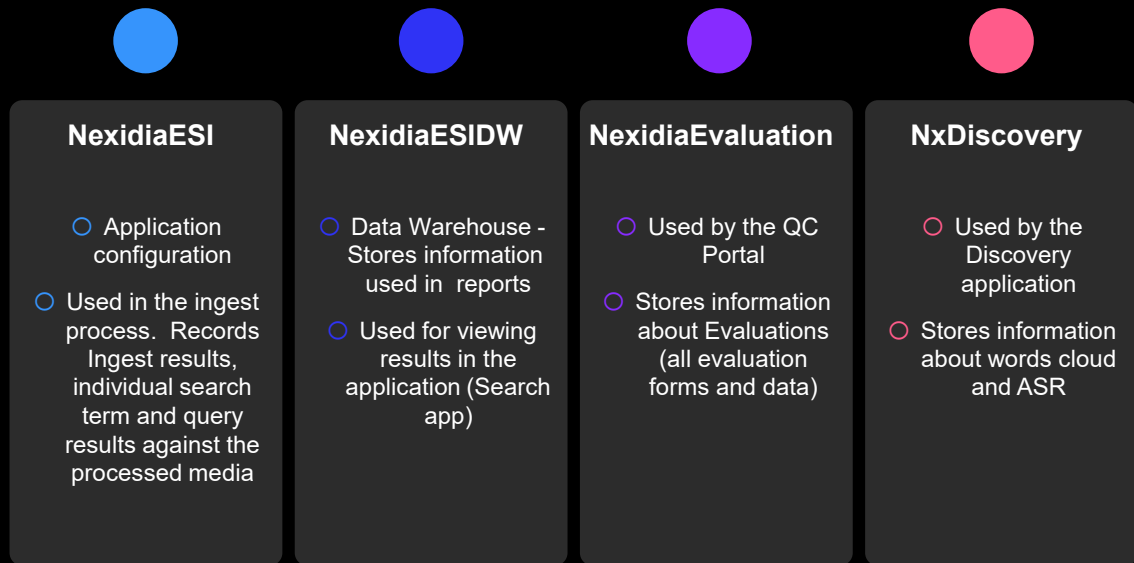
For example, DEF decrypts encrypted media files, performs a redaction of sensitive data, converts text interaction to a JSON file, and marks the different channels in a stereo call, etc.

Once the metadata and media files are in the correct format, DEF will place the media file in the production media folders and the metadata in the source media inbox table in the Interaction Analytics database.

Databases



Databases



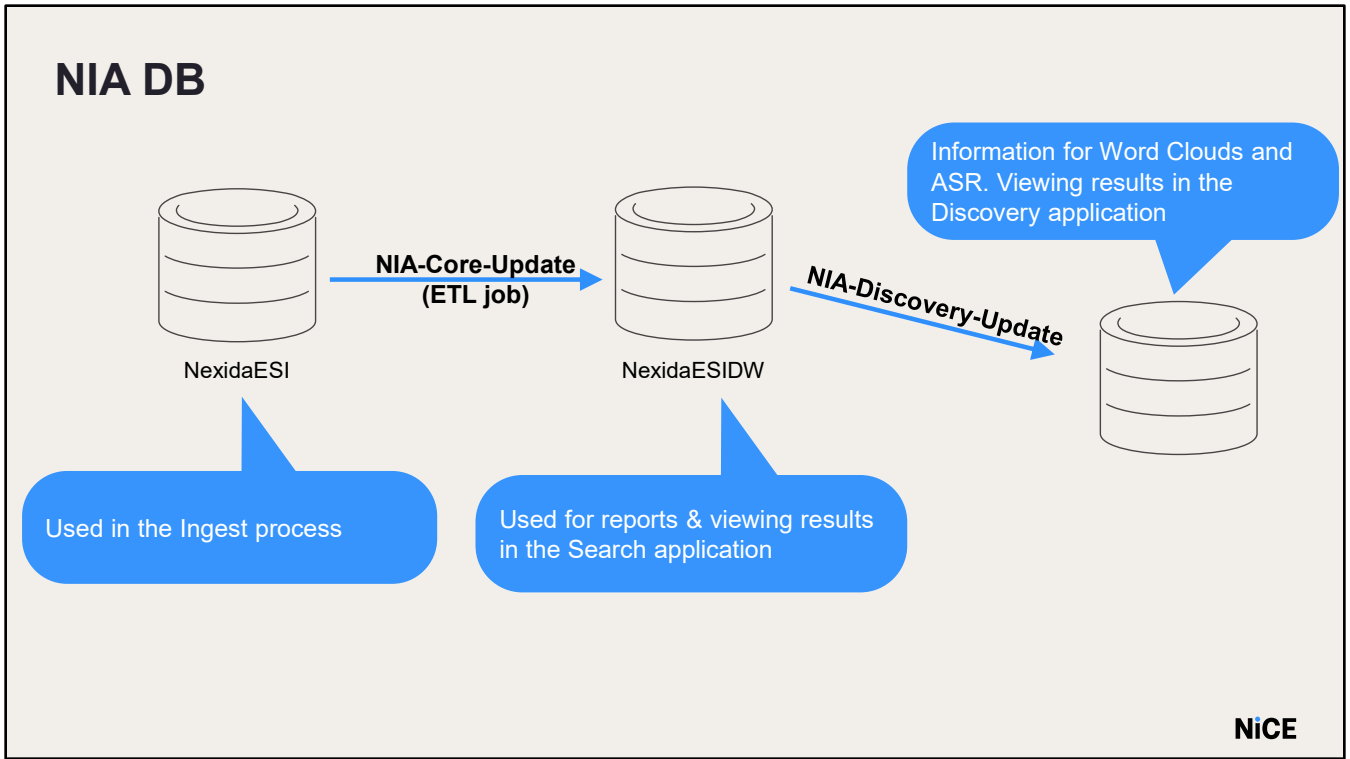
NexidiaESI - The results database records individual search term and query results against the processed media.

NexidiaESIDW – Provides rapid response at scale. The search service accesses this database to identify any pre-existing results against user-requested searches.

NexidiaEvaluation – Holds the information for storage and reporting on the evaluations

NxDiscovery – Holds the information for data processing through the Discovery engines





When you activate a session, the ingest process begins. The Reporting Database must be updated before you can view, search, and analyze your results.

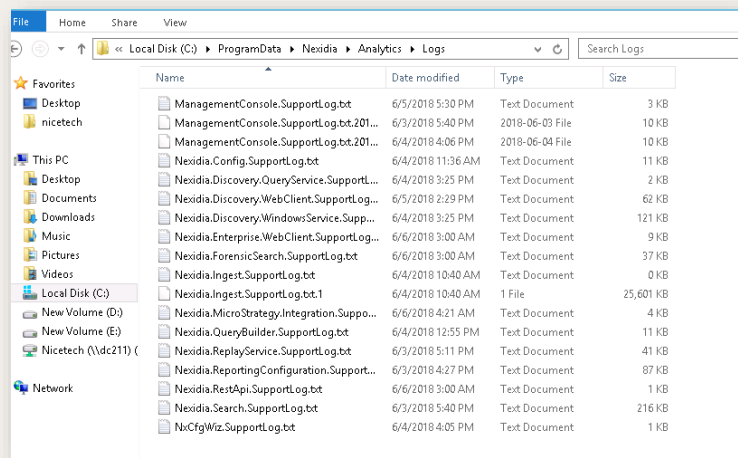
To view the TMS in the Discovery application, the NXDiscovery must be updated with the TMS definitions and results.

ASR = Advance Speech Recognition

Log Files &
Services



Log Files



Each Nexidia application and service has its own support log



Log Files

- Configuration files are in corresponding service folder
- Example: Config file for Ingest flow:
D:\Program Files\Nexidia\Nexidia Analytics\IngestServer\LoggingConfig.xml
- Name of the log file explains the information reported in the log file.

C:\ProgramData\Nexidia\Analytics\Logs

Name	Date modified	Type	Size
ManagementConsole.SupportLog.txt	6/5/2018 5:30 PM	Text Document	3 KB
ManagementConsole.SupportLog.txt.201...	6/3/2018 5:40 PM	2018-06-03 File	10 KB
ManagementConsole.SupportLog.txt.201...	6/4/2018 4:06 PM	2018-06-04 File	10 KB
Nexidia.Config.SupportLog.txt	6/4/2018 11:36 AM	Text Document	11 KB
Nexidia.Discovery.QueryService.SupportL...	6/4/2018 3:25 PM	Text Document	2 KB
Nexidia.Discovery.WebClient.SupportLog...	6/5/2018 2:29 PM	Text Document	62 KB
Nexidia.Discovery.WindowService.Supp...	6/4/2018 3:25 PM	Text Document	121 KB
Nexidia.Enterprise.WebClient.SupportLog...	6/6/2018 3:00 AM	Text Document	9 KB
Nexidia.ForensicSearch.SupportLog.txt	6/6/2018 3:00 AM	Text Document	37 KB
Nexidia.Ingest.SupportLog.txt	6/4/2018 10:40 AM	Text Document	0 KB
Nexidia.Ingest.SupportLog.txt.1	6/4/2018 10:40 AM	1 File	25,601 KB
Nexidia.MicroStrategy.Integration.Supp...	6/6/2018 4:21 AM	Text Document	4 KB
Nexidia.QueryBuilder.SupportLog.txt	6/4/2018 12:55 PM	Text Document	11 KB
Nexidia.ReplayService.SupportLog.txt	6/3/2018 5:11 PM	Text Document	41 KB
Nexidia.ReportingConfiguration.Support...	6/3/2018 4:27 PM	Text Document	87 KB
Nexidia.RestApi.SupportLog.txt	6/6/2018 3:00 AM	Text Document	1 KB
Nexidia.Search.SupportLog.txt	6/3/2018 5:40 PM	Text Document	216 KB
NxCfgWiz.SupportLog.txt	6/4/2018 4:05 PM	Text Document	1 KB

All the log files are in NIA- C:\Program Data\Nexidia\Analytics\Logs.

The logging configuration files are in corresponding service folder.

Example: The config file for Ingest flow: D:\Program Files\Nexidia\Nexidia Analytics\IngestServer\LoggingConfig.xml

The name of the log file explains what information is reported in the corresponding file.

Other Log Files

C:\ProgramData\Nexidia\Analytics\Logs

Name	Date modified	Type	Size
Nexidia.BackgroundWorkerService.Supp...	9/16/2021 12:52 PM	Text Document	0 KB
Nexidia.BackgroundWorkerService.Supp...	9/16/2021 12:52 PM	1 File	5,121 KB
Nexidia.MicroStrategy.Integration.Supp...	9/16/2021 8:30 PM	Text Document	1,036 KB
Nexidia.BackgroundWorkerService.Supp...	9/15/2021 8:28 PM	2 File	5,121 KB
Nexidia.BackgroundWorkerService.Supp...	9/15/2021 4:05 AM	3 File	5,121 KB
Nexidia.Enterprise.WebClient.SupportLog...	9/14/2021 11:41 AM	4 File	5,121 KB
Nexidia.ForensicSearch.SupportLog.bt	9/14/2021 3:00 AM	Text Document	236 KB
Nexidia.ForensicSearch.SupportLog.bt	9/14/2021 3:00 AM	Text Document	261 KB
Nexidia.ForensicSearch.SupportLog.bt	9/13/2021 7:19 PM	5 File	5,121 KB
Nexidia.ForensicSearch.MediaPrep.Supp...	9/13/2021 6:00 PM	Text Document	184 KB
Nexidia.RestApi.SupportLog.bt	9/13/2021 5:59 PM	Text Document	34 KB
Nx.Cfg.Wiz.SupportLog.bt	9/13/2021 5:59 PM	Text Document	53 KB
Nexidia.Config.SupportLog.bt	9/13/2021 5:53 PM	Text Document	1,069 KB
Nexidia.Engage.Streaming.SupportLog.bt	9/13/2021 5:53 PM	Text Document	6 KB
Nexidia.FilterService.SupportLog.bt	9/13/2021 5:53 PM	Text Document	6 KB
Nexidia.RtamClient.ActivityLog.bt	9/13/2021 5:53 PM	Text Document	6 KB
Nexidia.DataComplianceService.Support...	9/13/2021 5:52 PM	Text Document	3,695 KB
Nexidia.Discovery.ConfusableTerms Servi...	9/13/2021 5:52 PM	Text Document	5 KB
Nexidia.Discovery.WindowsService.Supp...	9/13/2021 5:52 PM	Text Document	1,585 KB
Nexidia.EvaluationsAging.SupportLog.bt	9/13/2021 5:52 PM	Text Document	15,042 KB
Nexidia.Search.QueryService.SupportLog...	9/13/2021 5:52 PM	Text Document	717 KB
Nexidia.Discovery.QueryService.Support...	9/13/2021 5:47 PM	Text Document	9 KB
Nexidia.ReplayService.SupportLog.bt	9/13/2021 5:47 PM	Text Document	2,180 KB
Nexidia.InterACT.Streaming.SupportLog...	9/13/2021 5:30 PM	Text Document	4 KB
Nexidia.DataExportComplianceSupportL...	9/13/2021 5:30 PM	Text Document	196 KB
Nexidia.Discovery.ConfusableTermsApp...	9/13/2021 5:27 PM	Text Document	2 KB
Nexidia.Discovery.WebClient.SupportLog...	9/13/2021 4:54 PM	Text Document	228 KB
Nexidia.BackgroundWorkerService.Supp...	9/12/2021 4:33 AM	6 File	5,121 KB
Nexidia.BackgroundWorkerService.Supp...	9/12/2021 12:04 PM	7 File	5,121 KB
ManagementConsole.SupportLog.bt	9/10/2021 11:36 PM	Text Document	2 KB
Nexidia.Ingest.SupportLog.bt	9/10/2021 11:11 PM	Text Document	7,977 KB

Nexidia.RestApi.SupportLog.bt	9/13/2021 5:59 PM	Text Document	34 KB
Nx.Cfg.Wiz.SupportLog.bt	9/13/2021 5:59 PM	Text Document	53 KB
Nexidia.Config.SupportLog.bt	9/13/2021 5:53 PM	Text Document	1,069 KB
Nexidia.Engage.Streaming.SupportLog.bt	9/13/2021 5:53 PM	Text Document	6 KB
Nexidia.FilterService.SupportLog.bt	9/13/2021 5:53 PM	Text Document	6 KB
Nexidia.RtamClient.ActivityLog.bt	9/13/2021 5:53 PM	Text Document	6 KB
Nexidia.DataComplianceService.Support...	9/13/2021 5:52 PM	Text Document	3,695 KB
Nexidia.Discovery.ConfusableTerms Servi...	9/13/2021 5:52 PM	Text Document	5 KB
Nexidia.Discovery.WindowsService.Supp...	9/13/2021 5:52 PM	Text Document	1,585 KB
Nexidia.EvaluationsAging.SupportLog.bt	9/13/2021 5:52 PM	Text Document	15,042 KB
Nexidia.Search.SupportLog.bt	9/13/2021 5:52 PM	Text Document	717 KB
Nexidia.Discovery.QueryService.Support...	9/13/2021 5:47 PM	Text Document	9 KB
Nexidia.ReplayService.SupportLog.bt	9/13/2021 5:47 PM	Text Document	2,180 KB
Nexidia.InterACT.Streaming.SupportLog...	9/13/2021 5:30 PM	Text Document	4 KB
Nexidia.DataExportComplianceSupportL...	9/13/2021 5:30 PM	Text Document	196 KB
Nexidia.Discovery.ConfusableTermsApp...	9/13/2021 5:27 PM	Text Document	2 KB
Nexidia.Discovery.WebClient.SupportLog...	9/13/2021 4:54 PM	Text Document	228 KB
Nexidia.BackgroundWorkerService.Supp...	9/12/2021 4:33 AM	6 File	5,121 KB
Nexidia.BackgroundWorkerService.Supp...	9/11/2021 12:04 PM	7 File	5,121 KB
ManagementConsole.SupportLog.bt	9/10/2021 11:36 PM	Text Document	2 KB
Nexidia.Ingest.SupportLog.bt	9/10/2021 11:11 PM	Text Document	7,977 KB
Nexidia.BackgroundWorkerService.Supp...	9/10/2021 8:52 PM	8 File	5,121 KB
Nexidia.EvaluationsWeb.SupportLog.bt	9/9/2021 3:00 AM	Text Document	24 KB

The name of the log file explains what information is reported in the corresponding file.



Log Files: Message level

- There are 5 logging levels:
 1. Fatal
 2. Error
 3. Warn
 4. Info
 5. Debug
- Fatal is the most severe level of log
- You can change the message level accordingly in the priority value.
- **Debug Mode:** If you change the message level to Debug mode, remember to change it back to the default mode when you are done troubleshooting.

NiCE

There are 5 logging levels:

1. Fatal
2. Error
3. Warn
4. Info
5. Debug

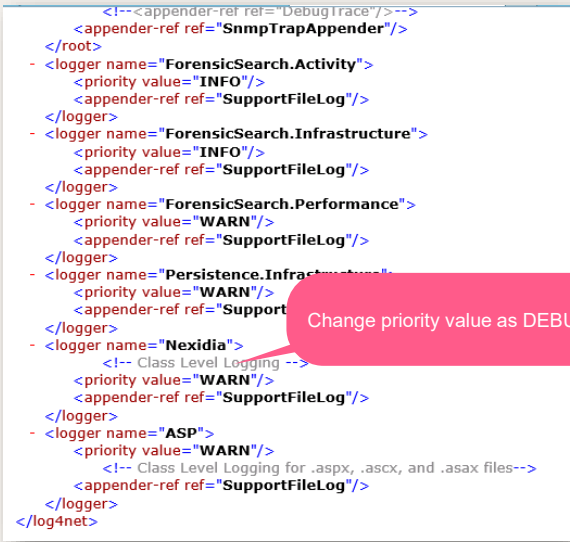
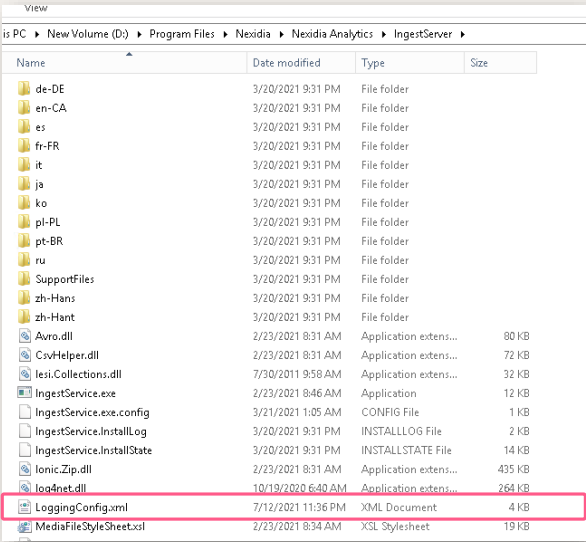
- Fatal indicates a critical error that causes the application or system to terminate immediately or become unusable and is the most severe level of log

You can change the message level accordingly in the priority value.

- Debug Mode: If you change the message level to Debug mode change it back to the default mode that is NOTICE, after you have checked the log file and got enough information
- This is because Debug mode will generate more data.



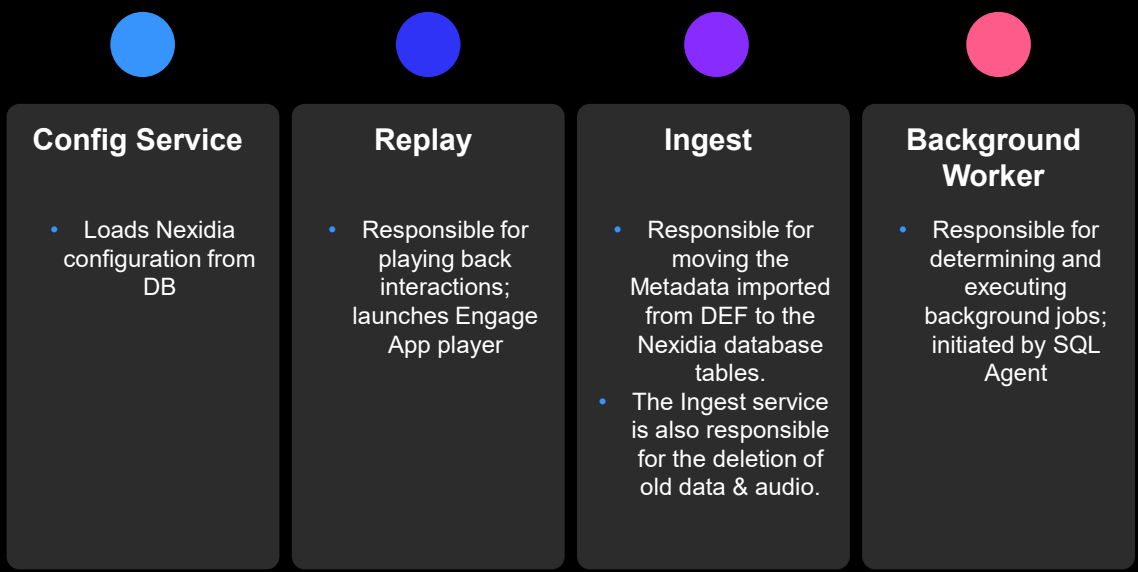
How to Change Log Levels



Say for example, in ingest flow, if you want to change the priority value =“WARN” to “DEBUG”, open the LoggingConfig.xml and change each Priority Value to “DEBUG”.



Analytics Services



Note: All services begin with the prefix "Nexidia" (e.g., *Nexidia Configuration*), except for MicroStrategy. While MicroStrategy includes multiple services, Nexidia only utilizes the Intelligence and Listening Services.

Ingest Service Overview:

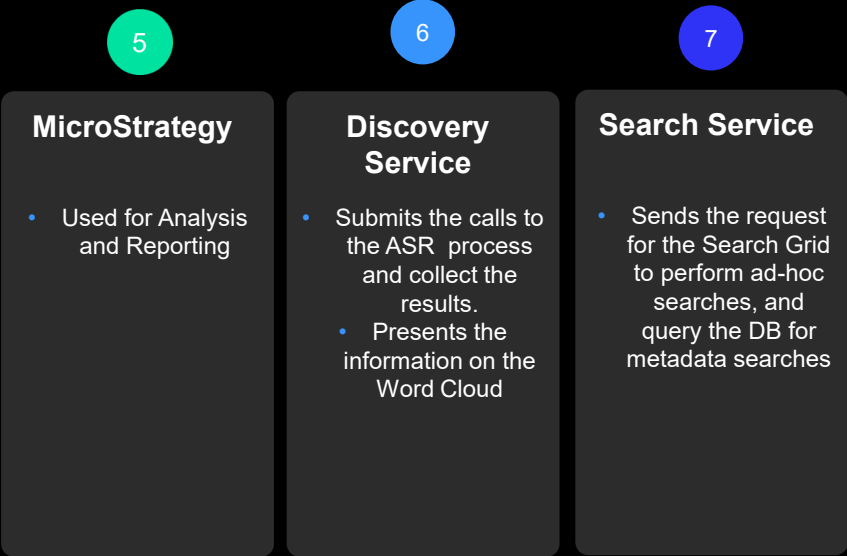
The Ingest Service monitors for newly available media and initiates ingestion by submitting requests to the Search Grid. Once results are returned, it updates the database with call metadata and query outcomes.

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Analytics Services



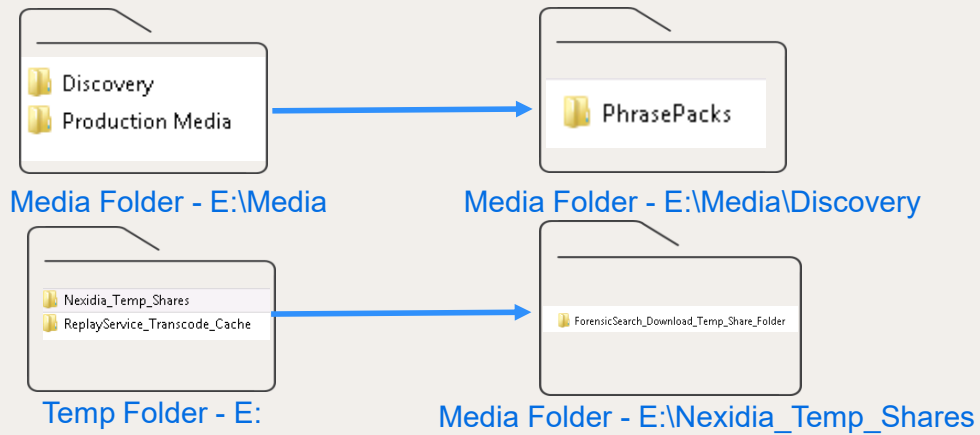
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Ingest Service Overview:

The Ingest Service monitors for newly available media and initiates ingestion by submitting requests to the Search Grid. Once results are returned, it updates the database with call metadata and query outcomes.



Server Prerequisites – Media Shared Folder



The Nexidia Services user must have read/write access to these folders

This is the recommended folder structure, and the folders should be shared

The production media folder contains all the interactions to be analyzed:

<Drive>:\Media

<Drive>:\Media\Production Media

The replay folder is used by the player to playback calls.

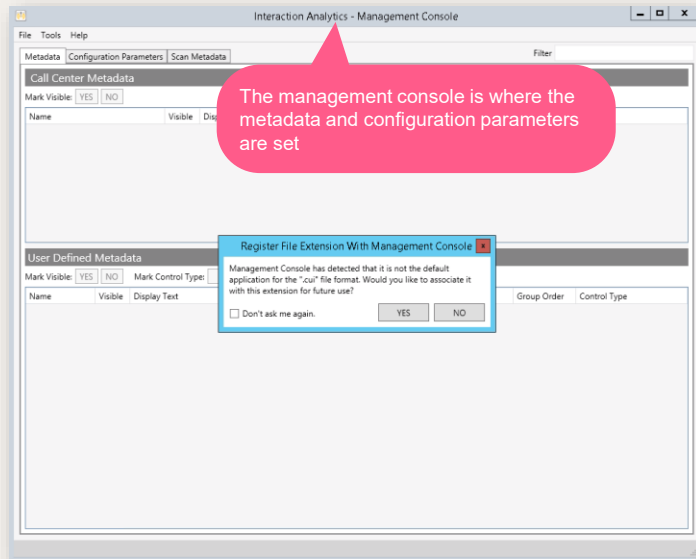
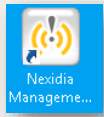
Temp sharers is used by the search application

In the training lab, the folders have been created on the E:\ drive

MANAGEMENT
CONSOLE



Management Console



The Management Console is used to modify application parameters and settings that are stored in the database, for example where the audio files are located or what is the data & audio retention (aging) period etc.

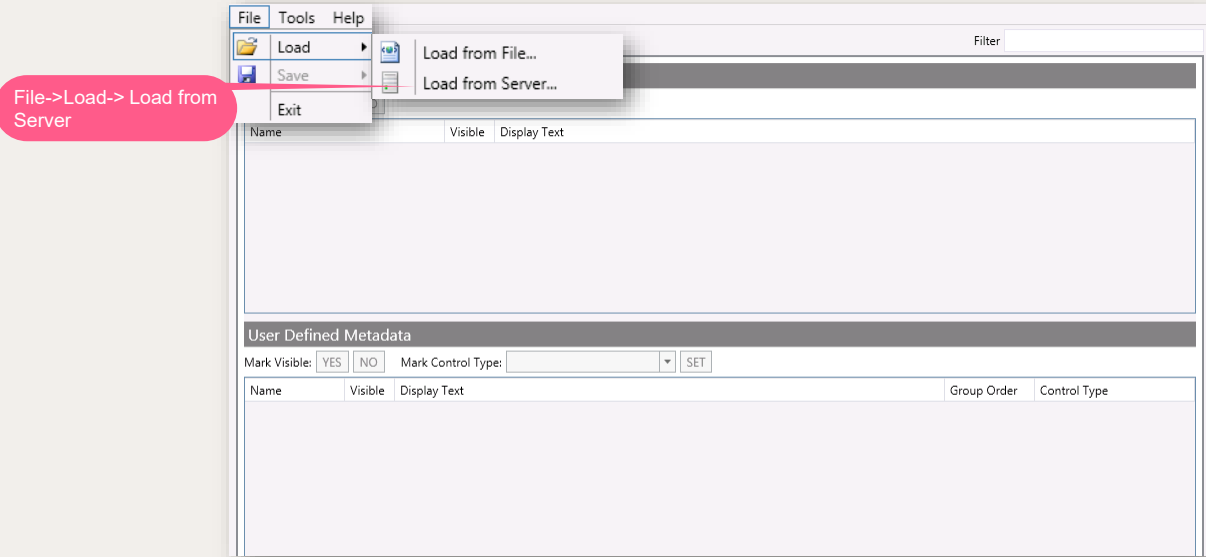
Open Management Console. There should be a shortcut on the desktop.

You will be asked if you want to associate the.cui extension with Management Console.

Click **“Yes”**.



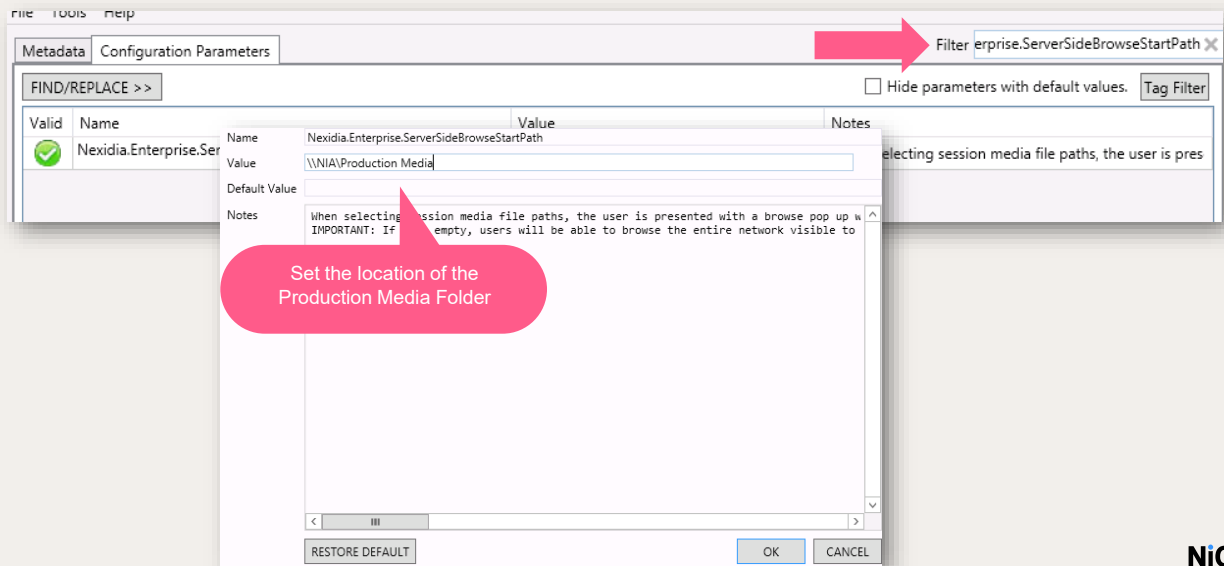
Management Console



First, the existing configuration needs to be loaded from the DB server into the console.



Management Console



Set the parameter Nexidia.Enterprise.ServerSideBrowseStartPath to the path of the production Media.

When selecting session media file paths, the user is presented with a browse pop up window. This value specifies the initial path to be browsed. Users will not be allowed to browse beyond this point. It can be either empty (users will have access to the entire network), a domain (IE "NEXIDIA"), a drive (IE "C:\"), a computer (IE "\\Server") or a path (IE "\\Server\Share", "\\Server\Share\Directory", "C:\directory").

IMPORTANT: If left empty, users will be able to browse the entire network.

Management Console

Administration > Session Management > New Session

STATUS

Session Status

Server Status

Job Status

SESSION MANAGEMENT

QUERIES

QUERY GROUPS

TERM LISTS

TARGET MEDIA SETS

EVALUATION SCORECARDS

BASIC SETTINGS

MEDIA FILE PA

New Session:

Save Cancel

Delete Add

Select Sources

Check (tick mark symbol) the file paths to be added

	NAME	SIZE	TYPE	DATE MODIFIED
☒	Additional Calls		Folder	7/6/2020 10:51:39 AM
☒	Billing		Folder	6/7/2018 1:42:39 PM
☒	CS Tier 1		Folder	6/7/2018 1:43:22 PM
☒	CS Tier 2		Folder	6/7/2018 1:45:56 PM
☒	Engage		Folder	7/6/2020 10:51:48 AM
☒	New Support Agents		Folder	7/6/2020 10:51:29 AM
☒	Stereo Calls		Folder	9/2/2019 3:13:55 PM
☒	Test2		Folder	7/29/2020 12:35:33 PM
☒	test		Folder	7/29/2020 12:26:59 PM

Add Cancel

!

If the "Nexidia.Enterprise.ServerSideBrowseStartPath" parameter is left empty users will not be able to browse to it from the session definition



Management Console

FileToolsHelp

MetadataConfiguration Parameters

FIND/REPLACE >>

Filter SearchGrid

Hide parameters with default values. Tag Filter

Valid	Name	Value	Description
✓	Nexidia.DiscoverySearchGrid.Password	DefaultTenant	The Discovery Search Grid's tenant password.
✓	Nexidia.DiscoverySearchGrid.SystemUri	http://GRID1:25002/	The Discovery Search Grid's System URI.
✓	Nexidia.DiscoverySearchGrid.Tenant	DefaultTenant	The Discovery Search Grid's tenant.
✓	Nexidia.Ingest.SearchGrid.MaxHttpConnectionsPerEndpoint	32	The maximum number of HTTP connections to Search G
✓	Nexidia.Search.SearchGrid.MaxHttpConnectionsPerEndpoint	16	The maximum number of HTTP connections to Search G
✓	Nexidia.SearchGrid.AdminPassword	admin	Search Grid Admin password
✓	Nexidia.SearchGrid.AdminTenant	admin	Search Grid Admin tenant
✓	Nexidia.SearchGrid.Password	DefaultTenant	Setting used to connect to SearchGrid
✓	Nexidia.SearchGrid.SystemUri	http://GRID2:25002/	Setting used to connect to SearchGrid
✓	Nexidia.SearchGrid.Tenant	DefaultTenant	Setting used to connect to SearchGrid

Discovery Search Grid

Phonetic Search Grid

Filter on SearchGrid change the following parameters two parameters:

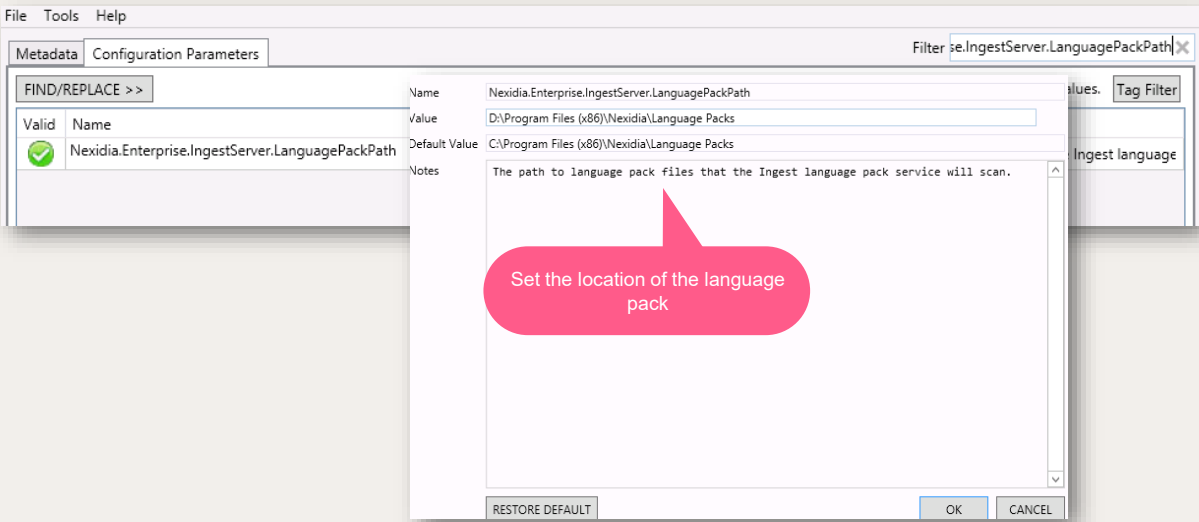
Nexidia.DiscoverySearchGrid.SystemUri

Nexidia.SearchGrid.SystemUri

In our lab the Discovery search grid installed on **Grid1** server and the Phonetics search grid is installed on the **Grid2** server, all other services are installed on the NIA server.



Management Console



Nexidia.Enterprise.IngestServer.LanguagePackPath – if you installed the language pack not on the default installation location.



Management Console

The screenshot shows the Management Console interface. A red callout bubble points to the 'File > Save > Save to Server' menu path. The 'Parameters' window is open, showing a table with columns 'Value' and 'Notes'. The first row has 'QC1' in the 'Value' column and 'The hostname of the machine that the Discovery service' in the 'Notes' column. A red arrow points to the 'Save to Server...' option in the 'Save' submenu. Below the 'Parameters' window, a dialog box asks 'Would you also like to synchronize metadata with Reporting?' with a note that Reporting must be fully configured. It has 'YES' and 'NO' buttons, with a red arrow pointing to 'NO'. To the right, a task completion window shows '4 of 4 Tasks have completed successfully' with a green progress bar and a 'DONE' button. Below this is a table of tasks.

Name	Progress
Update call center metadata settings	Complete
Update user-defined metadata settings	Complete
Update configuration parameters	Complete
Update Scan metadata fields	Complete

At the bottom left, there is a purple button with a warning icon and the text 'Restart the Services'.

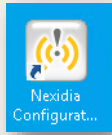
Once all the parameters are set save the configuration setting on the DB.

Click “File >> Save >> **Save to Server...**”

A dialog box will appear asking if you would like to synchronize metadata with reporting? Choose **NO** (since we have not yet created the reporter DB)

Close Management Console.

Interaction Analytics - Configuration Wizard



Configuration Service Connection Settings

Current Value:

NIA:47220|REST_port=47301,SSL=no

Update

Database

Edit Database Connection Settings

Web Configuration

Synchronize Validate

Reporting Web Integration

Remove Deploy

Services and Web Apps

Stop Start

Restart Update

☐ Include MicroStrategy Intelligence Server

Enable AutoStart Disable AutoStart

Undeploy Deploy

Redeploy

Done

Analytics Services & MSTR WEB Application service

Checkbox is disabled if the Intelligence Server is not installed on the local computer

The Services and Web Apps area of the Interaction Analytics Configuration Wizard are always enabled, and provides the functionality to Stop, Start, Restart and Update all the Interaction Analytics services and web applications including MicroStrategy Web.

The "Include MicroStrategy Intelligence Server" checkbox is disabled if the Intelligence Server is not installed on the local computer.



Review Question #1

If you don't see any results in the Search application – what would be the best DB to investigate?

NexidiaEvaluation

NexidiaESI

NexidiaESIDW

NexidiaDiscovery



Review Question#2

If you wanted to confirm the Ingest process is working – what would be the best DB to investigate?

NexidiaEvaluation

NexidiaESI

NexidiaESIDW

NexidiaDiscovery



summary

Describe

- Logs and Services of Interaction Analytics
- Interaction Analytics Databases
- Dataflow in Interaction Analytics
- Management Console





Thank You

SQL JOBS

NiCE Interaction Analytics & Quality
Central Troubleshooting Fundamentals

Create a
NiCE..
world ☺



objectives

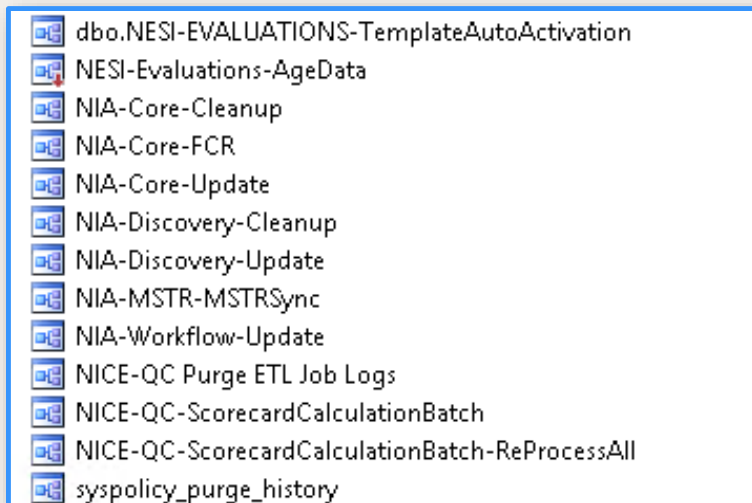
- Describe the SQL Jobs Structure
- Describe the Jobs Administration tab in the Management Console



Jobs Structure



SQL Jobs



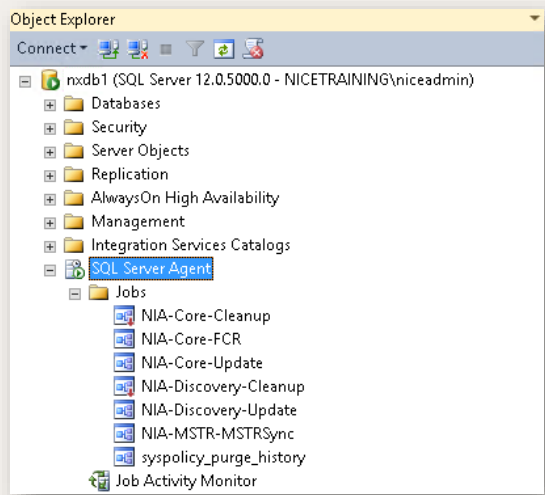
Jobs are grouped together according to the tasks they perform – update data, delete data, etc.

Background System Jobs have been remodeled to better deploy, manage and execute background tasks, to be able to deal with the increasing scale and volume of Interaction Analytics deployments.

In Interaction Analytics, most background tasks are now referred to as System Jobs and are automatically scheduled and executed by Interaction Analytics. These tasks include Ingest, ASR, MSTR Synchronization, and all Interaction Analytics jobs.



12.2 SQL Job Changes



Fewer Jobs

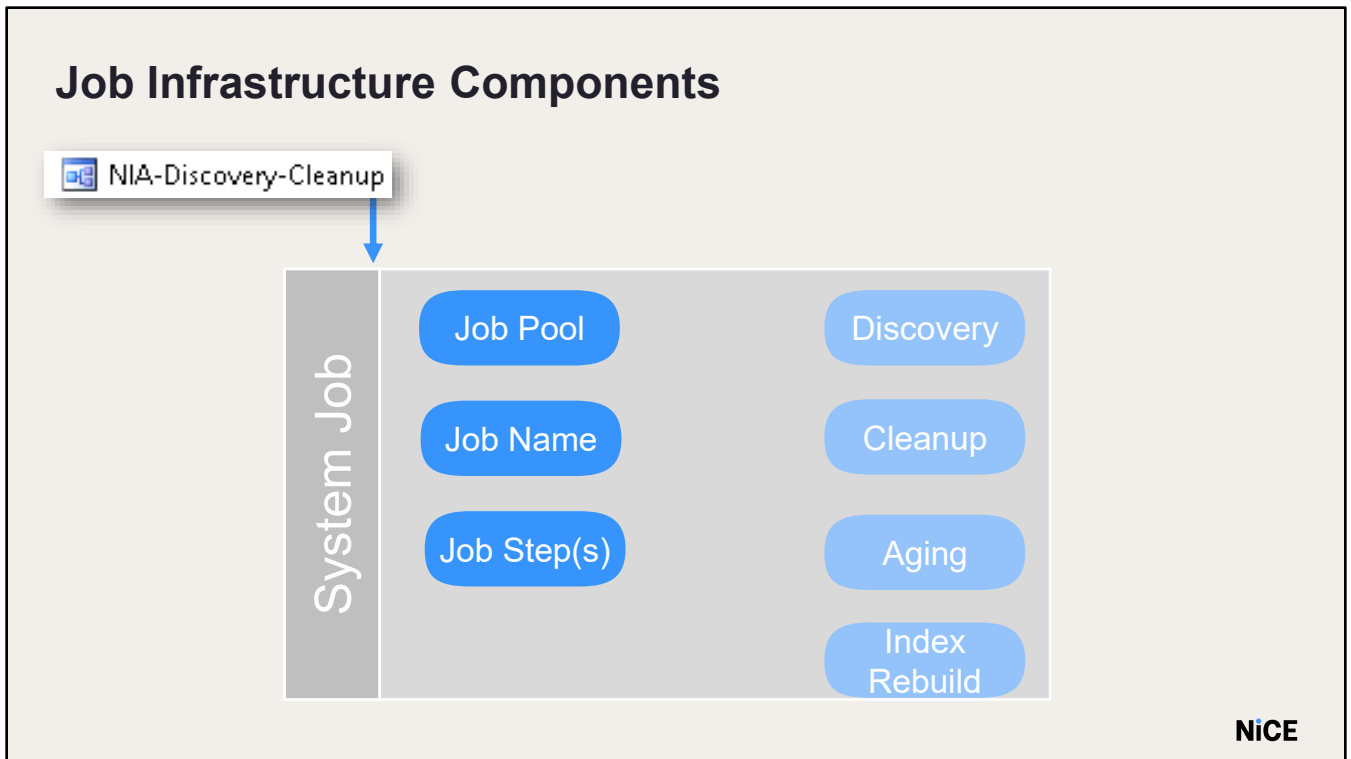
Default Schedule

Job Priority

In Interaction Analytics, most background tasks are now referred to as System Jobs and are automatically scheduled and executed by Interaction Analytics. SQL Jobs have been restructured to include five Job Pools. Each SQL job belongs to its own Job Pool.

- Schedules are set by Default
- Jobs have a Priority (Low, Medium, High)
- Jobs can belong to different Job Pools





In this structure, the SQL Agent Job “NIA-Discovery-Cleanup” requests the System Job to run. The System Job is composed of three components:

- Job Pool: A collection of one or more Jobs
- Job Name: The name of a specific job
- Job Step: Job Steps run in a job context
 - Only one Job can execute in a Job Pool at any given time
 - Some Job Steps must Pass for Job to Continue
 - Some Job Steps run periodically (e.g., once every five times a job runs)
 - Some Job Steps can run concurrently with other Job Steps in the same Job

Job Infrastructure Components

Name

Core.Cleanup

Enabled

False

Priority

40

Successful Executions

0

Operational Window

23:00:00 - 05:00:00

Actions

Name

Enabled

Sequence

Frequency

Service Name

Ignore Step Failure

Actions

Aging

False

1

1

Ingest

False

ESI Phy Delete

True

2

1

BG

False

ETL

True

3

1

BG

TMS Bridging

True

4

1

BG

DW Phys Delete

True

5

1

BG

ESI Index Rebuild

True

5

1

BG

DW Index Rebuild

True

6

1

BG

System Job Requests Retention

True

6

1

BG

Real Time Test Data Cleanup

True

7

7

BG

False

MediaReservation

True

7

1

BG

True

Core.FCR

True

30

2

N/A

Core.Update

True

10

3017

N/A

Name

Enabled

Sequence

Frequency

Service Name

Ignore Step Failure

Actions

ETL

True

1

1

BG

False

TMS Bridging

True

2

1

BG

False

Evals ETL

True

3

5

BG

False

QM Calc

True

4

5

BG

False

Evals Data Sync

True

4

1

BG

False

Associate Hierarchy

True

6

5

BG

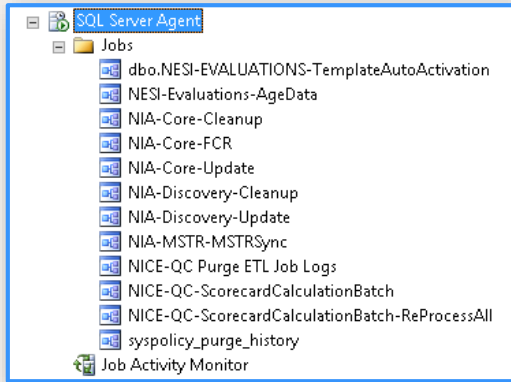
False

Jobs can belong to different Job

Jobs can belong to different Job Pools



SQL Jobs



Only 1 job per job pool can run at a time

Jobs have a Priority (Low, Medium, High)

Job Steps run in a job context

Naming Convention: NIA-[JOBPOOL]-[JOB]

SQL Jobs have been structured using Job Pools. Each SQL job belongs to its own Job Pool.

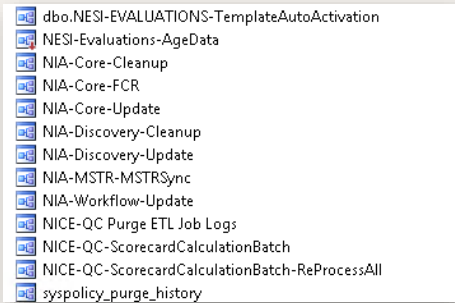
Schedules are set by Default.

Jobs have a Priority (Low, Medium, High).

Jobs can belong to different Job Pools.



SQL Jobs – Execution



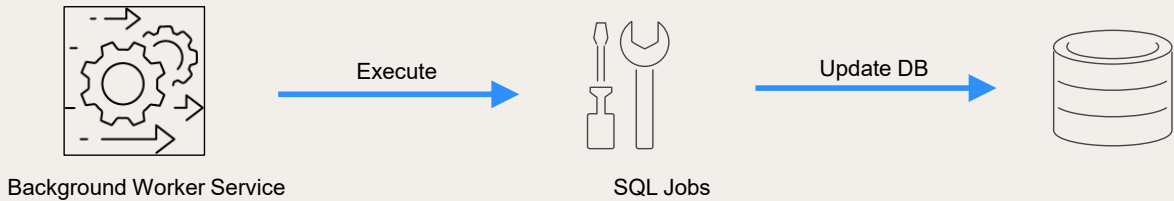
SSMS only makes requests for various JobPool – JobSteps to run and does not actually cause jobs to run, or more specifically, does not cause JobSteps to execute

It is important to note that the NIA jobs seen in SSMS only make requests for various JobPool-JobSteps to run and does not actually cause jobs to run, or more specifically, does not cause Job Steps to execute.

While these SSMS jobs can be scheduled and enabled, Job Step execution is determined by whether the Job Pool-Job containing it is enabled. The Job Pool also has scheduling ability within it.



SQL Jobs – Execution



Many of the Job Steps mentioned require the Windows Interaction Analytics Background Worker Service to be running. Some Job Steps are dependent on the NIA Ingest and Discovery Windows Services.

The Background Worker Service is classified as a Job executor service; other executor services are Ingest and the Discovery Service. This service is responsible for executing most of the background jobs in Interaction Analytics. The Background Worker Service polls for jobs that are fired from SQL Agent and other sources. It then determines if they should be executed. If the Background Worker Service determines a job can execute, it executes each Job Step in the job until the job completes.

Jobs
Administration



Jobs Administration

Only the job history for the last 24 hours is loaded, with a Maximum of 100 items.

Run Job

View Job history

Name	Successful Executions	Operational Window	Actions
Core.Cleanup		23:00:00 - 05:00:00	  
Core.FCR		N/A	  
Core.Update	True 10 4016	N/A	  
Discovery.Cleanup	False 30 0	23:00:00 - 05:00:00	  
Discovery.Upc			  
Ingest.Ingest			  
ASR.ASR			  
MSTR.MSTRS			  
DEU.DEU			  
Workflow.Upc			  

Name	Metadata	Request Date	Start Date	End Date	Duration (Secs)	Status
Update	N/A	7/16/2020 11:46:00 AM	7/16/2020 11:46:05 AM	7/16/2020 11:46:20 AM	15	Successful
ETL	N/A	7/16/2020 11:46:05 AM	7/16/2020 11:46:05 AM	7/16/2020 11:46:08 AM	3	Successful
TMS Bridging	N/A	7/16/2020 11:46:10 AM	7/16/2020 11:46:10 AM	7/16/2020 11:46:10 AM	0	Successful
Evals Data Sync	N/A	7/16/2020 11:46:15 AM	7/16/2020 11:46:15 AM	7/16/2020 11:46:15 AM	0	Successful
Update	N/A	7/16/2020 11:45:00 AM	7/16/2020 11:45:04 AM	7/16/2020 11:45:19 AM	15	Successful
Update	N/A	7/16/2020 11:44:00 AM	7/16/2020 11:44:03 AM	7/16/2020 11:44:28 AM	25	Successful
Update	N/A	7/16/2020 11:43:00 AM	7/16/2020 11:43:02 AM	7/16/2020 11:43:17 AM	15	Successful
Update	N/A	7/16/2020 11:42:00 AM	7/16/2020 11:42:01 AM	7/16/2020 11:42:16 AM	15	Successful
Update	N/A	7/16/2020 11:41:00 AM	7/16/2020 11:41:05 AM	7/16/2020 11:41:21 AM	15	Successful
Update	N/A	7/16/2020 11:40:00 AM	7/16/2020 11:40:04 AM	7/16/2020 11:40:20 AM	15	Successful
Update	N/A	7/16/2020 11:39:01 AM	7/16/2020 11:39:03 AM	7/16/2020 11:39:29 AM	25	Successful

The Jobs tab in the Management Console lists all the background system jobs.

From here, you can easily run the jobs. You can edit the job run details and view the job history.

Some jobs are license dependent, such as MSTR, Discovery & QC jobs, and only display with relevant valid license.



Jobs Administration

The screenshot shows a table of jobs with columns: Name, Enabled, Priority, Successful Executions, Operational Window, and Actions. A dialog box titled 'Enable /Disable Job' is open, showing options for 'Job Enabled', 'Operational Window (Local Time)', and 'Operational Window Enabled'. The 'Operational Window' is set from 23:00:00 to 05:00:00.

Name	Enabled	Priority	Successful Executions	Operational Window	Actions
Core.Cleanup	False	40	0	23:00:00 - 05:00:00	[Edit] [Refresh] [Info]
Core.FCR		30	4	N/A	[Edit] [Refresh] [Info]
Core.Update		10	4016	N/A	[Edit] [Refresh] [Info]
Discovery.Cleanup	False	30	0	23:00:00 - 05:00:00	[Edit] [Refresh] [Info]
Discovery.Update	True	10	403	N/A	[Edit] [Refresh] [Info]
Ingest.Ingest	True	1	267	N/A	[Edit] [Refresh] [Info]
ASR.ASR				N/A	[Edit] [Refresh] [Info]
MSTR.MSTRSync				N/A	[Edit] [Refresh] [Info]
DEU.DEU				N/A	[Edit] [Refresh] [Info]
Workflow.Update				N/A	[Edit] [Refresh] [Info]

Dialog Box: Enable /Disable Job

☐ Job Enabled

☐ Operational Window (Local Time)

☒ Operational Window Enabled

From: 23:00:00

To: 05:00:00

OK Cancel

Operational window, set in local time.

Jobs can be timebound by an Operational Window. Jobs that are defined as such can only be requested or executed within this operational window. The operational window is designed to prevent background-maintenance type jobs from running during business hours when their execution materially affects productivity or system responsiveness.

- If a job is requested within an operational window, it is queued for execution
- If a job is requested within an operational window, and is queued, but drifts outside of its window before execution, the job is cancelled
- Jobs can be executed manually to ignore operational window

Jobs Administration

The screenshot shows a 'Jobs Administration' window with a tree view on the left and a table of job steps on the right. A red arrow points to the 'Core.Cleanup' job in the tree. The table lists job steps with columns for Name, Enabled, Sequence, Frequency, Service Name, Ignore Step Failure, and Actions. Callouts provide additional context:

- Executed by the Ingest Service:** Points to the 'Ingest' service name for the 'Aging' step.
- Executed by the background service:** Points to the 'BG' service name for steps like 'ESI Phy Delete' and 'DW Index Rebuild'.
- Step runs concurrently:** Points to the 'Frequency' column for the 'Aging' step, which is set to 1.
- Step runs periodically once every 7 times a job runs:** Points to the 'Frequency' column for the 'Real Time Test Data Cleanup' step, which is set to 7.

Name	Enabled	Sequence	Frequency	Service Name	Ignore Step Failure	Actions
Aging	False	1	1	Ingest	False	[Icons]
ESI Phy Delete	True	2	1	BG	False	[Icons]
ETL	True	3	1	BG	False	[Icons]
TMS Bridging	True	4	1	BG	False	[Icons]
DW Phys Delete	True	5	1	BG	False	[Icons]
ESI Index Rebuild	True	5	1	BG	False	[Icons]
DW Index Rebuild	True	6	1	BG	False	[Icons]
System Job Requests Retention	True	6	1	BG	False	[Icons]
Real Time Test Data Cleanup	True	7	7	BG	False	[Icons]
Med...	True	7	1	BG	True	[Icons]
+	True	30		N/A		[Icons]
+				N/A		[Icons]
+				23:00:00 - 05:00:00		[Icons]
+				N/A		[Icons]
+				N/A		[Icons]
+				N/A		[Icons]
+	True	1	132	N/A		[Icons]
+	False	20	0	N/A		[Icons]
+	True	1	403	N/A		[Icons]

Expand the Job pool to view the job steps; for each step, you see which windows service is the executor of that step.

Sequence – order in which the job step will be executed, same sequence number means that the steps run concurrently



Jobs Administration

Name	Enabled	Priority	Successful Executions	Operational Window	Actions
Core.Cleanup	False	40	0	23:00:00 - 05:00:00	  

Name	Enabled	Sequence	Frequency	Service Name	Ignore Step Failure	Actions
Aging	False	1	1	Ingest	False	
ESI Phys Delete	True	2	1	BG	False	
ETL	True	3	1	BG	False	
TMS Bridging	True	4	1	BG	False	
DW Phys Delete					False	
ESI Index Rebuild					False	
DW Index Rebuild					False	
System Job Requests Retention					False	
Real Time Test Data Cleanup					False	
MediaReservation					True	

☐ Step Enabled

☐ Allow parent Job to succeed if this step fails

Frequency

OK

Cancel



Core.FCR



Core.Update



Discovery.Cleanup



Discovery.Update



Ingest.Ingest



ASR.ASR



MSTR.MSTRSync



DEU.DEU



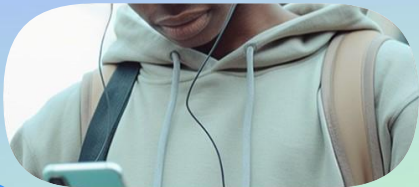
Workflow.Update

Ingest.Ingest	True	1	267	N/A	  
ASR.ASR	True	1	170	N/A	  
MSTR.MSTRSync	True	1	132	N/A	  
DEU.DEU	False	20	0	N/A	  
Workflow.Update	True	1	403	N/A	  

Allow parent job to succeed If this step fails - If this step is causing the parent job to fail, you can refresh the job and run again without this step.
The step will fail but the job is able to run successfully.



REVIEW
QUESTIONS



Review Question#1

Which Analytics service is responsible for executing the SQL jobs?

Replay Service

Config Service

Background
Worker Service

Search Service



Review Question #2

Which SQL Job needs to run to populate the NexidiaESIDW database?

NIA-Core-update (ETL step)



summary

- SQL Jobs Structure
- Jobs Administration tab in the Management Console





Thank You

INGEST FLOW

NiCE Interaction Analytics & Quality
Central Troubleshooting Fundamentals

Create a
NiCE..
world ☺



objectives

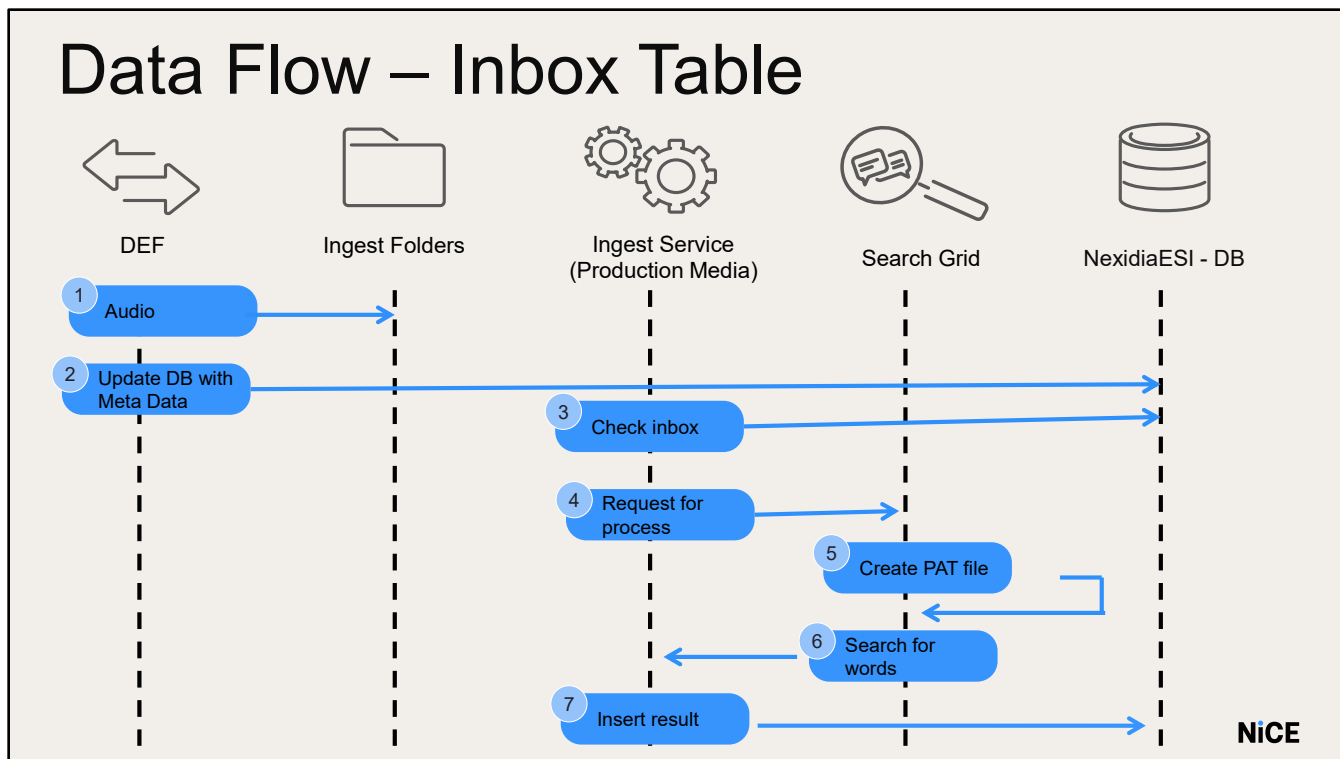
- Describe the Ingest Flow
- Describe the Ingest Main Tables in the Database
- Describe the Troubleshooting Steps for the Ingest Flow



INGEST FLOW



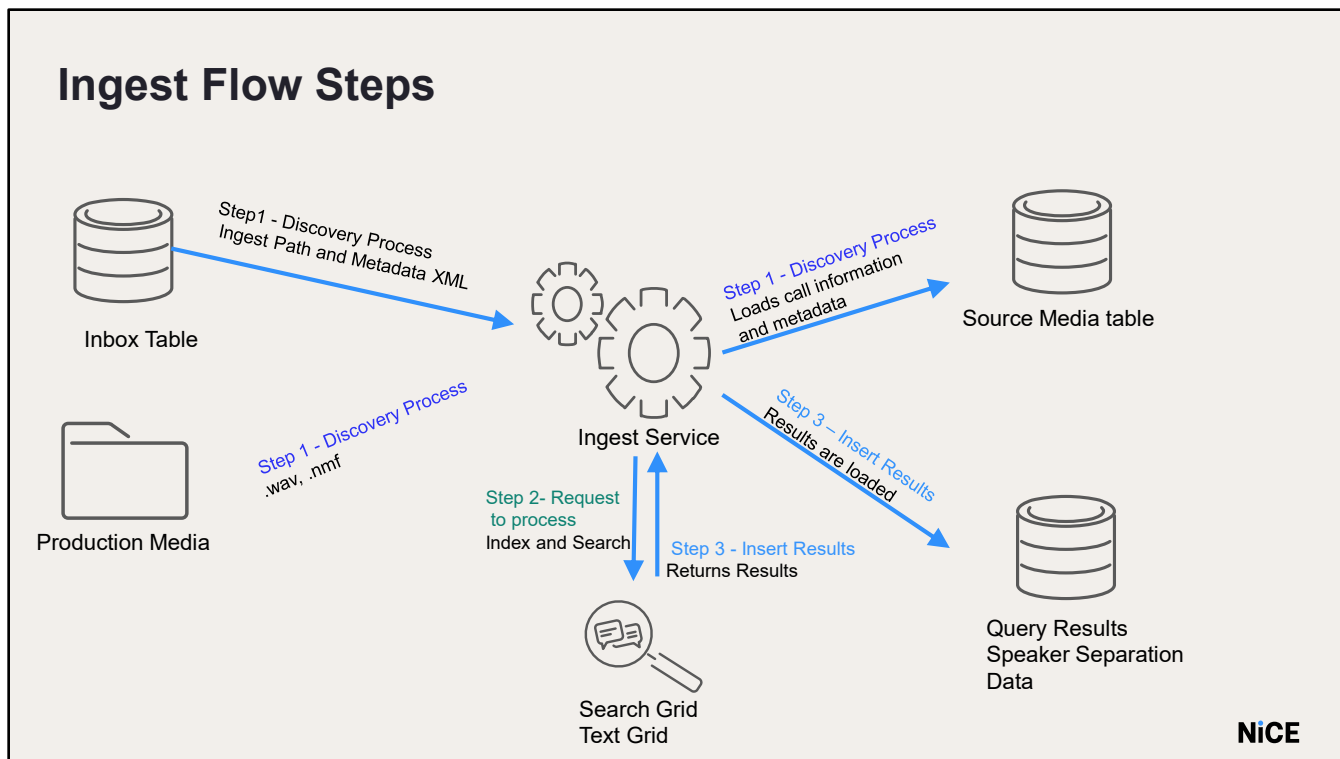
Data Flow – Inbox Table



DEF automates the retrieval of media files and metadata from external sources.

1. DEF places the audio in staging folders
2. DEF inserts the interaction's metadata to the database (NexidiaESI > NexidiaESI.SourceMediaInbox table)
3. Ingest service checks this table for new entries new interactions to process.
4. The ingest service sends a request to the search grid to process the interaction (create PAT file & search for words and phrases).
5. The search grid will create the phonetic index / PAT file and save them locally.
6. The search grid applies any search terms or queries to the PAT file. Results from those searches are sent to the ingest service.
7. The ingest service inserts the results from the search grid into the database.





The ingest flow can be broken down to 3 main steps:

1. Ingest discovery process- finds the files and loads them to the database. The ingest service checks if the media exists, insert a row to the source media table and load the call metadata from the XML to the different metadata tables. The source media table is an inventory of all the calls\interactions that were discovered by the ingest service.
2. Send a request to the search grid to process the interaction (creating PAT files and search for words and phrases)
3. Insert the search result to the database

Ingestion Flow- Logs

2021-07-13 16:00:02,931 167641	INFO	23	ScanningObserver.RunScans - initiating scan [(null)] Ingest.Infrastructure.MediaObservation
2021-07-13 16:00:02,931 167641	INFO	23	ScanningObserver.RunScans - scan in progress [(null)] Ingest.Infrastructure.MediaObservation
2021-07-13 16:00:02,934 167644	INFO	23	ScanningObserver.RunScans - inboxing declarations [(null)] Ingest.Infrastructure.MediaObservation
2021-07-13 16:00:02,943 167653	INFO	23	ScanningObserver.RunScans - scan completed [(null)] Ingest.Infrastructure.MediaObservation

Scanning the folder for new calls

2021-07-13 18:49:42,014 4326230	INFO	26	ScanningObserver.RunScans - initiating scan [(null)] Ingest.Infrastructure.MediaObservation
2021-07-13 18:49:42,014 4326230	INFO	26	ScanningObserver.RunScans - scan in progress [(null)] Ingest.Infrastructure.MediaObservation
2021-07-13 18:49:42,014 4326230	DEBUG	26	scanCompleted: path=\\NIA\Production Media\Billing\, duration=00:00:00, filesPerSecond=Infinity, files=51, subdirectories=0 [(null)] Ingest.Infrastructure.MediaObservation
2021-07-13 18:49:42,014 4326230	DEBUG	26	scanCompleted: path=\\NIA\Production Media\CS Tier 1\, duration=00:00:00, filesPerSecond=Infinity, files=106, subdirectories=0 [(null)] Ingest.Infrastructure.MediaObservation

Scan is completed

2021-07-13 18:49:43,521 4327737	DEBUG	25	Found Media via Inbox; RowId=156, ExternalMediaId=\\NIA\Production Media\CS Tier 1\982.wav [(null)] Ingest.Infrastructure.MediaObservation

Source Media Inbox table is updated

NiCE

- Let us see the Ingest Service Log file.
- Ingest Service Log file location:
C:\ProgramData\Nexidia\Analytics\Logs\Nexidia.Ingest.SupportLog.txt file
- Note: Some of these messages are DEBUG(since the message priority level is DEBUG)
1. When there are no calls, you can see that the messages “Ingest service is waiting and scanning the folders for new calls” are repeating
 2. Indicates that scanning is completed
 3. Media is found in the source media folder and the Source Media Inbox table is updated.



Ingestion Flow- Logs

2021-07-13 18:49:44,446 4328663	DEBUG	FileEntrySink	Persisting Media: \\NIA\Production Media\CS Tier 1\982.wav [ingestFilePath=\\NIA\Production Media\CS Tier 1\982.wav]
2021-07-13 18:49:44,446 4328663	DEBUG	FileEntrySink	Attempted to load MediaStatistics from Metadata for externalMediaId=\\NIA\Production Media\CS Tier 1\982.wav [ingestFilePath=\\NIA\Production Media\CS Tier 1\982.wav]
2021-07-13 18:49:44,462 4328679	DEBUG	FileEntrySink	Attempted to load tags from Metadata sourceMediaId=156 [ingestFilePath=\\NIA\Production Media\CS Tier 1\982.wav]
2021-07-13 18:49:44,462 4328679	DEBUG	FileEntrySink	Attempted to create detector track from media statistics for sourceMediaId=156 [ingestFilePath=\\NIA\Production Media\CS Tier 1\982.wav]
2021-07-13 18:49:44,462 4328679	DEBUG	FileEntrySink	mediaDiscovered: ingestFilePath=\\NIA\Production Media\CS Tier 1\982.wav, externalMediaId=\\NIA\Production Media\CS Tier 1\982.wav, savedState=MediaIdNotExist [ingestFilePath=\\NIA\Production Media\CS Tier 1\982.wav]
2021-07-13 18:49:44,526 4328743	DEBUG	52	NewWork SourceMediaId 156 [(null)] Ingest.Infrastructure.WorkExecution
2021-07-13 18:49:44,526 4328743	DEBUG	52	WorkMaintBatch=4115, SourceMediaId=157 [(null)] Ingest.Infrastructure.WorkExecution
2021-07-13 18:49:44,538 4328754	DEBUG	53	Processing SourceMediaId 156 [(null)] Ingest.Infrastructure.WorkExecution
2021-07-13 18:49:44,538 4328754	DEBUG	53	WorkToBeDone for SourceMedia=156: SearchTermCount=1, StructuredQueryCount=25 [(null)] Ingest.Infrastructure.WorkExecution
2021-07-13 18:49:46,668 4330885	DEBUG	MediaStMgr-1	Media declared to Search Grid: searchGridId=204 [sourceMediaId=156, ingestFilePath=\\NIA\Production Media\CS Tier 1\982.wav]

Loading of metadata

New request identified

Call is processed

Sending declaration
to Search Grid

1. Loading of metadata for call in Interaction Analytics
2. New processing request identified for call ingestion (per call basis)
3. Processing call and identifying work steps to be completed by search grid
4. Sending declaration to search grid to create new task for processing inside the Phonetics engine



Ingestion Flow- Logs

2021-07-13 18:49:47,575 4331791 DEBUG PatCrtnMgr-3 Requested pat creation:
mediaOperationUri=<http://10.1.159.38:25022/tenant/1/media/204/mediaOperationInventory/9e2c5764-815d-422c-aa42-e39fb91eed16> [sourceMediaId=156, ingestFilePath=\\NIA\Production Media\CS Tier 1\982.wav] Ingest.Infrastructure.WorkExecution

2021-07-13 18:50:32,141 4376357 DEBUG SrchRsltMon Completed search [localSearchId=e99af47d-2011-4989-bdd3-5050504f81d0
<http://grid2:25002/tenant/1/search/ef252cdd-886e-4663-97bc-8756693af875>] Ingest.Infrastructure.WorkExecution

2021-07-13 18:50:48,970 4393186 DEBUG ResultWorker Ingest work succeeded: leaseDuration=00:01:04.4520000 [sourceMediaId=156, ingestFilePath=\\NIA\Production Media\CS Tier 1\982.wav] Ingest.Activity

PAT file creation

Success Message
from Search Grid

Ingestion is
successful

1. Task request for indexing of call by search grid (PAT file creation)
2. Receiving success message from search grid for wordsearch
3. Message saying call was successfully ingested after the results were inserted into the DB- END



INGEST FLOW
DATABASE



Ingest Main Tables



Source Media Inbox

- Inventory of all the interactions that were imported by the DEF server
- Contains interactions that have been ingested (processed) to the system, and interactions that are waiting to be ingested
- DEF populates this table



Source Media

- Inventory of all the interactions that were discovered by the ingest service



Source Media Tick

- Track the status of each interaction from the ingest perspective

We are going to focus on the following three main tables to trace the ingest process :

Source media inbox

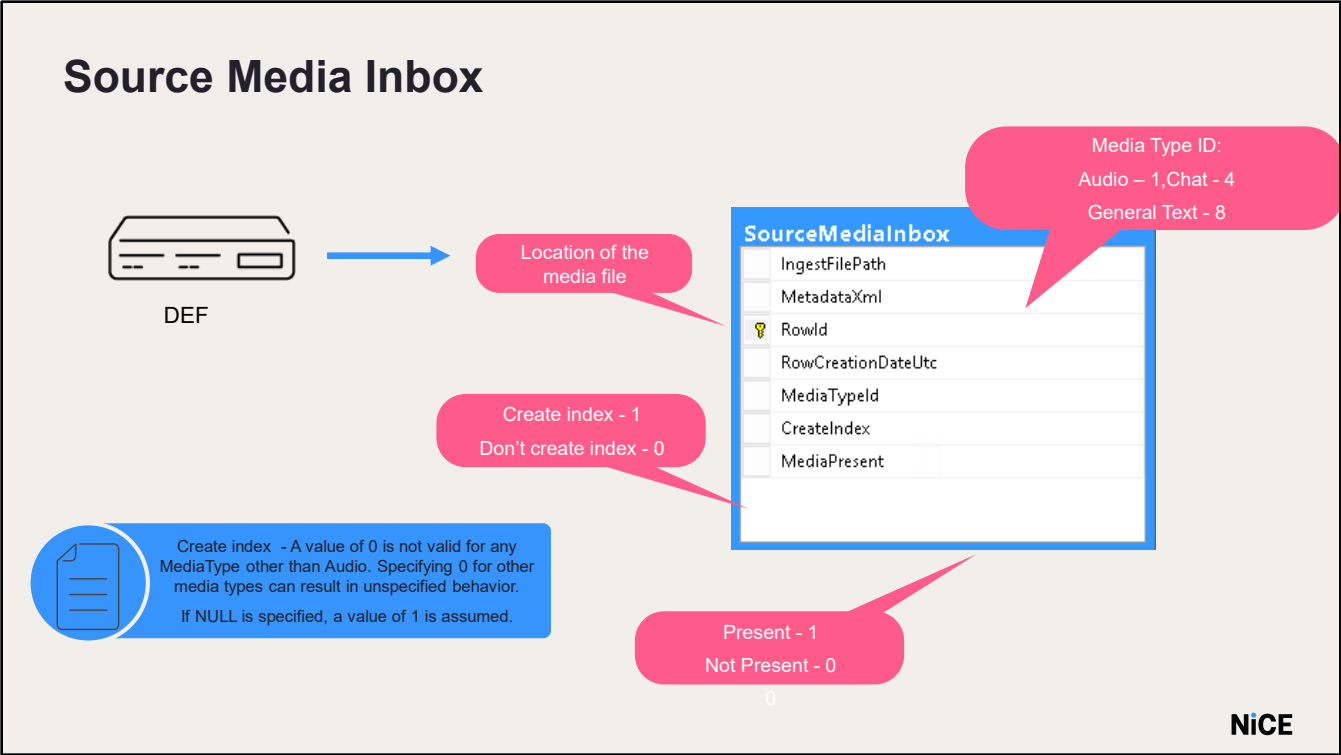
Source media

Source media tick



SOURCE MEDIA
INBOX TABLE





DEF populates the source media inbox table

Inbox Table - contains all interactions that were imported by the DEF server. It includes the path to the audio location and an XML with the interaction metadata.

Media Type Id :Audio - 1, Chat - 4, General Text - 8

Create index - if we don't need an index file like in QC standalone - 0

Note: Create index - A value of 0 is not valid for any MediaType other than Audio. Specifying 0 for other media types can result in unspecified behavior.
If NULL is specified, a value of 1 is assumed.



Source Media Inbox Table

```
SELECT * FROM [NexidiaESI].[dbo].[SourceMediaInbox]
```

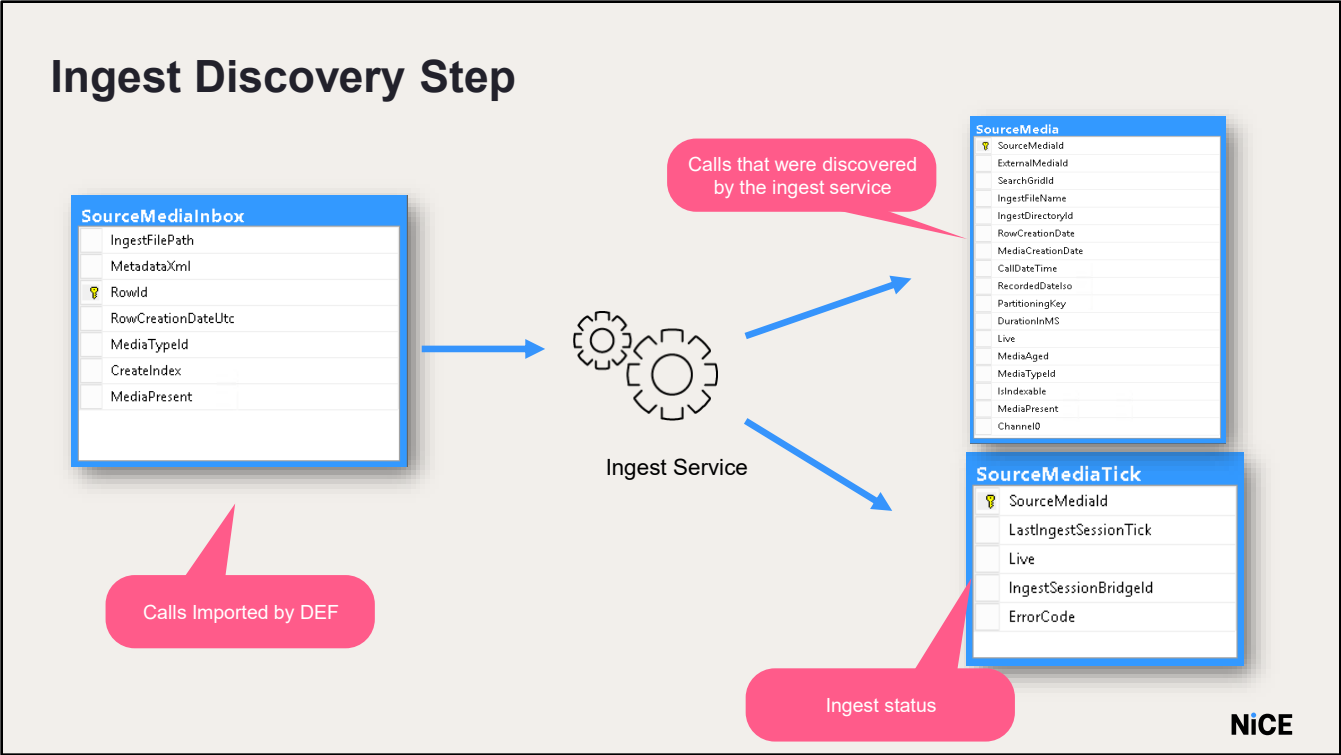
Results		Messages				
	IngestFilePath	MetadataXml	RowId	RowCreateDateUtc	Media TypeId	CreateIndex
1	\\od107appdb01\Production Media\AIG\Valic Financial Advisors (VFA)\2017\06\0...	<CallData><AgentId>7166</AgentId><AgentLogin>U54BD1D</AgentLo...	287316	2017-06-05 16:36:39	1	1
2	\\od107appdb01\Production Media\AIG\Valic Financial Advisors (VFA)\2017\06\0...	<CallData><AgentId>9713</AgentId><AgentLogin>jawade</AgentLogin>...	287315	2017-06-05 16:36:39	1	1
3	\\od107appdb01\Production Media\AIG\Valic Financial Advisors (VFA)\2017\06\0...	<CallData><AgentId>9713</AgentId><AgentLogin>jawade</AgentLogin>...	287314	2017-06-05 16:36:39	1	1
4	\\od107appdb01\Production Media\AIG\Valic Financial Advisors (VFA)\2017\06\0...	<CallData><AgentId>9713</AgentId><AgentLogin>jawade</AgentLogin>...	287313	2017-06-05 16:36:39	1	1
5	\\od107appdb01\Production Media\AIG\Valic Financial Advisors (VFA)\2017\06\0...	<CallData><AgentId>7166</AgentId><AgentLogin>U54BD1D</AgentLo...	287312	2017-06-05 16:36:39	1	1
6	\\od107appdb01\Production Media\AIG\Valic Financial Advisors (VFA)\2017\06\0...	<CallData><AgentId>9714</AgentId><AgentLogin>jgrossma</AgentLogi...	287311	2017-06-05 16:36:39	1	1
7	\\od107appdb01\Production Media\AIG\Valic Financial Advisors (VFA)\2017\06\0...	<CallData><AgentId>7161</AgentId><AgentLogin>U54BD07</AgentLog...	287310	2017-06-05 16:36:39	1	1
8	\\od107appdb01\Production Media\AIG\Valic Financial Advisors (VFA)\2017\06\0...	<CallData><AgentId>7166</AgentId><AgentLogin>U54BD1D</AgentLo...	287309	2017-06-05 16:36:39	1	1
9	\\od107appdb01\Production Media\AIG\Valic Financial Advisors (VFA)\2017\06\0...	<CallData><AgentId>9295</AgentId><AgentLogin>harris</AgentLogin><...	287308	2017-06-05 16:36:39	1	1
10	\\od107appdb01\Production Media\AIG\Valic Financial Advisors (VFA)\2017\06\0...	<CallData><AgentId>9295</AgentId><AgentLogin>harris</AgentLogin><...	287307	2017-06-05 16:36:39	1	1

Contains all interactions that were imported by the DEF server

How the Ingest service “knows” which interactions it already discovered, and which are new interactions.

In our example the next time the Ingest service runs, it will handle interactions from row ID 287312-287316.

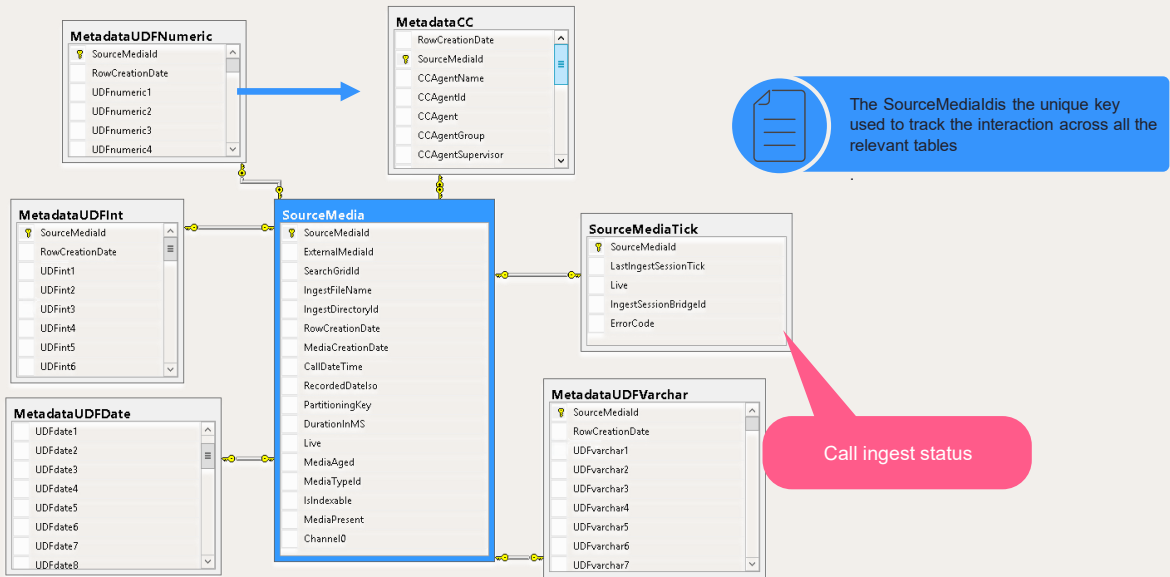




Once the ingest service discovers a call from the inbox table it will create an entry for that call in the source media table and the source media tick table



Ingest Discovery Process -Tables



Once the ingest service starts to process a call from the source media inbox. It will create an entry for that call in the source media table and will enter the data from the metadata XML into the relevant metadata tables.

At the same time, the ingest service will add the call to the source media tick table, this table track the call progress\status during the ingest process.

Media – Metadata

```
SELECT * FROM [NexidiaESI].[dbo].[MetadataCC]
```

	RowCreationDate	SourceMediaId	CCAgentName	CCAgentId	CCAgentGroup	CCAgentSupervisor	CCSiteName	CCCallDNIS	CCCallExtension	CCCallDirection
1	2017-04-30 05:06:28.7568169	1	Patricia Genovese	11990	East	Stephanie Ervin	Atlanta	2265	2939	Inbound
2	2017-04-30 05:06:28.8958022	2	Patricia Genovese	11990	East	Stephanie Ervin	Atlanta	2265	2939	Inbound
3	2017-04-30 05:06:28.9088059	3	Patricia Genovese	11990	East	Stephanie Ervin	Atlanta	2355	2939	Inbound
4	2017-04-30 05:06:28.9308189	4	Patricia Genovese	11990	East	Stephanie Ervin	Atlanta	2289	2939	Inbound
5	2017-04-30 05:06:28.9498116	5	Patricia Genovese	11990	East	Stephanie Ervin	Atlanta	2277	2939	Inbound
6	2017-04-30 05:06:28.9718062	6	Patricia Genovese	11990	East	Stephanie Ervin	Atlanta	2277	2939	Inbound

The SourceMediaID is the key to finding the metadata that is associated with a call.



SOURCE MEDIA
TABLE



Source Media Table

```
SELECT * FROM [NexidiaESI].[dbo].[SourceMedia]
```

	SourceMediaId	ExternalMediaId	SearchGridId	IngestFileName	IngestDirectoryId	RowCreationDate
146	146	\\NIA\Production Media\6.0.c Telco Data.1\05212005-307.wav	143	05212005-307.wav	1	2017-04-30 05:06:40.0041643
147	147	\\NIA\Production Media\6.0.c Telco Data.1\05212005-308.wav	147	05212005-308.wav	1	2017-04-30 05:06:40.0831748
148	148	\\NIA\Production Media\6.0.c Telco Data.1\05212005-309.wav	148	05212005-309.wav	1	2017-04-30 05:06:40.3341662
149	149	\\NIA\Production Media\6.0.c Telco Data.1\05212005-310.wav	149	05212005-310.wav	1	2017-04-30 05:06:40.3691765
150	150	\\NIA\Production Media\6.0.c Telco Data.1\05212005-311.wav	150	05212005-311.wav	1	2017-04-30 05:06:40.4912049
151	151	\\NIA\Production Media\6.0.c Telco Data.1\05212005-312.wav	151	05212005-312.wav	1	2017-04-30 05:06:40.6381840

Unique ID to be used to track the call in all tables

When the interaction was processed by the ingest service

Path to the audio file

Null - the call was not yet processed by the SG

This table contains all the interactions that have been ingested to the system.

Source Media ID - Unique Id to be used to track the call in all tables.

External Media ID -path to the audio file

Search Grid ID - Media Id in the search logs & DB. If the column is populated with information, then the search Grid processed this interaction, created a PAT file and searched for phrases.

Ingest Directory ID- Numerical representation of the path to Ingest directory (Refers to IngestDirectory table)

Row creation date - when the interaction was processed by the ingest service.



Source Media Table

```
SELECT * FROM [NexidiaESI].[dbo].[SourceMedia]
```

MediaCreationDate	CallDateTime	DurationInMS	Live	Fetches	MediaAged	LastReplay	MediaTyped
2017-04-25 13:41:43.2570000	2016-02-11 23:00:00.0000000	65850	1	0	0	NULL	1
2017-04-25 13:41:43.7100000	2016-02-11 22:00:00.0000000	49410	1	0	0	NULL	1
2017-04-25 13:41:44.0400000	2016-02-11 21:00:00.0000000	101470	1	0	0	NULL	1
2017-04-25 13:41:44.6530000	2016-02-11 20:00:00.0000000	124890	1	0	0	NULL	1
2017-04-25 13:41:45.3430000	2016-02-11 19:00:00.0000000	119760	1	0	0	NULL	1
2017-04-25 13:41:46.0400000	2016-02-11 18:00:00.0000000	89610	1	0	0	NULL	1

When the interaction was inserted to the ingest folder

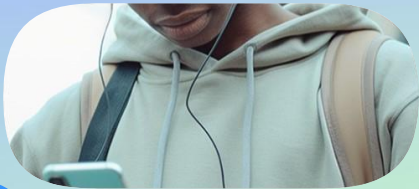
Audio Present : 1
Audio not Present : 0

Media creation date - when the interaction was inserted to the ingest folder.

Live – 1 means audio is in the system & can be playback, 0 means audio was deleted.



SOURCE MEDIA
TICK TABLE



Source Media Tick

SourceMediaTick	
SourceMediaId	
LastIngestSessionTick	
Live	
IngestSessionBridgeId	
ErrorCode	

Call ingest status

-1 new interaction has not been ingested

LastIngestSessionTick	Calls
-1	4
4007	54479212

Processed Interactions

Inactive interactions

LastIngestSessionTick	Calls
-1	122
5	18
6996	110159145
6997	188099527
2147483647	412257

Processed Interactions

```
SELECT Distinct LastIngestSessionTick,Count(*) calls from [NexidiaESI].[dbo].[SourceMediaTick]
group by LastIngestSessionTick
```

The numbers of the session tick are the priority for the ingest flow; it will process from the lowest number to the highest number.

We run the script(SELECT Distinct LastIngestSessionTick,Count(*) calls from [NexidiaESI].[dbo].[SourceMediaTick]group by LastIngestSessionTick)

to see the volume of calls in each unique LastIngestSessionTick values . You can see the output results here.

Last ingest session tick: -1 -> new interaction has not been ingested First, it will process -1 then 5.

How many calls are waiting to be processed is -1 (122). The calls between 5-6996 are in the middle of processing, 6997 are processed

Session tick id in the billion means that those interactions are not alive any more from the ingest perspective.



Ingest Flow Error Codes

Error Code	Description
0	Success
1	Unknown ingest error
10	Unknown pat creation error
11	Cannot convert media
12	Cannot access media
13	No audio in media
20	Media too short
21	Non-speech percentage too high
30	Unknown search error
31	Search error
40	Structured query management error

	SourceMediaId	LastIngestSessionTick	Live	IngestSessionBridgeId	ErrorCode
312	312	9	1	4	0
313	313	9	1	4	0
314	314	9	1	4	0
315	315	9	1	7	0



Error code 0 means that the search grid successfully processed the interaction

Error code – if the SG (search grid) failed to process the interaction, in the error code column, we will see the error code reason for that failure. Error code 0 means that the search grid successfully processed the interaction



Relevant Tables for the Ingest Flow

```
SELECT * FROM [NexidiaESI].[dbo].[SourceMediaTick]
```

	SourceMediaId	LastIngestSessionTick	Live	IngestSessionBridgeId	ErrorCode
312	312	9	1	4	0
313	313	9	1	4	0
314	314	9	1	4	0
315	315	9	1	7	0

```
SELECT * FROM [NexidiaESI].[dbo].[IngestSessionBridge]
```

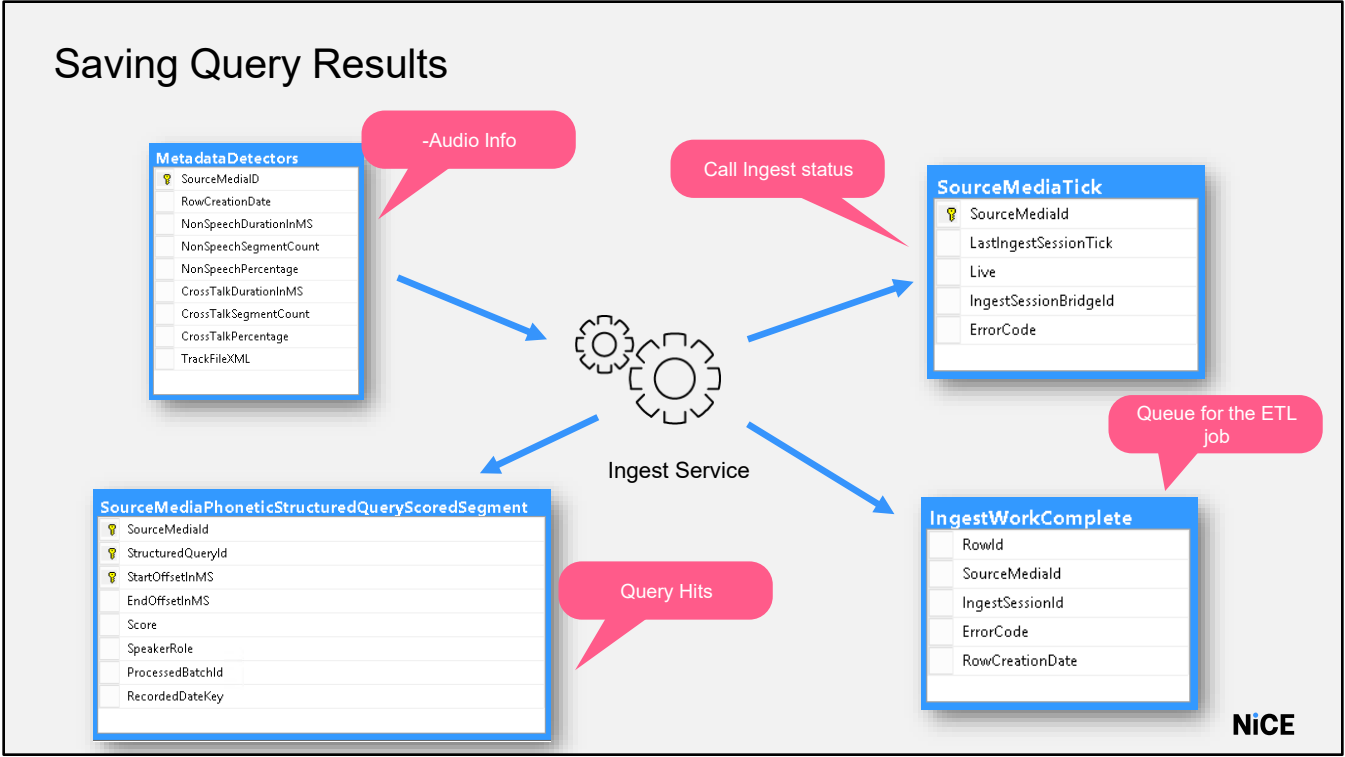
	IngestSessionBridgeId	IngestSessionId
1	2	6
2	3	6
3	3	7
4	4	6
5	4	8

It's important to identify which session is ingesting the call. Each interaction is attached to a session.
We can find that session using the IngestSessionBridge Table.



SAVING QUERY
RESULTS





Once the ingest service retrieves the search results, it will update the query hits for each media type, the error code if there were errors during the ingest flow, etc. and it will update the ingest work complete table.

The following tables will be updated:

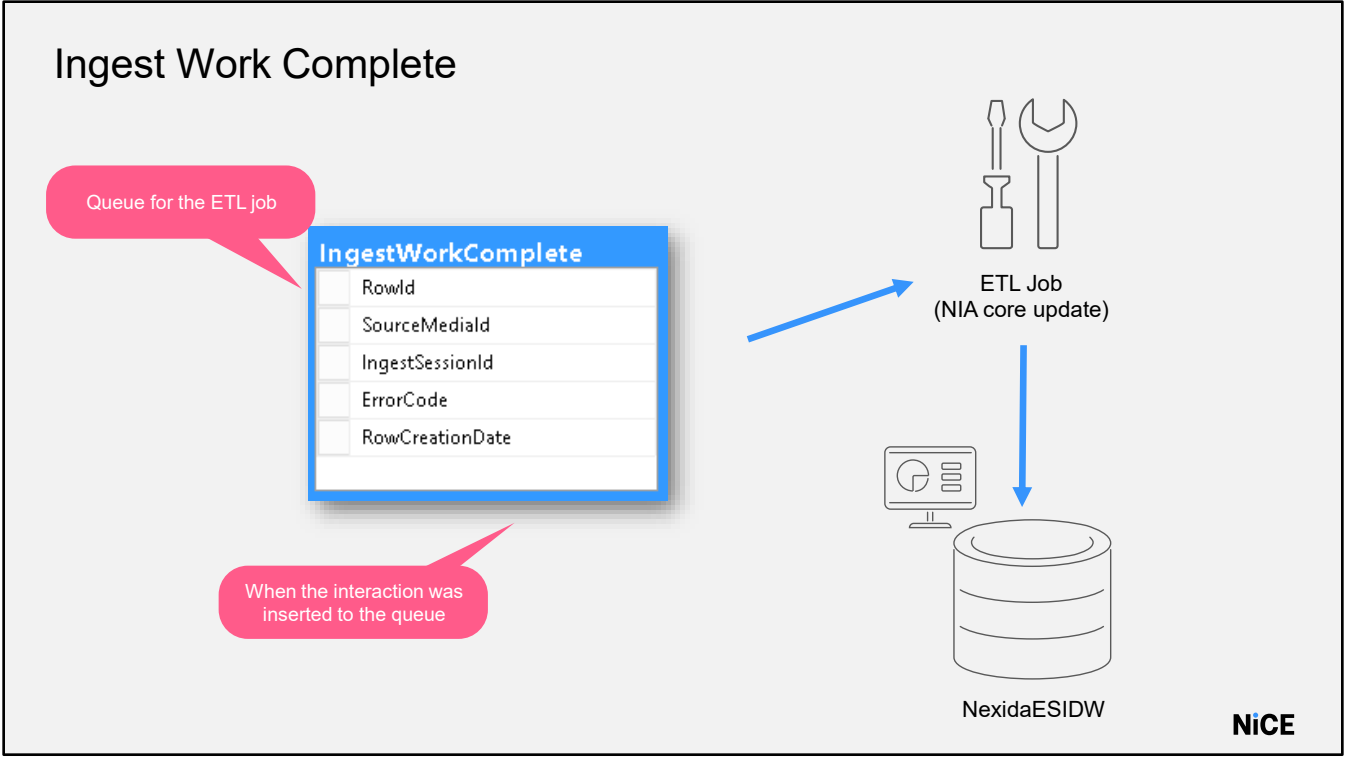
Metadata detector- contains information on the audio such as speech duration cross talk, etc.

Source media phonetics structured query – contains the query hits that were found on the interaction media.

Source Media Tick- the ingest service will update the session tick id to the latest one and update the error code.

IngestWorkComplete – this is the queue for the ETL job part of the NIA core update SQL job





This table is the queue for the ETL job- the ETL job will move all the interactions with error code 0 (meaning successful) to the NexidiaESIDW DB.

Once the ETL job moves the interaction, it will delete it from this table.

Row creation data- we can see how long the interaction is waiting to be process by the ETL job.

If the ETL job processed all the interactions, this table would be empty.



Do it Yourself

- Stop the Nexidia Search Grid Agent Service located on the Grid3 server
- Start the Analytics services in NIA Server
- Create a new session, and point it to the “**Billing**” media folder, select queries to apply on the interactions
- Stop the Nexidia Ingest Service (Nia Server)
- How many interactions appear in the **SourceMediaInbox** table (NexidiaESI DB)?



Do it Yourself

- How many interactions appear in the **SourceMedia** table (NexidiaESI DB)?
- Start the Ingest Service, login to the Session Status tab in the application, and wait until the ingest is done processing all the interactions in the session(click on the update result and reactivate the session option).
- How many interactions appear in the **SourceMedia** table (NexidiaESI DB)?
- What is the value of the 'Search Grid ID 'column in the **SourceMedia** table?
- What is the value of the 'last ingest Session Tick 'column in the **SourceMediaTick** table?



Do it Yourself

- Start the Nexidia Search Grid Agent Service located on the Grid3 server
- What is the value of the 'Search Grid ID 'column in the **SourceMedia** table?
- What is the value of the 'last ingest Session Tick 'column in the **SourceMediaTick** table?



Troubleshooting Scenario:

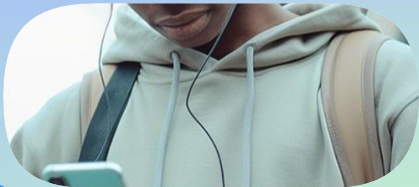
- ACME is a call center with NIA+QC. Steven Mills, who is working as a Team lead, needs to import all the calls in **TS1Lab** media folder in NIA. When he tried to do so, he was not able to find the “Session management” option in the administration menu. Make sure that he can import all calls in the media folder. If he is not able to do so, identify which calls were not imported and why?_____

Credentials
Username :smills
Password :password

You may need to reset the password. You can reset to the same “**password**”.



TROUBLESHOOTING STEPS



Unable to Access Media File – Ingest Service

ERROR - Ingest.Infrastructure.WorkExecution

Thread: FetchWorker1

Context: [sourceMediaId=9693551, ingestFilePath={full path and filename}, leaseId=43819448]

Details:

MediaFetching: **Could not fetch media:** destinationFileName={full path and filename}.fetch

System.IO.IOException: **The specified network name is no longer available.**

at System.IO. _Error.WinIOError(Int32 errorCode, String maybeFullPath)

at System.IO.File.InternalCopy(String sourceFileName, String destFileName, Boolean overwrite)

at System.IO.File.Copy(String sourceFileName, String destFileName)

at Nexidia.Core.MediaFetcher.UncMediaFetcher.FetchMedia(String externalMediaId, String destFileName)

at

Nexidia.Core.Ingest.Pipeline.IngestWorkExecutionStage.Internal.MediaFetchingFilter.FetchMedia(IEnumerafetchers, WorkExecutionContext context)

Log File: Ingest Log

This is usually reported, if there is an issue attempting to access the media file. This could be due to a network error or access issue to the folder where the media file is located.



Unable to Open Media File - Ingest Service

ERROR - Ingest.Activity

Thread: ResultWorker

Context: [resultBatchId=027b5593-35aa-4db7-84d5-f1da75c7e8b0
sourceMediaId=9658033, ingestFilePath={full path and
filename}, leaseId=43783930]

Details:

processingUnsuccessful: errorCode=**NXE_FILE_OPEN_ERROR**,
leaseDuration=00:00:10.1780000

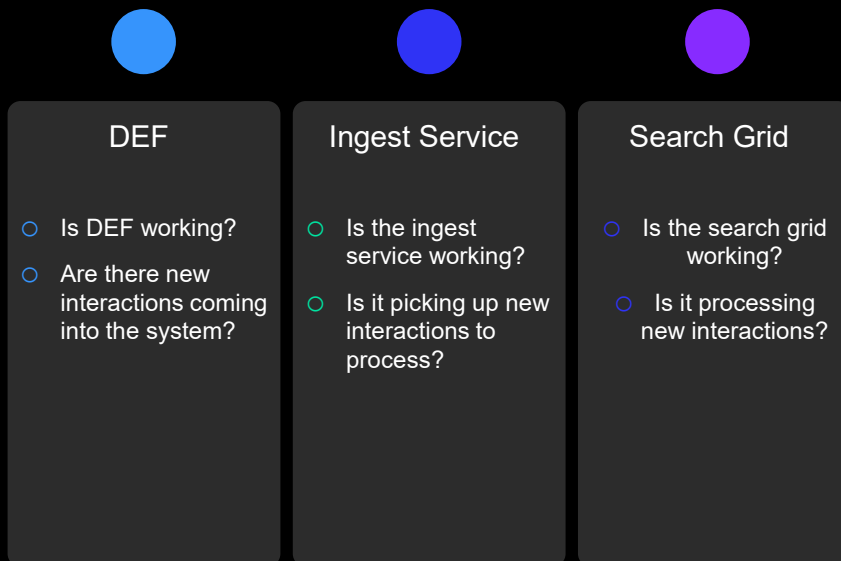
Log File: Ingest Log

In this case the media file was found by the ingest service. However, the application was not able to open the file, either because it is not a valid audio file or because is not a CODEC that the application is able to read.





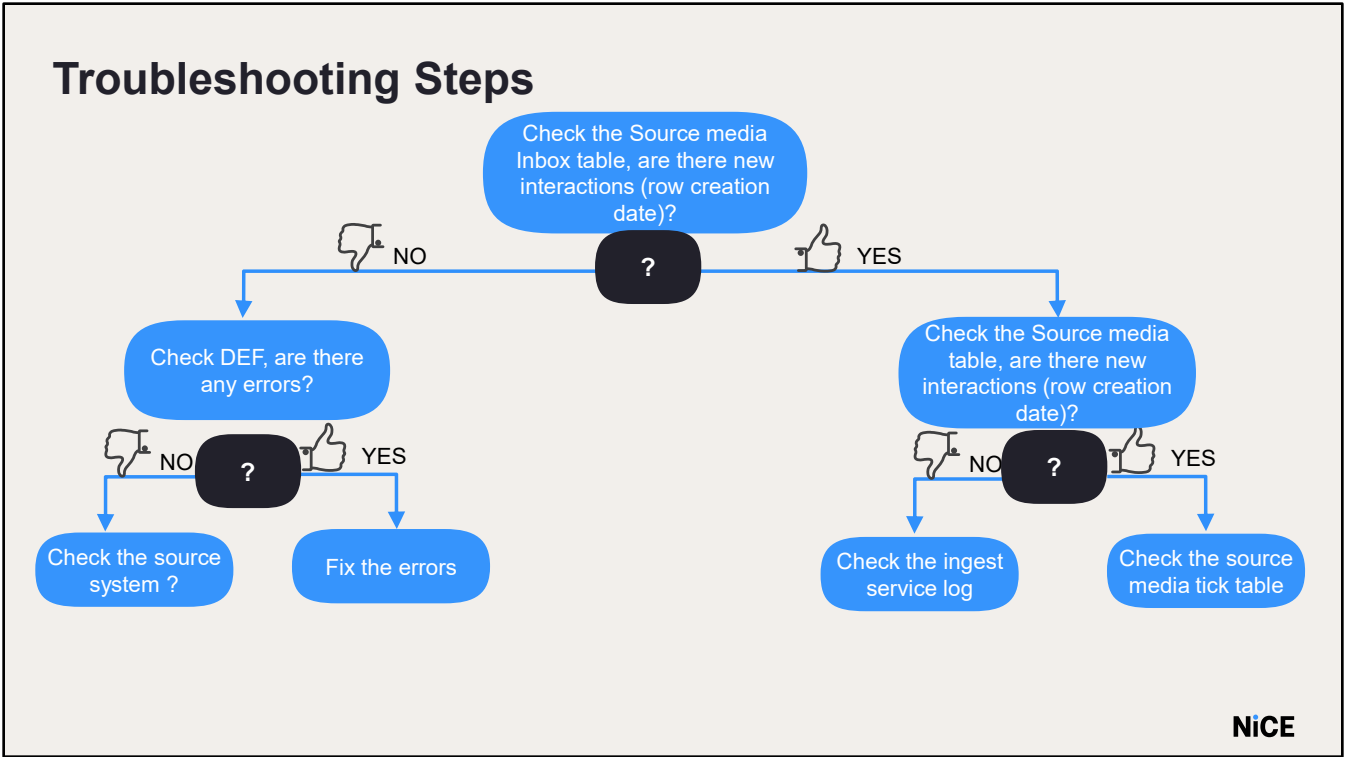
Troubleshooting Steps – Ingest Flow



When the customer reports that there are no new data in the system, you should start the troubleshooting process by investigating the three main parts of the ingest flow:

1. Is DEF working? Are there new interactions coming into the system?
2. Is the ingest service working? Is it picking up new interactions to process?
3. Is the search grid working? Is it processing new interactions?



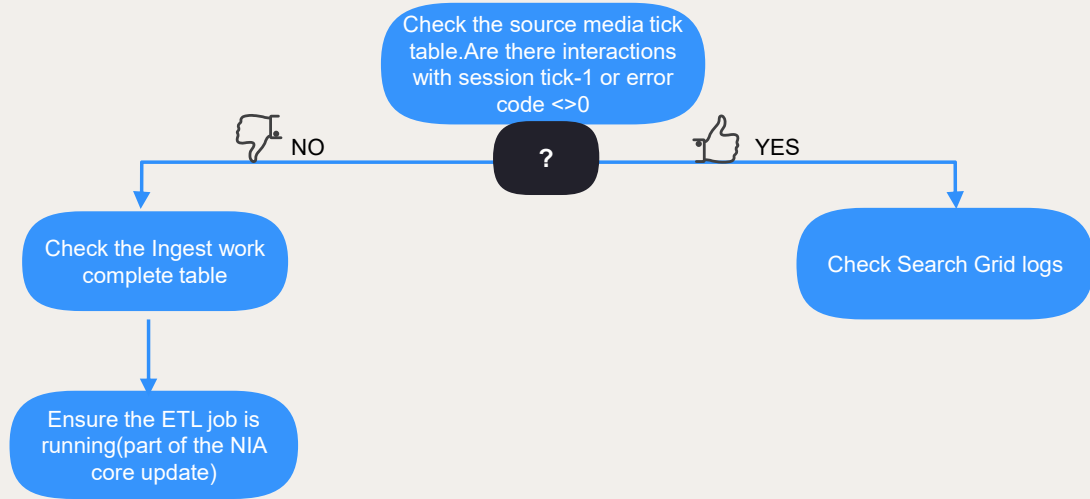


Start by checking the Source media inbox table to check if DEF imports new interactions to the system. You can use the raw creation date column to check when was the last time that DEF updated this table. If you see that there are no new interactions in this table, continue the investigation by checking the DEF server. If there are no errors on the DEF server, most likely, the problem is with the source system.

If the inbox table contains new interactions, check the source media table, did the ingest service picked up new interactions to process. If the answer is no, check the ingest service log for errors.



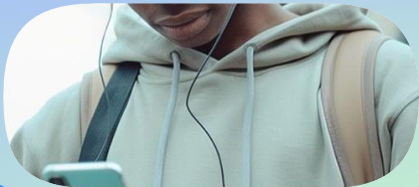
Troubleshooting Steps



If the ingest service picked up new interaction to process, check the source media tick table/ if you find many interactions with error code <> 0 or with session Tick -1 continue the investigation by looking at the search grid logs.

If everything seems to be Ok check that the ETL job is running, you can also verify it by looking at the ingest work complete table use the row creation date to see how long the interaction is “waiting” to be proceeded by the ETL job

REVIEW
QUESTIONS



Review Question #1

What is the Source Media Inbox Table?

Track the status of each interaction from the ingest perspective

Inventory of all the interactions that were imported by the DEF server (processed interactions, and waiting to be processed)

Inventory of all the interactions that were discovered by the ingest service

Review Question #2

What is the Source Media Table?

Track the status of each interaction from the ingest perspective

Inventory of all the interactions that were discovered by the ingest service

Inventory of all the interactions that were imported by the DEF server (processed interactions, and waiting to be processed)



Review Question #3

What is the Source Media Tick Table?

Track the status of each interaction from the ingest perspective

Inventory of all the interactions that were discovered by the ingest service

Inventory of all the interactions that were imported by the DEF server (processed interactions, and waiting to be processed)



Review Question #4

What does ' Last Ingest Session Tick' -1 means?

New call that is waiting to be
ingested

summary

- Ingest Flow
- Ingest Main Tables in the Database
- Troubleshooting Steps for the Ingest Flow





Thank You

DISCOVERY (ASR) FLOW

NiCE Interaction Analytics & Quality
Central Troubleshooting Fundamentals

Create a
NiCE..
world 

ASR – Advance Speech Recognition

objectives

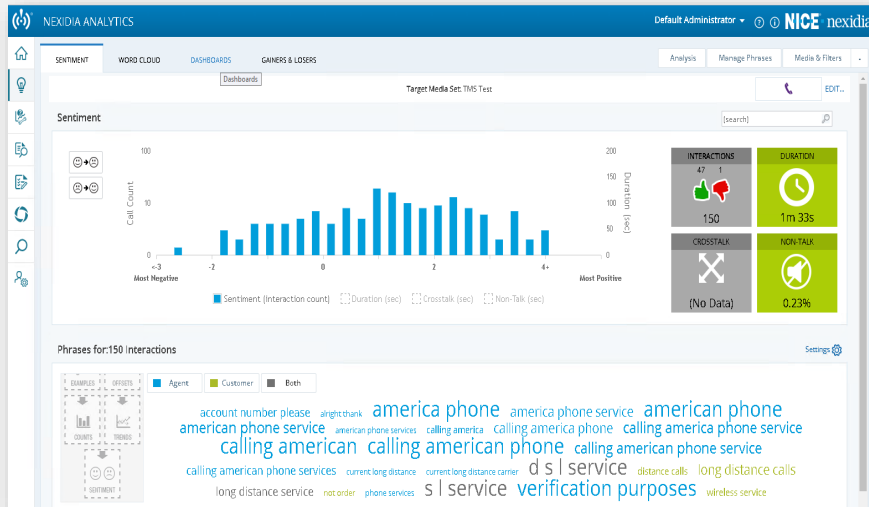
- Describe the Discovery Flow
- Describe the Sampling Process
- Describe the Discovery Flow Main Databases Tables
- Describe the Discovery Flow Common Issues



DISCOVERY FLOW
OVERVIEW



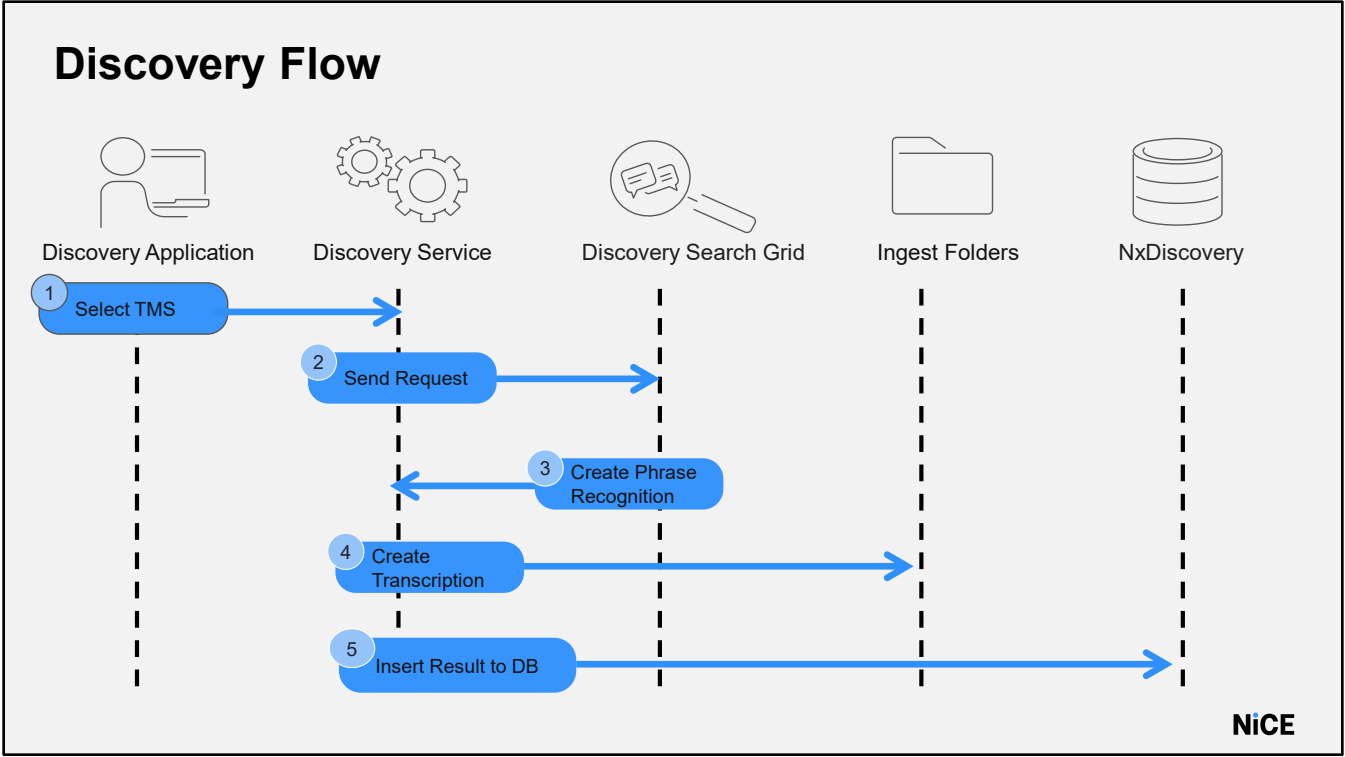
Discovery – What You Don't Know



Discovery uses a speech to text. Discovery results are not based on query results; the insights that are provided are based on the transcriptions of calls. Therefore, we don't need to build a query upfront, and instead the words cloud will show the phrases occurring on a specific set of calls.

Discovery in Interaction Analytics is the starting point for discovering and understanding interactions between your agents and customers. It is used to identify topics, trends, and themes. Based on your processed interactions, Word Clouds provide insight into the relationship between topics and their relative occurrence. The Word Clouds show the top phrases and apply sentiment analysis to them.





Key Outputs:

Topic & Trend Detection: Automatically identifies themes for proactive analytics.

Related Phrases: Maps topic relationships for root cause analysis.

Word Cloud & Sentiment Analysis: Visualizes frequent terms and sentiment trends.

Call Transcription: Generates and stores transcripts (NxPr files).

Workflow:

User selects a TMS in the Discovery app.

Discovery service sends request to the Search Grid (ASR).

ASR creates a Phrase Recognition Index and returns it.

Discovery service uses it to generate transcription.

Results are stored in the Discovery database.





1. The process starts with the NexidaESIDW database. The calls need to be ingested and ETL to the NexidaESIDW.
2. The sampling process looks for calls that are candidates for the Discovery process according to the TMS definitions and the sampling rate. The process will collect those calls and insert them into NXDiscovery DB.
3. The Discovery service will check the NxDiscovery database for calls to process and will send a request to the ASR\Discovery SG to process those calls.
4. The Discovery service will update the NXDiscovery database and will generate the sentiments score and will create the transcription file.
5. The sentiment score will be stored in NexidaESIDW so they will be available in the reports



Discovery Flow Logs

```

2021-07-13 22:38:30,880 255828 DEBUG 25 DataAgingHandler - Beginning to process a batch from the Discovery aging cleanup queue [(null)] DxLog.Infrastructure
2021-07-13 22:38:30,880 255828 DEBUG 25 DataAgingHandler - Finished processing a batch from the Discovery aging cleanup queue [(null)] DxLog.Infrastructure
2021-07-13 22:38:31,666 256614 DEBUG 6 Cannot start ASR Job because already running with request ID 146484. [(null)] Nexidia.Esi.ServiceExecutionFramework.ContinuousJobs.SimpleJobStepRunner
2021-07-13 22:38:31,666 256614 DEBUG 6 PausePipeline:False [(null)] DxLog.Infrastructure
2021-07-13 22:38:32,181 257129 DEBUG 24 No requestId was found [(null)] Nexidia.Esi.ServiceExecutionFramework.ServiceExecutionStateManager
2021-07-13 22:38:36,693 261641 DEBUG 6 Cannot start ASR Job because already running with request ID 146484. [(null)] Nexidia.Esi.ServiceExecutionFramework.ContinuousJobs.SimpleJobStepRunner
2021-07-13 22:38:36,693 261641 DEBUG 6 PausePipeline:False [(null)] DxLog.Infrastructure
2021-07-13 22:38:37,202 262158 DEBUG 24 No requestId was found [(null)] Nexidia.Esi.ServiceExecutionFramework.ServiceExecutionStateManager

```

No requestId Found

Request phrase recognition

```

2021-07-13 22:44:10,939 595887 INFO 6 Declaring media: \\NIA\Production Media\CS Tier 2\05212005-594.wav [(null)] DxLog.Infrastructure
2021-07-13 22:44:11,032 595900 INFO 6 Requested phrase recognition: ExternalMediaId=\\NIA\Production Media\CS Tier 2\05212005-594.wav, RequestId=f0ea312f-f2ae-4f5b-bc41-

```

Let us see about the Discovery Log files:

The Discovery flow logs can be found at:

C:\ProgramData\Nexidia\Analytics\Logs\Nexidia.Discovery.WindowService.SupportLog.txt

1. When the queue is empty and there are no calls available for discovery processing you will be seeing No requestId was found message
2. Once new calls have been identified for discovery processing, the discovery service will see those new calls in the queue and send a task to the search Grid declaring the new call media for phrase recognition



Discovery Flow Logs

Exporting results for media

2021-07-13 22:45:05,275 650223 DEBUG 5 Exporting results for media: ExternalMediaId=\\NIA\Production Media\CS Tier 2\05212005-594.wav [(null)] DxLog.Infrastructure

Phrase recognition completed

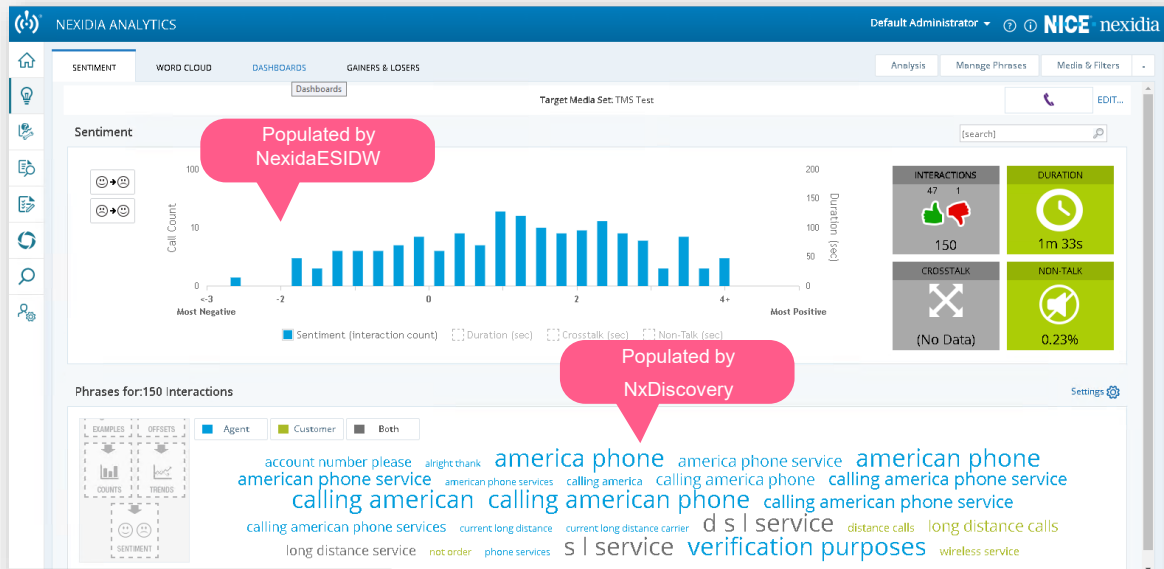
2021-07-13 22:45:06,018 650966 INFO 23 Phrase recognition completed.: ExternalMediaId=\\NIA\Production Media\CS Tier 2\05212005-594.wav, RequestId=f0ea312f-f2ae-4f5b-bc41-9d7854f4af06, Purpose=Discovery, Stereo=False [(null)] DxLog.Infrastructure

NiCE

1. Exporting the results for media for phrase recognition to be saved along with the audio in the production media folder(Nxpr file)
2. Once the discovery is completed, Success message will be shown



Discovery



NICE

Each part of the Discovery application is populated by different databases.

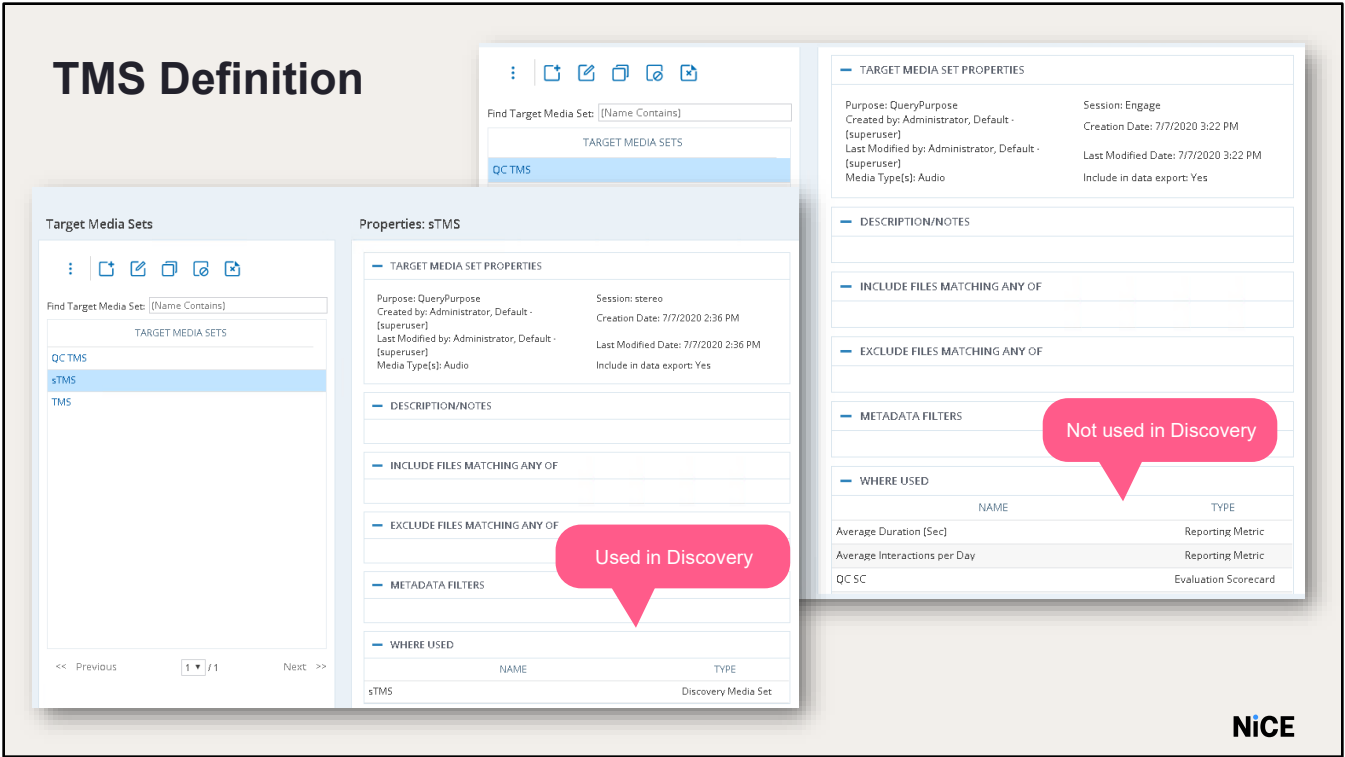
The word cloud is populated with data from NxDiscovery database, and the sentiment graph is populated with data from NexidaESIDW database.

If there is data in sentiment graph but not in the word cloud this means that Discovery DB is not updated with the result, but the NexidiaDW is updated



TARGET MEDIA
SET





Everything starts with the TMS, for Discovery to process calls, at least one TMS should be defined and enable for Discovery(check “Where Used”).

If the TMS is not set to be used in discovery, it means that the sampling process will not use it to send calls for discovery.

In the TMS definition, we can see the session associated with the TMS if we need to check if calls are ingested.

If it is being used for QC it must have a reporting metric selected.



TMS Definition

SENTIMENT WORD CLOUD DASHBOARDS GAINERS & LOSERS

Media Set Administration

Save

In the table below, pick the Target Media Sets to sample, pick a sampling rate for each one and activate the Media Sets before clicking Save.

Target Media Set Name	Sample Rate	Active	Dispositioning Enabled	Phrase Pack	Phrase Group
QC TMS	0	☐	☐		
sTMS	100	☒	☐	Telco 0.0.0	GG
TMS	100	☐	☐	Telco 0.0.0	GG



You can only apply one phrase pack on a set of calls. If two TMS are using the same set of calls, they should use the same phrase pack

The “Media Set Administration” tab in the Discovery application, will present all the TMS that are defined in the system.

Enable the relevant TMS for ASR or disposition. Disposition enables the query builder

When you enable a target media set for query dispositioning, it consumes approximately 15GB per set.

Avoid enabling dispositioning on overlapping data sets or sets not used for query building.

Note: you can only apply one phrase pack on a set of calls. If two TMS are using the same set of calls, they should use the same phrase pack as well.



SAMPLING
PROCESS



TMS in Discovery DB



NIA-Discovery-Update
job(Discovery ETL)

TargetMediaSet

TargetMediaSetId

ParentTargetMediaSetId

Name

Description

IsAsrEnabled

AsrSamplingRate

MaxSampledMediaFileKey

MaxSampledMediaForQueryBu...

LanguagePackId

PhraseGroupId

DispositioningEnabled

PhrasePackId

Deleted

MediaType

IsCjsGenerated

RowCreationDateUtc

NamedSetDeleted

MediaReservationIdentity

	TargetMediaSetId	ParentTargetMediaSetId	Name	Description	IsAsrEnabled	AsrSamplingRate	MaxSampledMediaFileKey
1	1	1	TMS		0	100	100
2	2	2	sTMS		1	100	166
3	3	3	QC TMS		0	0	0

1- Enabled for ASR
0- Not enabled for ASR

The Discovery ETL job will update the TMS table in the NxDiscovery database with information on which TMs are enabled to be used in Discovery.

The ‘IsAsrEnabeled’ filed- will identify which TMSs are enabled for ASR, and the ‘DispositioningEnabled’ identify which TMSs are enabled for the VQB.

‘AsrSamplingRate’ - shows what percentage of the calls from this TMS will be sampled for ASR.

Max sample Media file key -is used by the sampling process, and it will look for interactions ID greater than that ID.



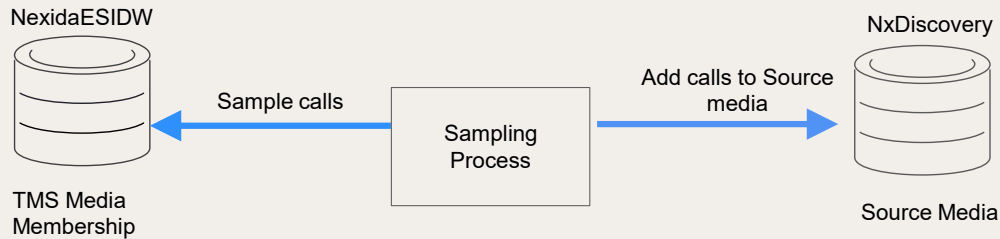
Sampling process



When the background service executed the NIA Discovery Update job, the job will check if there are any TMS that are set for ASR or Disposition.

if it finds such TMS, the NIA Discovery Update job will initiate the sampling process

Sampling process



Sampling process searches the tables in NexidaESIDW.

It will get the list of media files that are associated with the TMS that is enabled for ASR or Distortion, and based on the sampling rate, it will choose the media that will be processed in Discovery.

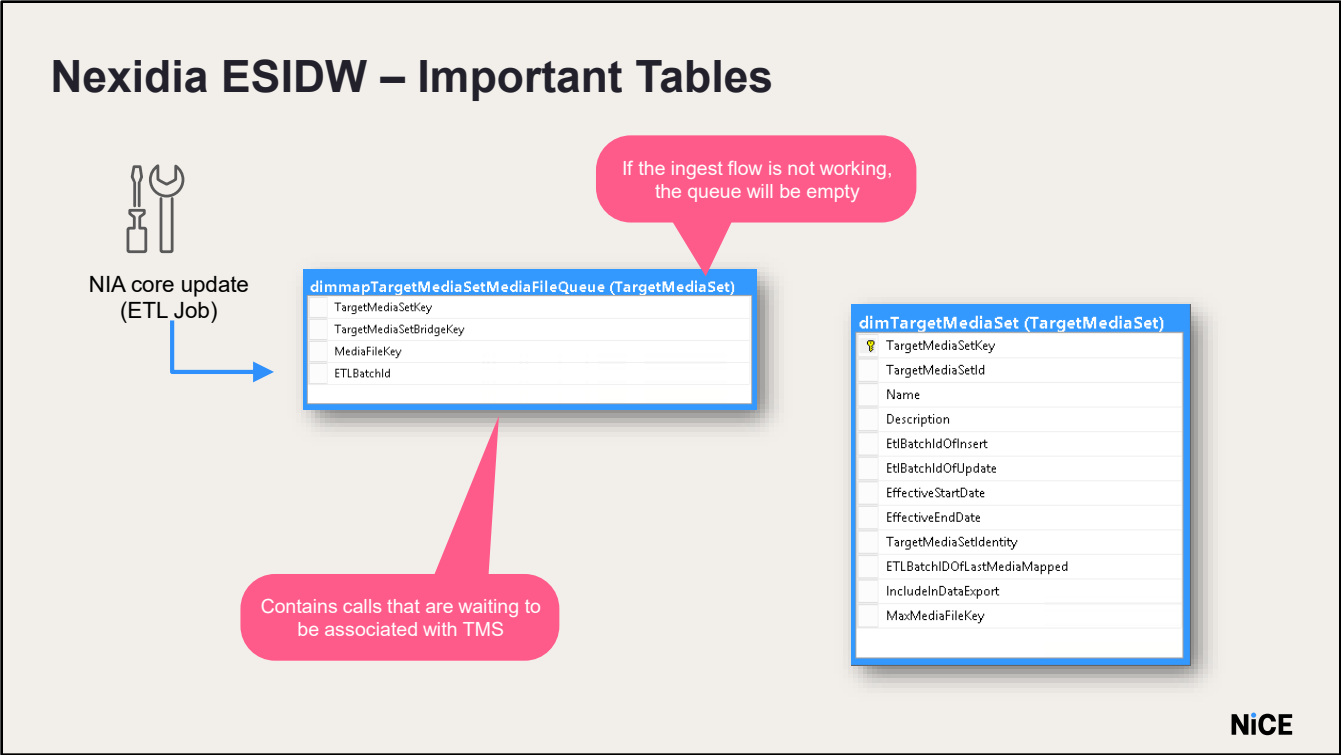
The candidate calls will be added to the Source media and Target Media Set Source Media, tables.

If you see calls assigned to TMS in NexidaESIDW but not in Discovery DB, then check if the populate is running.

If the job is running, check the background service log file – it will report any error in the sampling process

DATABASE





When the NIA Core Update (ETL job) runs, it's checking for each ingested call, should it be included in any of the TMS (according to the TMS definitions).

If a call is a match to one of the TMS, the ETL job will create an entry in the '[NexidiaESIDW].[TargetMediaSet].[dimmapTargetMediaSetMediaFileQueue]' table.

Target media Set key- to which TMS this call should be associated (Got from table '[NexidiaESIDW].[TargetMediaSet].[dimmapTargetMediaSet]' table)

ETL batch ID – ETL that created this entry in the queue.

The queue contains calls that are waiting to be associated with TMS. If the ingest flow is not working, the queue will be empty.



NexidiaESIDW – Important tables

dimmapTargetMediaSetMediaFileQueue (TargetMediaSet)	
TargetMediaSetKey	
TargetMediaSetBridgeKey	
MediaFileKey	
ETLBatchId	

Before 12.2 the Replay Service is responsible for moving the calls from the queue to the tables

From 12.2 the Background Service is responsible for moving the calls from the queue to the tables

dimmapTargetMediaSetBridgeMediaFile (TargetMediaSet)	
MediaFileKey	
TargetMediaSetBridgeKey	
RecordedDateKey	
ETLBatchID	
InsertedETLBatchID	
TMSExportTrackingKey	
TMSInsertedExportTrackingKey	

dimmapTargetMediaSetBridgeTargetMediaSet (TargetMediaSet)	
TargetMediaSetBridgeKey	
TargetMediaSetKey	
ETLBatchID	



If there is a backlog in the queue investigate the relevant log (Replay Service\ Background Service)

Once calls are associated with the TMS, Queue entry is deleted, and an entry is added to the following tables:

‘dimmapTargetMediaSetBridgeTargetMediaSet’

‘dimmapTargetMediaSetBridgeMediaFile’

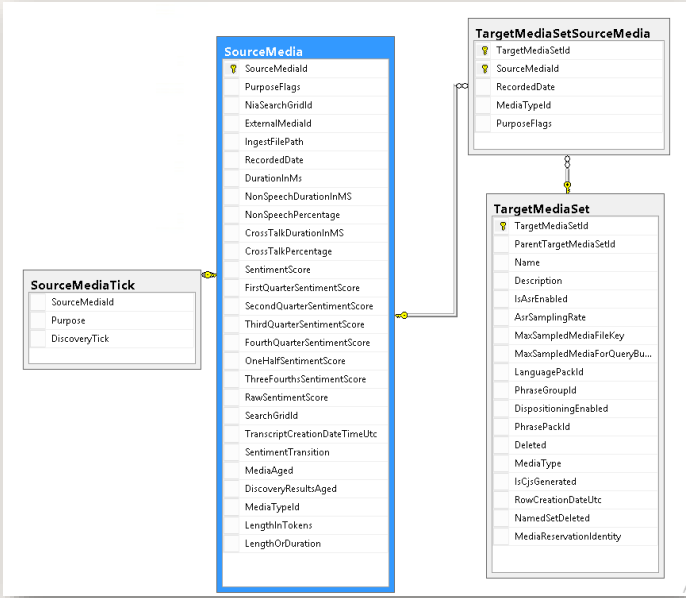
If the call is not in those tables, the discovery service will not handle the call.

Background Service is responsible for moving the calls from the queue to the tables

If there is a backlog in the queue investigate the relevant log (Replay Service\ Background Service)



NxDiscovery Tables



These are the main tables



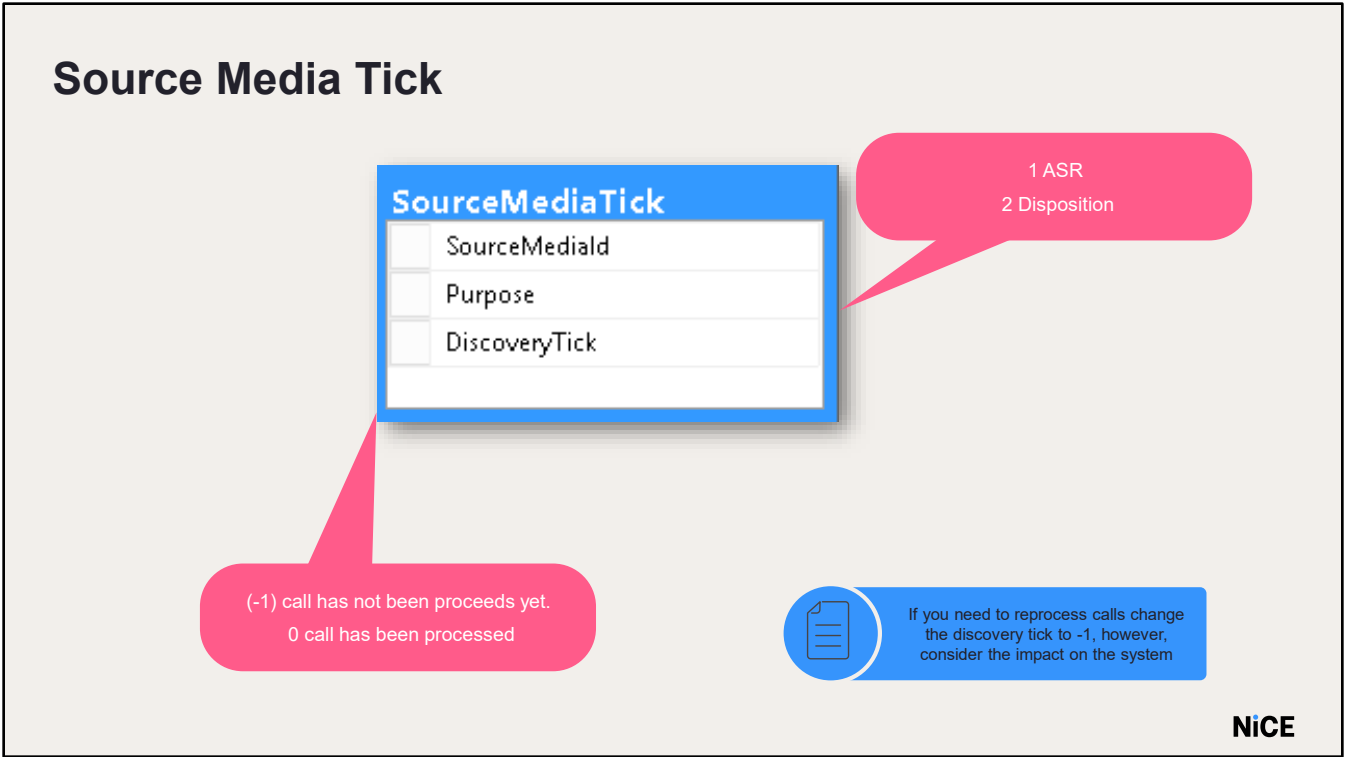
Source Media Tables

SourceMedia
SourceMediaId
PurposeFlags
NiaSearchGridId
ExternalMediaId
IngestFilePath
RecordedDate
DurationInMs
NonSpeechDurationInMS
NonSpeechPercentage
CrossTalkDurationInMS
CrossTalkPercentage
SentimentScore
FirstQuarterSentimentScore
SecondQuarterSentimentScore
ThirdQuarterSentimentScore
FourthQuarterSentimentScore
OneHalfSentimentScore
ThreeFourthsSentimentScore
RawSentimentScore
SearchGridId
TranscriptCreationDateTimeUtc
SentimentTransition
MediaAged
DiscoveryResultsAged
MediaTypeId
LengthInTokens
LengthOrDuration

If the field is null, the transcription file was not created and did not calculate sentiment score

Source media -all the calls that have been sampled for discovery includes info relevant for discovery, for example, sentiment score. Transcription creation time- if the field is null, the transcription file was not created and did not calculate sentiment score





Discovery tick (-1) call has not been processed yet, 0 call has been processed

We can have a call with process 0 transcription null, maybe we have problem with the transcription process (check log)

Purpose – 1 ASR, 2 disposition if the call is set for both ASR and Disposition, we will have two records, the same MediaId different purpose.

If you need to reprocess calls – change the discovery tick to -1
But consider the impact on the system



Target Media Set Source Media

Which media associated with which TMS

TargetMediaSetSourceMedia

- TargetMediaSetId
- SourceMediaId
- RecordedDate
- MediaTypeId
- PurposeFlags

Make sure that the TMS is active otherwise the calls will not be processed

Media Set Administration

Save

In the table below, pick the Target Media Sets to sample, pick a sampling rate for each one and activate the Media Sets before clicking Save.

Target Media Set Name	Sample Rate	Active	Dispositioning Enabled	Phrase Pack	Phrase Group
QC TMS	0	☐	☐		
sTMS	100	☒	☐	Telco 0.0.0	GG
TMS	100	☐	☐	Telco 0.0.0	GG

Target media set source media – tells us which media belong\associated to which TMS.

Make sure that the TMS is active otherwise the calls will not be processed



Do it Yourself

- Create TMS
- How many interactions appear in the dimmapTargetMediaSetBridgeMediaFile table (NexidiaESIDW)
- Navigate to the discovery application and set the TMS you created in the previous step for ASR, with a sample rate of 100%, don't active the TMS, and save it.
- How many interactions appear in the Source Media table (NxDiscovery)



Do it Yourself

- Activate the TMS and save.
- Click on the Sentiment tab, in the Select Target Media set and filter window select the TMS and set the date range to start from the year 2000 click ok
- How many interactions appear in the Source Media table (NxDiscovery)

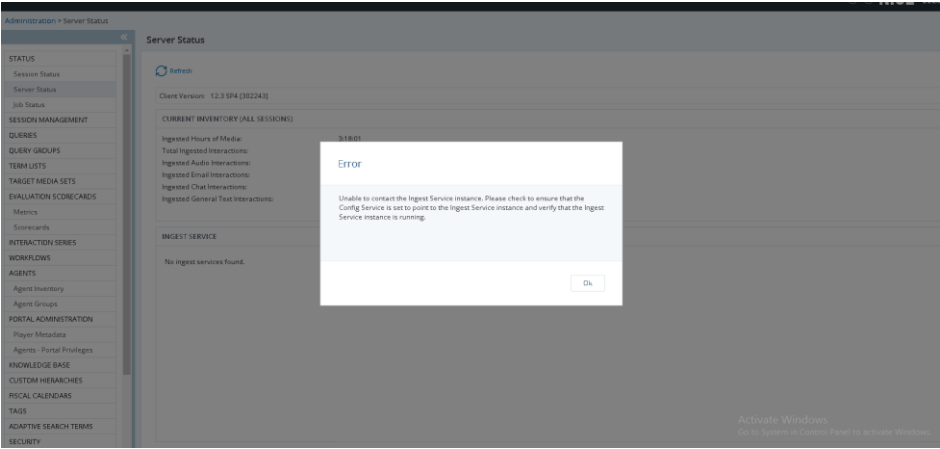
Troubleshooting Scenario:

- Steven Mills received a report from his team about an issue when they tried to create a new TMS for the interactions in “Billing” media folder. They wanted to remove some phrases from the phrase group. When they tried to view the word cloud in Discovery application and the sentiment, they could not see any data. But another TMS created for the same “Billing” media folder was producing results. The report also mentioned that there is an error in the server status indicating a problem with the ingest service as below. Steven Mills tried to verify that issue, but he was not able to do so.



Do it Yourself

Identify and fix the issues so that Steven Mills and his team can see the discovery word cloud and resolve the problem with the ingest service.



COMMON ISSUES





Can't see new result in Discovery

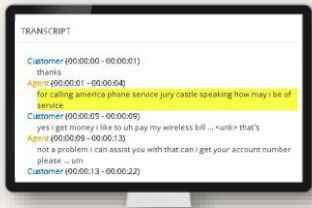
- Check if calls are ingested- if not this is ingestion issue and not Discovery issue.
- Check that the ETL Job runs, and calls are in NexidiaESIDW
- Is the TMS enabled for Discovery
- Is there a backlog in the dimmapTargetMediaSetMediaFileQueue' table? If yes investigate the relevant log (Background Service)
- Do the calls exist in NxDiscovery DB (Source Media) – if not check that the Discovery ETL job run
- Have the calls been processed by the Search Grid (Source Media Tick)
- Investigate the Discovery service log
- Investigate the Search Grid log



Can't see sentiment score

- Have the calls been processed by the Search Grid (Source Media Tick)
- Did the Discovery ETL job run and update the sentiment score in NexidiaESIDW





Can't See Transcription

- Have the calls been processed by the Search Grid (Source Media Tick)
- Investigate the Discovery log files for failure to create the transcription file



REVIEW
QUESTIONS



Review Question #1

What creates the transcription file and the sentiment score?

Discovery Search
Grid

Discovery ETL job

ASR

Discovery Service

Multiple Choice. Incorrect answers should have Fade Out animations.



Review Question #2

You log in to the Discovery application; you can see the word cloud but not the sentiment result. What might be the problem?

Either the discovery service failed to create the sentiment score, or the Discovery ETL job failed to run\did not run, and as a result, the NexidiaESIDW is not updated



Review Question #3

There are no new results in the Discovery app, and you concluded that the sampling process is not working, which log should you investigate?

Replay service log

Discovery Service
Log

Ingest Service log

Background service
log



Review Question #4

During an investigation of the Discovery process, you noticed that the 'dimmapTargetMediaSetMediaFileQueue' table is empty what may be the reason (Select all that apply)?

The Discovery (ASR) search grid is not working

The Discovery service failed to run

The Ingest flow is not working

The NIA Core update (ETL job) failed to run



Review Question #5

What is the Source Media table in the NxDiscovery DB?

Inventory of all the interactions that were associated with TMS

Inventory of all the interactions that are waiting to be processed by the Discovery Search Grid

Inventory of all the interactions that were sampled for Discovery

None of the answers is correct



summary

- Discovery Flow
- Sampling Process
- Discovery Flow Main Databases Tables
- Discovery Flow Common Issues





Thank You

SEARCH GRID

NiCE Interaction Analytics & Quality
Central Troubleshooting Fundamentals

Create a
NiCE..
world ☺



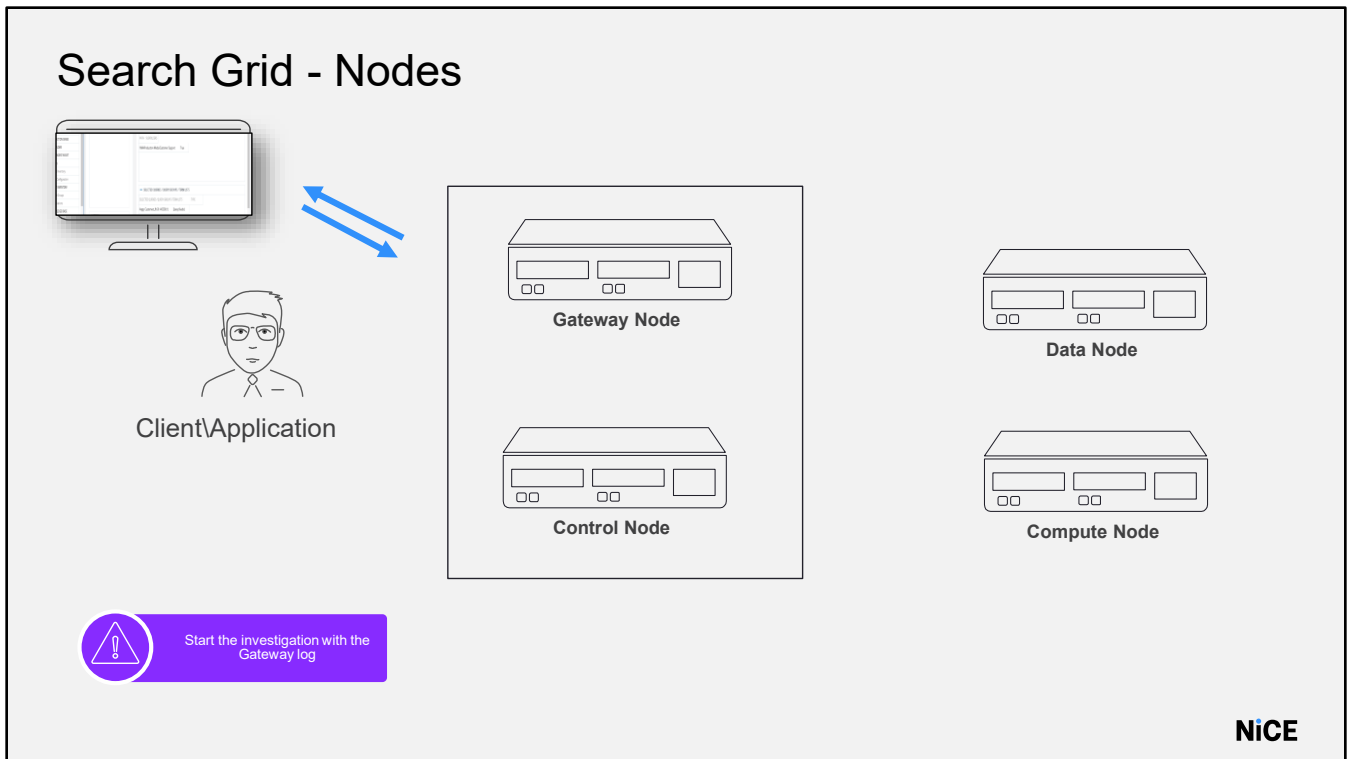
objectives

- Describe the Phonetics and Discovery Search Grid



NIA & SEARCH
GRID





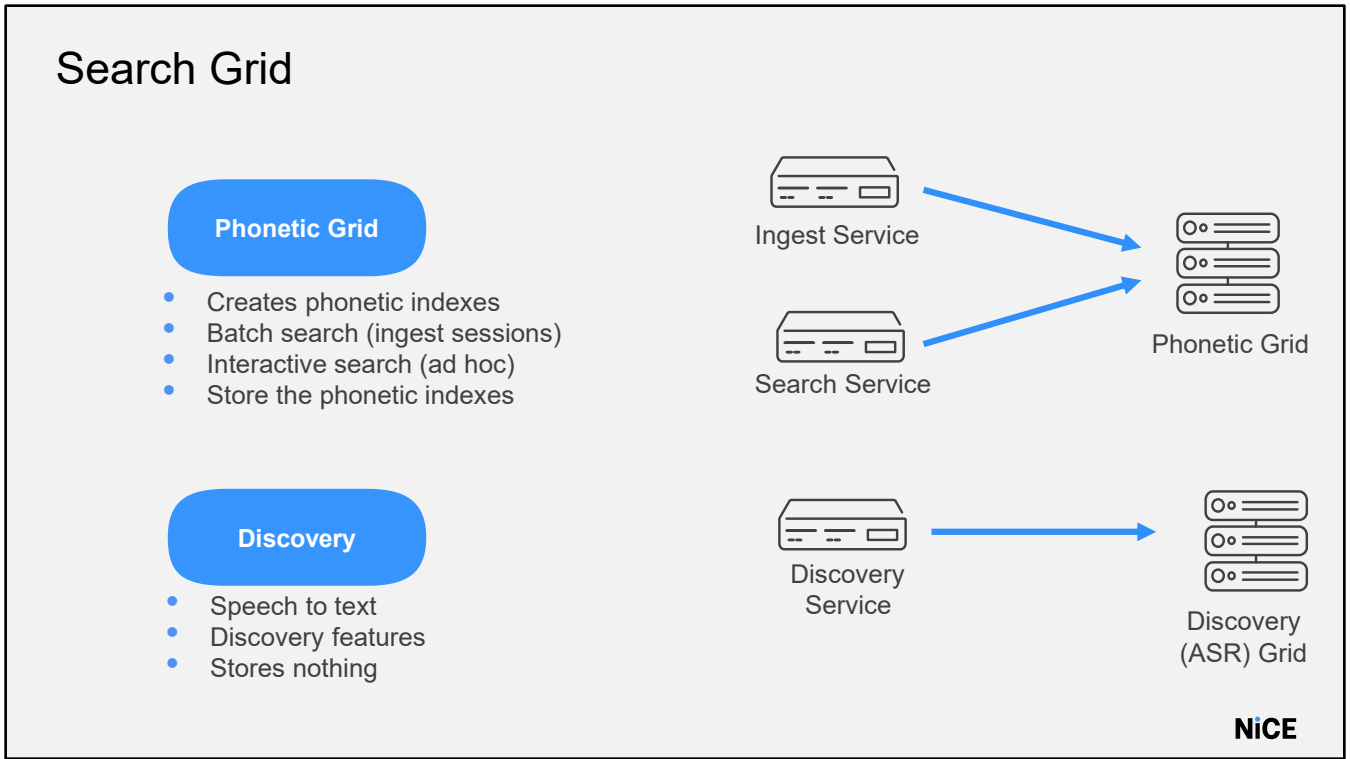
Search grid participates in the Ingest (Session) , Ad – Hoc Search and Discovery

Search Grid is the analytics engine. It is a distributed system for analyzing and searching recorded audio.

There are three types of search grids

The Search Grid is organized into fundamental deployment units called Grid Nodes or simply Nodes. There are four different types of Nodes, each type performing a different set of system functions:

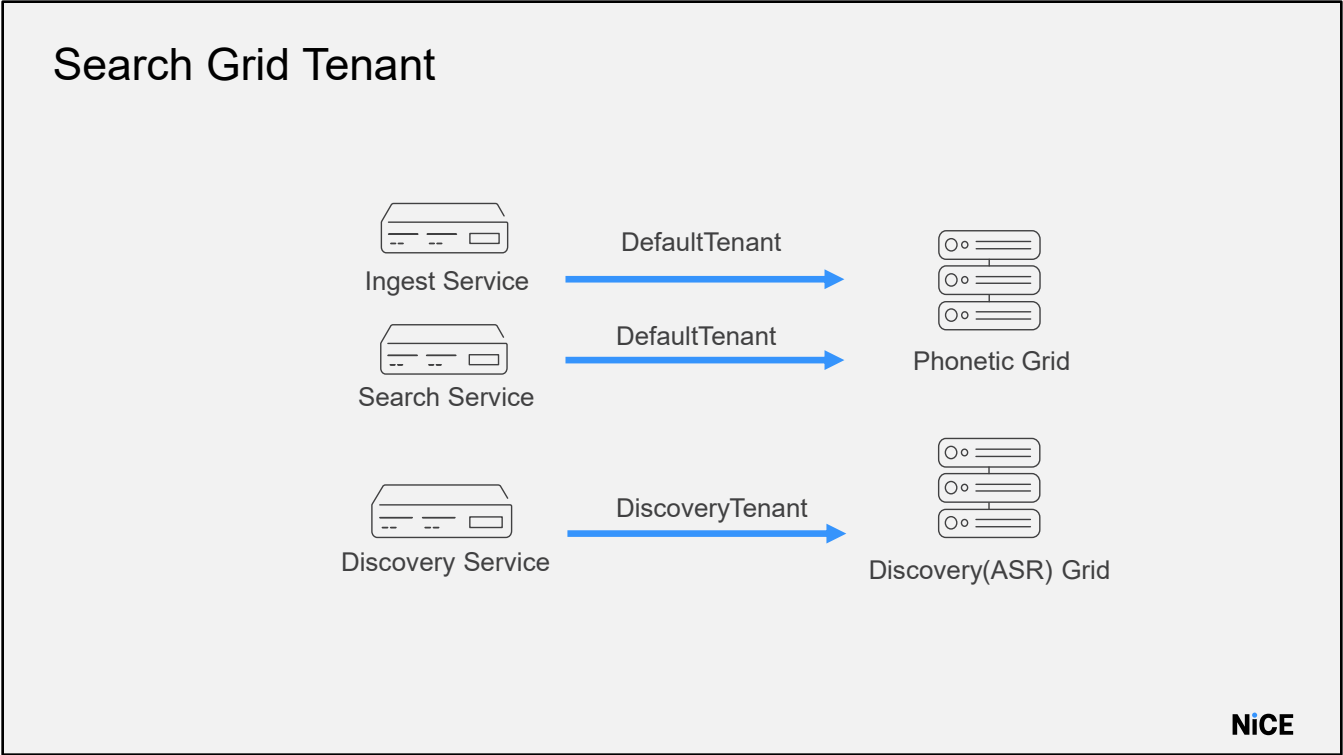
1. Gateway Node - Communicates with the client/application. Receives requests for analysis and routes the requests to the relevant Node.
2. Control Node - Stores global configuration data such as Node locations, etc..
3. Data Node - Creates the index file and performs phonetics searching, advance speech recondition action, etc.
4. Compute Node - Can be added and remove as needed, provides extra CPU power to the data node.



As seen before, two NIA services communicate with the Phonetic Search Grid: the ingest services send requests to create a pat file for new interactions and perform searches on those index files according to the queries definition.

The search service sends a request to the Phonetic Search Grid to perform ad hoc searches

The Discovery service communicate with the Discovery Grid



For performance reasons, the search grid uses different tenants for Phonetics & Discovery. Phonetics use the default tenant, and the ASR\Discovery uses the DiscoveryTenant



Search Grid - Logs

Media is declared to the gateway

gateway-service.log

```
2021-07-12 23:27:50,342 8128481144 INFO MsgHandler5 Declared media; tenant=1, id=204, externalMediaId=\\NIA\Production Media\CS Tier 1\979.wav, uri=http://10.1.159.38:25022/tenant/1/media/204 [correlationId=8e8b416f-5490-4d7e-9215-58c1eb9cd86d] c.n.j.g.r.m.c.MediaCreationTask
```

Request Media Operation

agent-service.log

```
2021-07-12 23:28:35,759 8128484804 INFO Scheduler-3 Beginning media operation [mediaLocation=file://nia/Production %20Media\CS%20Tier%201\979.wav, trackId=null, media=MediaKey [tenantId=1, mediaId=204], leaseUri=http://10.1.159.38:25122/lease/2be36221-e322-4b14-b1e3-ea4a00a8f970, mediaAccessId=57f2032d-7821-4020-9d38-2fa7d7b15172, channelId=null] c.n.j.w.c.e.MediaAccessExecutor
```

Completed Media Operation

agent-service.log

```
2021-07-12 23:28:37,103 8128486148 INFO Scheduler-3 Completed media operation; 2:02.950; 91.41xRT [mediaLocation=file://nia/Production%20Media\CS%20Tier%201\979.wav, trackId=null, media=MediaKey [tenantId=1, mediaId=204], leaseUri=http://10.1.159.38:25122/lease/2be36221-e322-4b14-b1e3-ea4a00a8f970, mediaAccessId=57f2032d-7821-4020-9d38-2fa7d7b15172, channelId=null] c.n.j.w.c.e.MediaAccessExecutor
```

Logs Locations:

- You can find the gateway logs on your GRID2 Server at D:\ProgramData\Nexidia\Search Grid\gateway-node\logs\gateway-service.log
- You can find the agent service log on your GRID3 Server at D:\ProgramData\Nexidia\Search Grid\data-node\logs\agent-service.log

1. Gateway log is the first log you will check. You can see that media is declared to the gateway service and the now the request have been given to the Data node.
2. Open agent service log on the Data node. You can see that the agent service have requested to process call
3. Agent service is processing the call



Search Grid - Logs

PAT file Creation

```
2021-07-12 23:28:51,649 8128500694 INFO Scheduler-1 Created monogest; 410ms [languagePackId=1, pat=PatKey[tenantId=1, mediaId=204, patId=1], leaseUri=http://10.1.159.38:25122/lease/f0fc07f0-0bf6-4b53-af43-2d51c68975bb] c.n.j.s.r.MonogestExecutor
```

Data Node receives the request

```
2021-07-12 23:27:50,467 8128453933 INFO Jetty-135112 Declared media; tenant=1, externalMediaId=\\NIA\Production Media\CS Tier 1\979.wav [nxRequestId=f83b61d1-3710-482f-8e5f-d9295164fbb6, remoteIp=10.1.159.28, tenantId=1, requestUri=/private/tenant/1/media/204, httpMethod=POST, mediaId=204, nxRequesterUri=nxgrid://gateway@gateway] c.n.j.s.r.MediaResource
```

Call processed successfully

```
2021-07-12 23:28:37,103 8128500569 INFO Jetty-135113 Media access transaction succeeded; 12ms [nxRequestId=9c093cf2-83bf-4c43-91a2-bbae3ff2bd0e, remoteIp=10.1.159.38, tenantId=1, requestUri=/mediaAccess/tenant/1/media/204/mediaAccess/57f2032d-7821-4020-9d38-2fa7d7b15172, httpMethod=POST, mediaId=204, nxRequesterUri=nxgrid://agent@DataNode1.data, mediaAccessId=57f2032d-7821-4020-9d38-2fa7d7b15172] c.n.j.b.r.MediaAccessTransaction
```

PAT file was accepted by Base service

```
2021-07-12 23:28:51,649 8128515115 INFO Jetty-135113 Accepted monogest [nxRequestId=94ae43de-b576-4888-98c8-eda6c177421b, remoteIp=10.1.159.38, patId=1, tenantId=1, requestUri=/monogest/tenant/1/media/204/pat/1, httpMethod=PUT, mediaId=204, nxRequesterUri=nxgrid://agent@DataNode1.data] c.n.j.s.r.MonogestResource
```

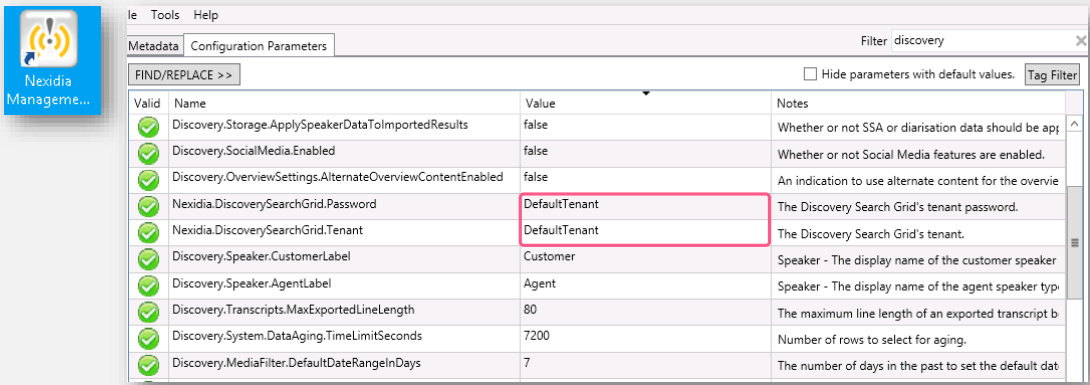
NICE

You can find the base service log on your GRID3 Server at D:\ProgramData\Nexidia\Search Grid\data-node\logs\base-service.log

1. Agent service have created the index file(PAT file)
2. The request been received by Data node.
3. Call was processed successfully
4. Base service has accepted the PAT file from the agent service



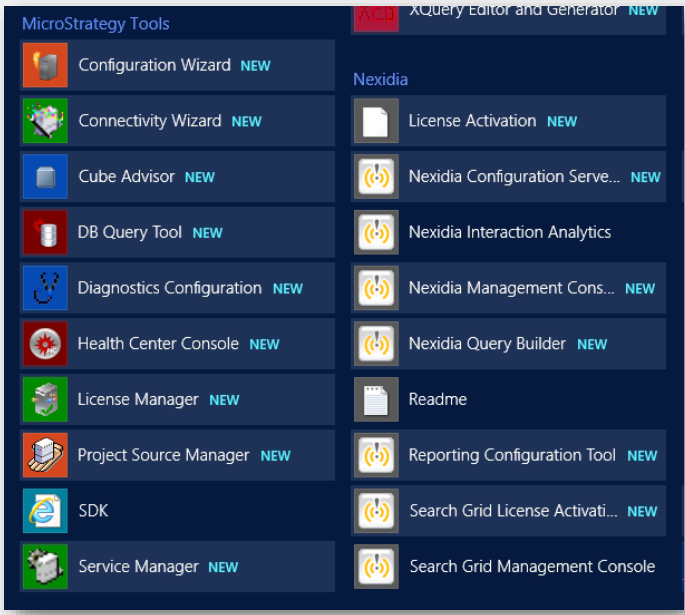
Management Console



In the Management console - filter on “discovery”
Make sure that the discovery tenant user and password are correct



SG Management Console



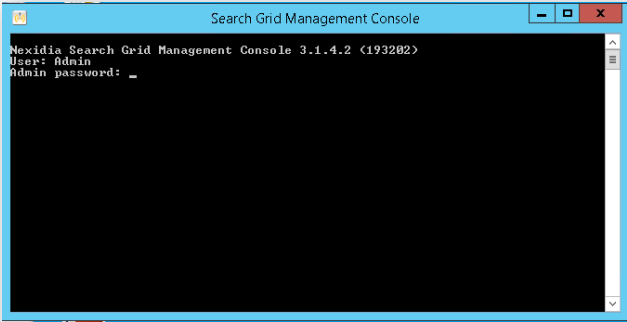
From the start menu run the Search Grid Management Console

The Search Grid includes a Management Console application that allows users to create and delete tenants, change passwords, and other useful functions.

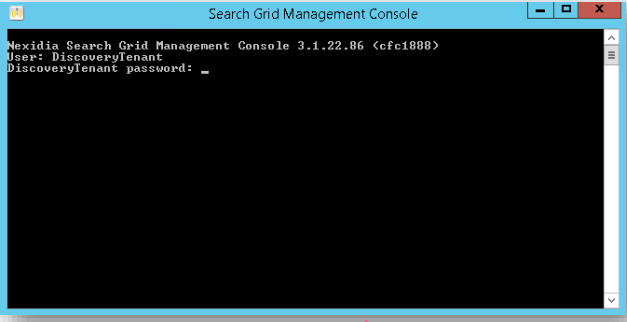
The Management Console is located on all Grid servers -<Install location>\bin\management-console.cmd.



SG Management Console



Username: admin
Password: admin



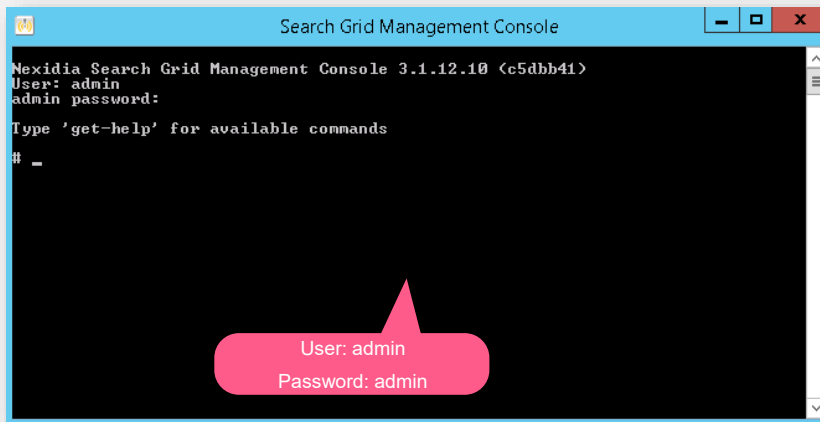
Username: DiscoveryTenant
Password: DiscoveryTenant

 Case Sensitive

- 1. Default Username and password: admin



Admin Mode



get-tenant

get-dmet

get-systop

get-cmet

NiCE

To enter Admin Mode, enter user "admin" and the admin user's password. After logging in, the prompt will be a hash symbol, '# '.

get-tenant - lists the currently defined tenants

get-systemtopology – (get-systop) shows the list of Nodes and their statuses.

get-computemetrics - (get-cmet) shows status of CPU-intensive tasks across the grid

get-datametrics - (get-dmet) shows status of persistent data storage, including searchable volume (get-dmet)



Log Files

```

</triggeringPolicy>
</appender>
- <appender class="ch.qos.logback.core.FileAppender" name="STARTUP">
  <File>${nexidia.searchgrid.log.dir}/control-service-startup.log</File>
  <Append>false</Append>
  - <encoder>
    <pattern>%d\t%-8r\t%-6p\t%-15.15t\t%m\t[%X]\t%c{1}%n%rEx</pattern>
  </encoder>
  - <filter class="ch.qos.logback.core.filter.EvaluatorFilter">
    - <evaluator class="ch.qos.logback.classic.boolex.OnMarkerEvaluator">
      <marker>STARTUP</marker>
    </evaluator>
    <onMatch>ACCEPT</onMatch>
    <onMismatch>DENY</onMismatch>
  </filter>
</appender>
<logger name="com.nexidia" level="INFO"/>
<logger name="com.nexidia.jyrus.hosting.beans.properties.GridPropertySourcesPropertyResolver" level="INFO"/>
<logger name="com.sun.jersey.spi.spring" level="WARN"/>
<logger name="org.apache" level="INFO"/>
<logger name="org.apache.curator" level="WARN"/>
<logger name="org.apache.zookeeper" level="ERROR"/>
<logger name="org.eclipse.jetty" level="WARN"/>
<logger name="org.springframework" level="WARN"/>
<logger name="org.springframework.jdbc.core.metadata.TableMetaDataProvider" level="ERROR"/>
- <root level="INFO">
  <appender-ref ref="FILE"/>
  <appender-ref ref="STARTUP"/>
</root>
</configuration>

```

Change the log level

Each service has its own log file - log location:<install location>\ProgramData\Search Grid\<Node Name>\logs

The log configuration file for each service is stored in <install location>\etc\<Node name>\<service name>-log-config.xml.

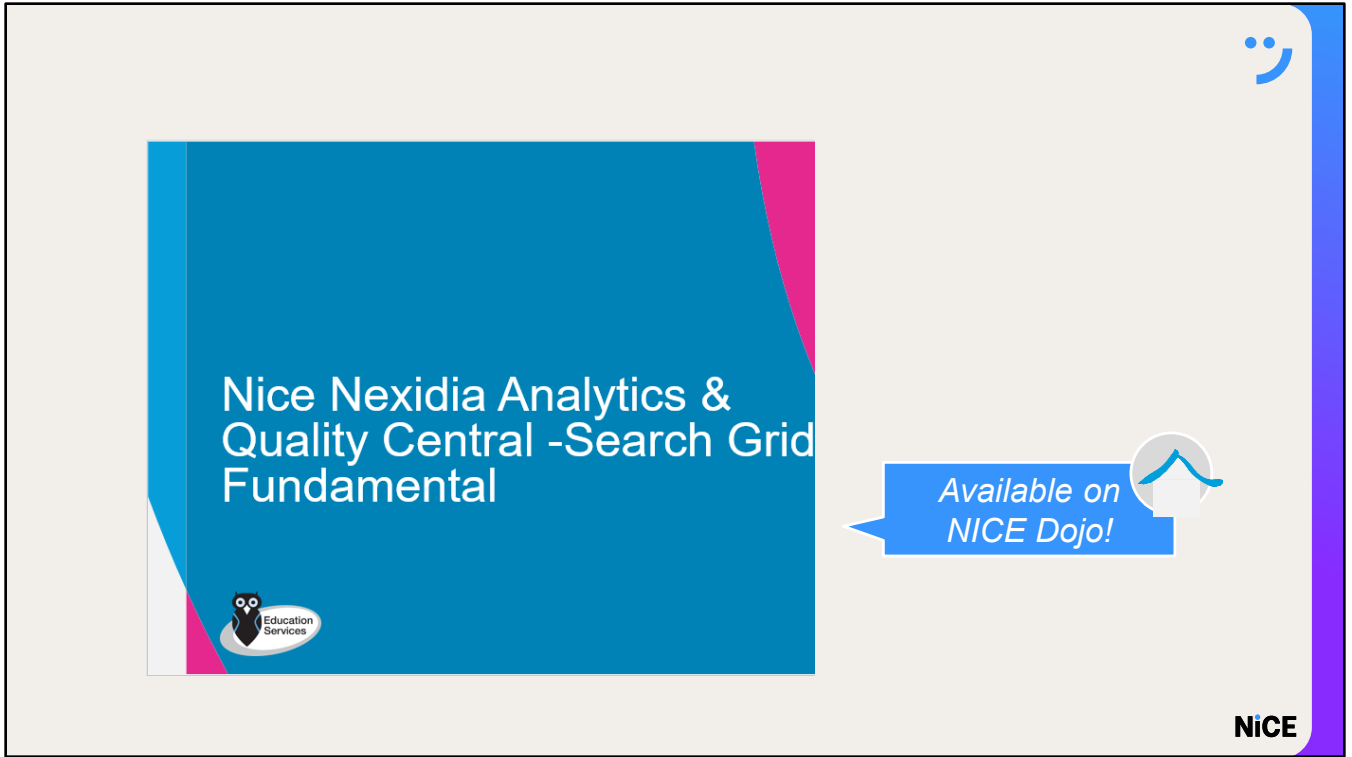
To change the log level, edit the <logger name="com.nexidia" level="INFO"/> parameter



Do it Yourself

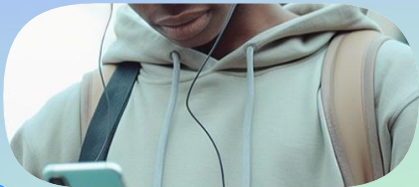
- Login to Grid 1 server, from the start menu, run the Search Grid Management console
- Login to SG Management Console with user admin & password admin
- Run the get-tenant command, how many tenants are there in this Search Grid?
- Run the get-systop command. This will show the locations of all the nodes.
- Stop the Nexidia Search Grid Agent Service, run the test-sys command
- Start the Nexidia Search Grid Agent Service, run the test-sys command





There is a Web-Based Training for the Search Grid

REVIEW
QUESTIONS



Review Question #1

The customer's system has a problem with the Ingest flow; after investigation, you conclude that the problem is with the search grid. Which search grid log should you investigate first?

Data node log

Gateway node log

Control node log

Compute node log





summary

- Phonetics and Discovery Search Grids





Thank You

Playback Flow

NiCE Interaction Analytics & Quality
Central Troubleshooting Fundamentals

Create a
NiCE
world 



objectives

- Describe the Playback Flow

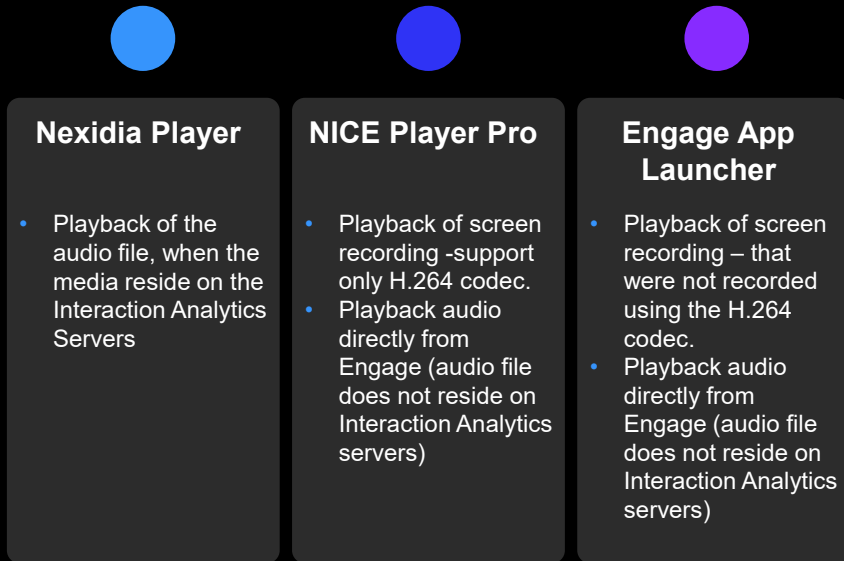


PLAYBACK
OPTIONS





Playback Options



Nexidia Player + NICE Player Pro: When Engage has Integration with Interaction Analytics, then Audio playback happened over Nexidia Player whereas ScreenRecording is getting played over NICE Player Pro

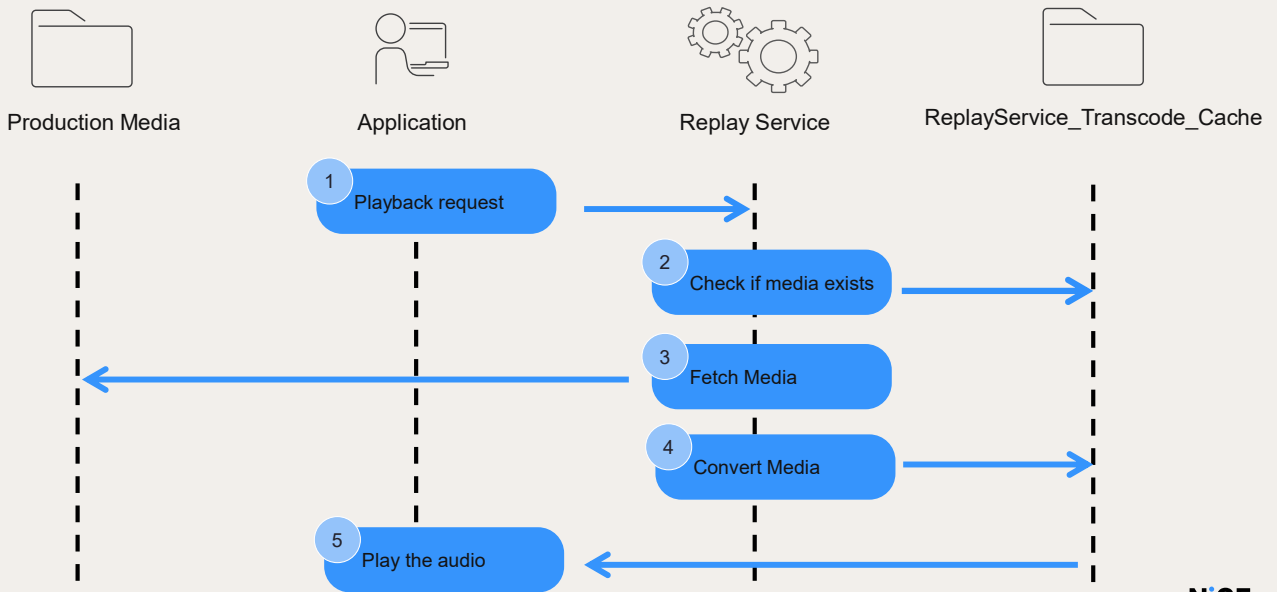
Unless NIA is integrated with Engage, only the Nexidia player will be available for use.



NEXIDIA PLAYER



Replay Flow – Quality Central Player



NiCE

1. When the user clicks on the Play button, the request is sent to the replay service.
2. The replay service checks if the audio file already exists in the “replayService_Transcode_Cache.” If so, the replay service will play the audio file.
3. If the audio file doesn't exist in the “ReplayService_Transcode_Cache” folder, the replay service will fetch it from the production media location
4. The replay service will convert the audio file to MP4 format and will create a PNG file.



Playback - Logs

No playback Files- Cache files are deleted

ReplayService.Infrastructure				
2021-07-14 16:28:31,107	64811904	INFO	18	Beginning file cache reconciliation process.: CurrentCacheGB=0.00 [(null)] ReplayService.Infrastructure
2021-07-14 16:28:31,107	64811904	INFO	18	Ending file cache reconciliation process.: CurrentCacheGB=0.00 [(null)] ReplayService.Infrastructure
2021-07-14 17:28:31,368	68412165	INFO	18	Beginning file cache trim process.: CurrentCacheGB=0.00, TargetCacheGB=10.00, DiskFreeGB=28.99, DiskTotalGB=30.00 [(null)]
ReplayService.Infrastructure				
2021-07-14 17:28:31,368	68412165	INFO	18	Completed file cache trim process.: CurrentCacheGB=0.00, TargetCacheGB=10.00, DiskFreeGB=28.99, DiskTotalGB=30.00 [(null)]

Generated job send to replay service

2021-07-14 17:49:05,581	69646378	INFO	79	Queueing job: Id=03b55361-b170-4a89-ae82-ac0493b1e391 [(null)] ReplayService.Infrastructure
-------------------------	----------	------	----	---

Log File: ReplayService.SupportLog

Log file location: NIA Server:

C:\ProgramData\Nexidia\Analytics\Logs

- 1. When the replay service is running and there is no playback request, all the cache files will be deleted
- 2. When we click playback , a job is generated and send to replay service



Playback - Logs

Transcode call to MP4, generate PNG file and place both in transcode media cache

```
2021-07-14 17:49:05,653 69646450 INFO 13 Started processing job: Job processor Id=13, Job Id=03b55361-b170-4a89-ae82-ac0493b1e391 [(null)] ReplayService.Infrastructure
2021-07-14 17:49:05,675 69646472 INFO 17 File added to Transcode Cache. SourceMediaId: 470 PathHash: 90e7272ed3836d10dc6138190003677c36bf3659c7ca841ada68b243dc27e3b0.mp4 [(null)]
ReplayService.Activity2021-07-14 17:49:07,388 69648185 INFO 17 File added to Transcode Cache. SourceMediaId: 470 PathHash:
115fe40aa4d857eec089c606e8bf572ec1f6f309878a90e8419858da7e2a1b76.png [(null)] ReplayService.Activity
2021-07-14 17:49:09,693 69650490 INFO 17 Job ID: 1, \\NIA\Production Media\CS Tier 2\05212005-593.wav, FFmpeg will be used to transcode. [(null)] WaveformGeneration.Activity
2021-07-14 17:49:09,728 69650525 INFO 17 Job ID: 1, \\NIA\Production Media\CS Tier 2\05212005-593.wav, Transcode Rates Calculated: SourceSamplingRate=8000 OutputSamplingRate=8000
OutputBitRate=16000 [(null)] WaveformGeneration.Activity
2021-07-14 17:49:10,732 69651529 INFO 13 Finished processing job: Job processor Id=13, Job Id=03b55361-b170-4a89-ae82-ac0493b1e391 [(null)] ReplayService.Infrastructure
```

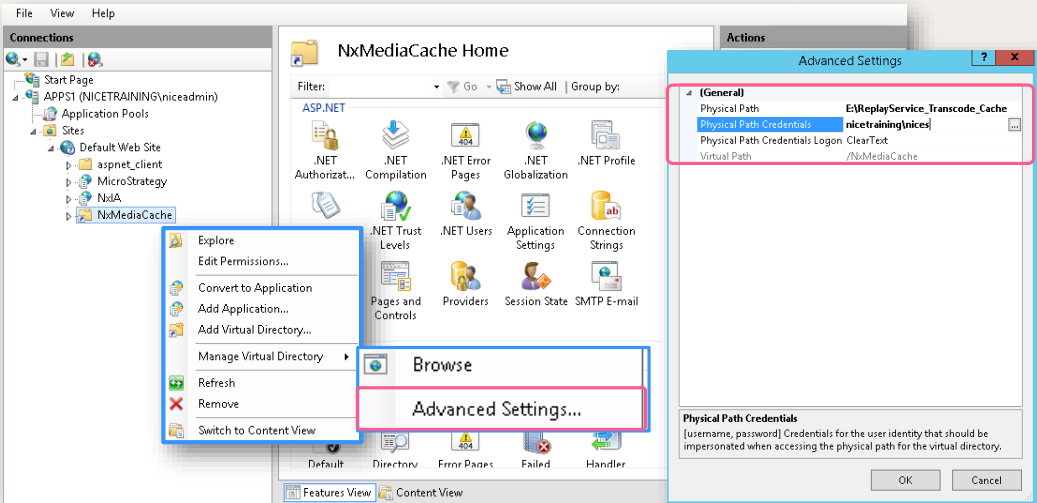
Log File: ReplayService.SupportLog

Replay service will

1. Get the call from the production media folder
2. Transcode it to an MP4
3. Generate a PNG file of the audio wave and
4. Place both files in the transcode media cache



NxMediaCache Virtual Directory



The NxMediaCache virtual directory must have both read/write privileges on the ReplayService_Transcode_Cache, on both the physical folder and the shared folder.

In the IIS Manager - Default Web Site - NxMediaCache, right-click select Manage Virtual Directory, advance setting, make sure that Interaction Analytics services user, has access to this folder, and the path to the folder is correct.



ReplayService_Transcode_Cache Folder

✓	ReplayService.Transcode.BitRate	Auto	Configuration for the bit rate (in bps) used when tran
✓	ReplayService.Transcode.Cache.CleanupInterval	60	The frequency at which the transcoded media cache :
✓	ReplayService.Transcode.Cache.ImportMP4	true	Determines if existing mp4 files should be copied din
✓	ReplayService.Transcode.Cache.MaxSize	10GB	Target maximum size for the service side transcoded
✓	ReplayService.Transcode.Cache.RetryFailureInterval	60	The number of seconds that should elapse before a f.
✓	ReplayService.Transcode.Cache...		ied to stream media. If empty, a default port is

Folder Size

Target maximum size for the service side transcoded media cache. This cache is persis

The target size can be specified in Megabytes, Gigabytes, Terabytes and also as a per

Valid values are in the form of [size][units], where units is one of MB, GB, TB or %.

Examples:

100MB = 100 Megabytes.

10GB = 10 Gigabytes.

15% = Grow until there is only 15% free space left on the disk. This takes quotas int

The default is 10 Gigabytes.

RESTORE DEFAULT

OK

CANCEL

The cache by default is set to a limit of 10GB. It will fill the 10GB and then remove calls as needed to keep it at 10GB.

The size of the ReplayService_Transcode_Cache folder is configurable in the Management Console.

Essentially the folder LRU(Least Recently Used) clean triggered by the cache size exceeding its configured limits. The MaxSize can be configured in units of MB/GB/TB or as a % of available drive space

To fix this about 90% of the time perform the following steps

1. Stop the Replay Service
2. Clear the ReplayService_Transcode_Cache
3. Start the Replay Service
4. Open a fresh page of NIA, and test replay capability.





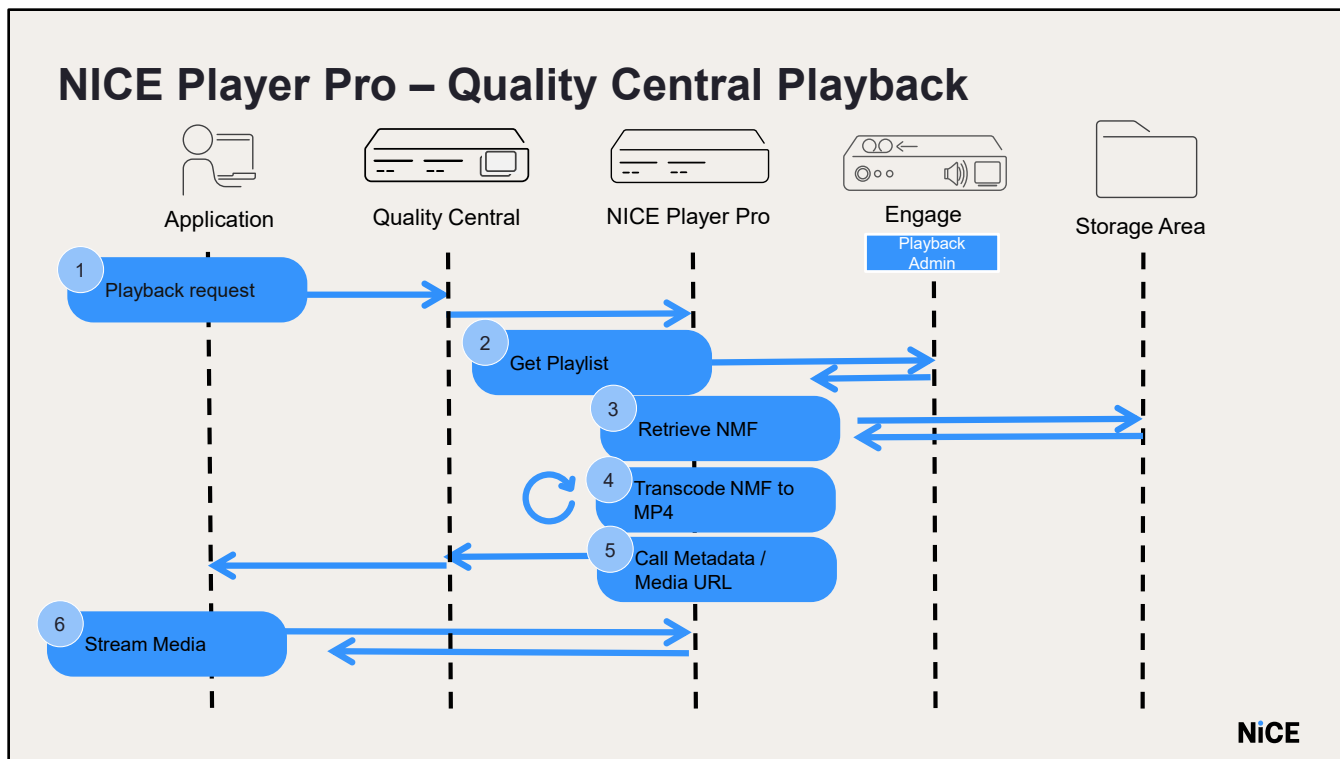
Can't playback Interaction

- Navigate to the call location and try to playback using Windows Media player. This will verify that the file is not corrupted
- Restart the replay service
- Check that the ReplayService_Transcode_Cache folder is not full
- Verify that the media file exists in the Production Media folder
- Check the Replay service log
- Check the folder path and Analytics user permission on the NxMediaCache virtual directory



NICE PLAYER
PRO (NPP)





Let's now look at the Playback Flow of NPP when integrated with Engage and QC.

1. It starts when the client sends a playback request which goes to Quality Central and then the NiCE Player Pro.
 2. The NiCE Player Pro communicates with the Playback Administrator on Engage to get the play list containing the call meta data and path to the .NMF recording files.
 3. The Playback Administrator sends the play list and path to NPP.
 4. NPP communicates with the storage area and retrieved the .NMF file.
 5. The Player Pro Media Server then starts retrieving the .NMF files from the storage area.
 6. Once the Player Pro Media Server gets the .NMF file packets, it starts transcoding the data from the packets to create a MP4 media file.
 7. When the transcoding starts, the NPP sends a URL and the meta data back to the client.
- While the media file is being processed, the client starts playing back the call without needing the fully processed MP4 file.





Can't playback Interaction

- Restart the replay service
- Check that the user have privilege to play this interaction in Engage
- Check the Replay Service log
- Check the Engage Streaming Support log
- Check the Nice Player Pro Logs (located on the NPP server)



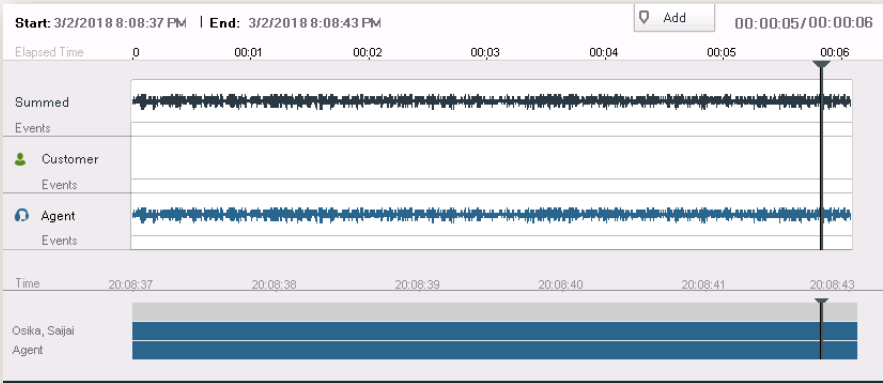
NiCE



ENGAGE APP
LAUNCHER



Engage App Launcher



App Launcher will be used to playback legacy screen recording (RCXP).

This appears in a separate tab with an independent player.
NOTE: only Internet Explorer is supported for the Engage App Launcher.



Do it Yourself

Troubleshooting Scenario:

Steven Mills is trying to review calls with his team which handles the “Billing calls”. He was reporting that he got some random playback issues. Make sure that he was able to playback **all** calls successfully.



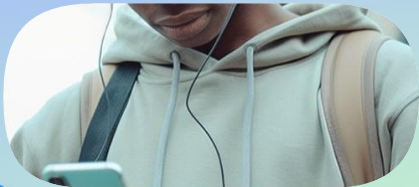


Can't playback Interaction

- Restart the replay service
- Check that the user has the privilege to play this interaction in Engage
- Check the Replay Service log



REVIEW
QUESTIONS



Review Question #1

The user failed to playback a call using the Nexidia player what might be the reason?

The
ReplayService_Trans
code_Cache
folder is full

The media file is
corrupted

Analytics services
user doesn't have
access to the
NxMediaCache virtual
directory and the
ReplayService_Trans
code_Cache

All the answers are
correct





summary

- Playback Flow





Thank You

QUALITY CENTRAL

NiCE Interaction Analytics & Quality
Central Troubleshooting Fundamentals

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world 

objectives

- Describe the QC Logs & Databases
- Describe the QC Ingest Flow
- Describe the Difference between Interaction SC and QC SC
- Describe the Workflow Flow



DATABASE &
LOGS FILES



QC Log-Files

Log file names	Details
Nexidia.AgentPortalWeb.SupportLog	QC web requests from Evaluations Portal to exposed APIs
Nexidia.EvaluationsAging.SupportLog	Nexidia Evaluations Aging Windows service logs
Nexidia.EvaluationsWeb.SupportLog	QC application Evaluations page logs from the Main portal
Nexidia.RestAPI.SupportLog	QC logs for Rest API calls hit by Nexidia Application modules



QC Databases

NexidiaESI

- Application configuration: QC App Configuration (Users, Agents, Workflows, Call Data)

NexidiaESIDW

- Data Warehouse - Stores information used in reports and in app. (Forensic Search and my evaluation)

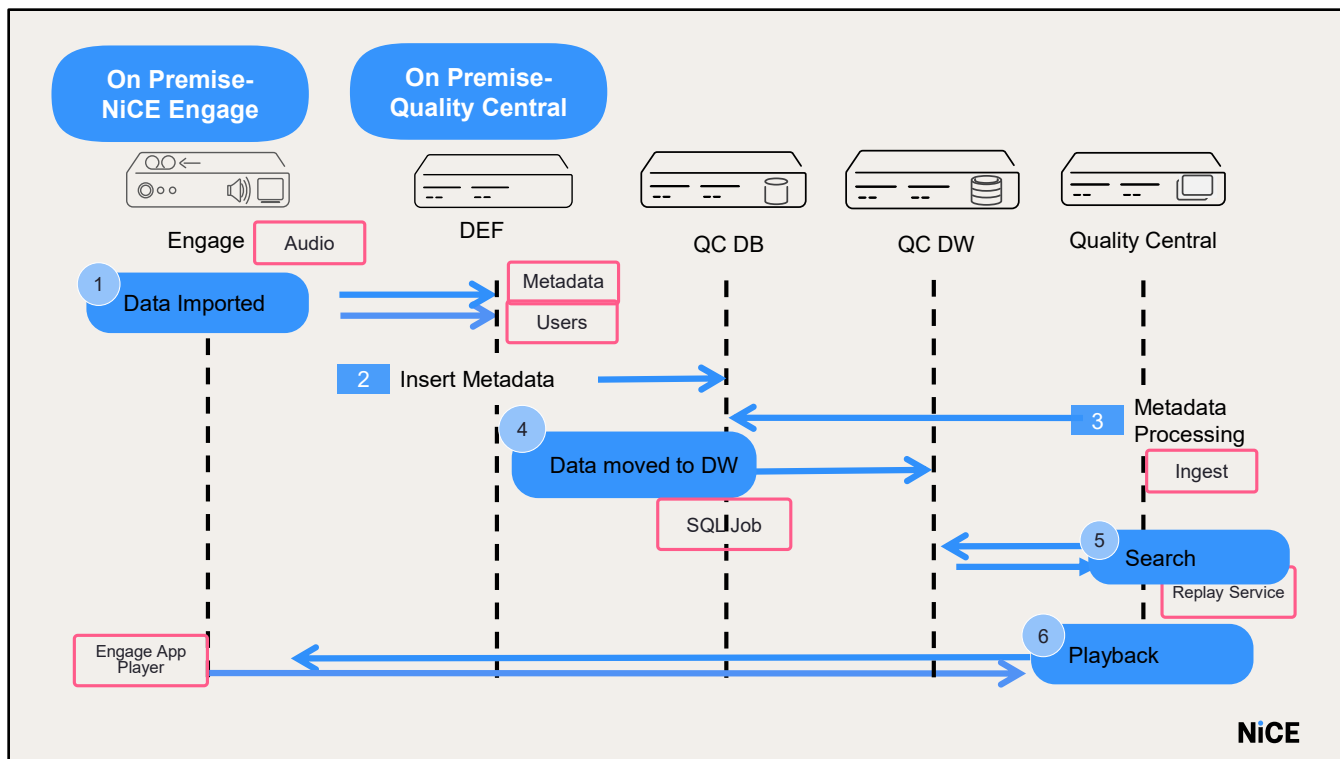
NexidiaEvaluations

- Used by the QC Portal
- Stores information about Evaluations (all evaluation forms and data)



PROCESS FLOW





1. DEF is used to import information from the Engage system. Metadata files (XML) are stored in the staging folders. NOTE: Only Metadata is imported; audio remains on the storage location.
2. DEF inserts data into the QC DB (NexidiaESI > NexidiaESI.SourceMediaInbox table).
3. Metadata is processed in the QC system by the Ingest service.
4. SQL Job NIA-Core-Update moves the data to the QC DW (NexidiaESIDW). This job runs every 10 minutes by default.
5. End-user Application searches for call by querying the QC DW (or it is sent as part of the Workflow Queue). Results are sent back to the application. This includes the Interaction ID. Note: this query uses Javascript.
6. When User clicks play, Replay Service requests the media source. NOTE: The Replay service will first look at the local media source and then query the Engage App Player.



1. DEF is used to import information from the Engage system. DEF also copies the audio from the 3rd-party Recorder to the Media Storage folder (on QC Apps). NOTE: Users / Agents are not imported from 3rd-party Recorders.
2. DEF inserts data into the QC DB (NexidiaESI > NexidiaESI.SourceMediaInbox table)
3. SQL Jobs move the data to the QC DW (NexidiaESIDW). This job runs every 5 minutes by default.
4. Quality Central Application is used to Search for interactions (queries the QC DW).
5. When User clicks play, Replay Service requests the link to the media source. Media is streamed from the Media Storage Location.



Ingest Flow

If NIA is installed audio must be imported into Nexidia

Users / Agents are not imported from 3rd-party solutions

In QC stand-alone system installed on-premise, audio files will not be imported into QC server



AGENTS & USERS

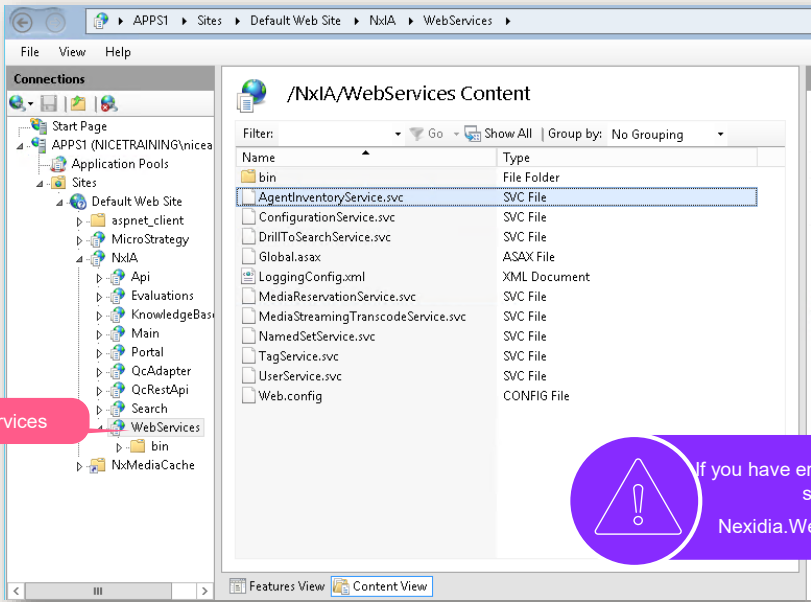


Agent and Users Imports Using DEF



DEF load the agents and users into the system
 DEF communicate with NIA webservices, Nia webservices will
 insert/update the agents and users in NexidiaESI database

NIA Webservices



Webservices



If you have errors in the import users\agents flow, start by investigating the Nexidia.WebServices.SupportLog.txt log file



Agent -Supervisor and Site

New/Edit Agent in Inventory

Photo

Select Photo (Max size: 2 MB)

*Evaluations Portal Login ID

pbritt

*First Name

Peter

*Last Name

Britt

*Agent ID

11945

*Account type

Internal Authentication

*Password

*Re-enter Password

*Active

E-mail

Verify E-mail

External User ID

Unique Agent ID

Site

Reno

Supervisor

Steven Mills

Agent Groups

* Indicates a required field

OK

Cancel

<?xml version="1.0" encoding="UTF-8"?>

<CallData>

<HubId>1</HubId>

<InteractionID>6536201056524763141</InteractionID>

<HasVideo>True</HasVideo>

<IsH264Embedded>True</IsH264Embedded>

<AgentName>Peter Britt</AgentName>

<AgentId>11945</AgentId>

<Group>West</Group>

<Supervisor>Steven Mills</Supervisor>

<Site>Reno</Site>

<DNIS>2265</DNIS>

<Extension>2946</Extension>

<CallDirection>Inbound</CallDirection>

<CallType>Did Not Order</CallType>

<CustomerId>507901</CustomerId>

<CustomerCity>San Jose</CustomerCity>

<CustomerState>CA</CustomerState>

<CallDateTime>2020-7-4T11:05:09+05:30</CallDateTime>

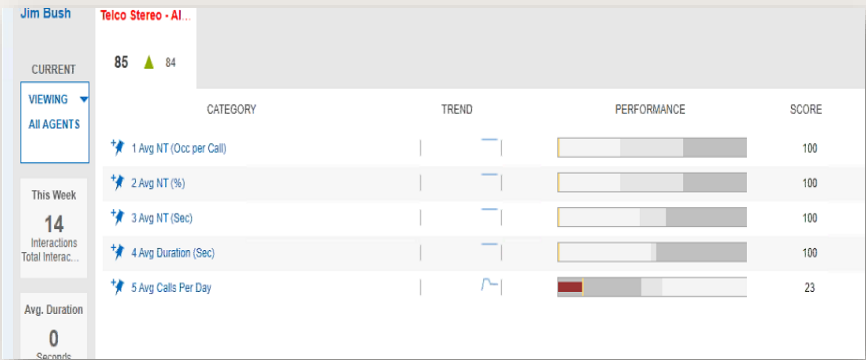
</CallData>

The agent's supervisor and site details are populated from the call metadata

The agent’s supervisor and sites details are populated from the call metadata



User -Supervisor and Site (metadata association)



“Viewing All Agents” scorecard will not present any data if the user metadata association was not configured.

For example, in the Portal Dashboard, the “Viewing All Agents” scorecard will not present any data if the user metadata association was not configured.



SCORECARDS



Scorecards Types – QC Scorecards



Defined in the Administration Tab

Based on ingested metadata

Link to TMS

Calculation is done in NexidaESIDW database

Evaluation Scorecards are also known as interaction scorecard
Need to define access rights per user\profile\agent



Scorecards Types – QC Scorecards

NICE QUALITY CENTRAL												
ID Contains		From 10/01/2022 to 10/31/2022		Row Contains		☐ Self-Evaluations only		View	Print	Evaluate		
ID	EVALUATOR	VERSION	TEMPLATE	AGENT	SCORE	AGENT SCORE	PUBLISHED	ACKNOWLEDGED ON	NOTES	COACHING REQUEST	COACHED	STATUS
4	David Voss	2	EngageQuality	Peter Britt	50.00 / 50 - (100.00%)		10/31/2022		Verify Billing Information			Published

Defined in the QC Portal

Based on Evaluation

Link to Evaluation Template

Calculation is done in NexidiaEvaluation database

QC scorecards are accessible to all users and agents



Can't see new results in Evaluation (Interaction) SC

- Verify that the option of "Use in Portal Dashboards" is enabled in the scorecard
- Verify that the scorecard column "Included" is marked Yes
- Check that Session, TMS and scorecard are well connected
- Check access rights to User/profile/agent
- Ingested metadata date should be in the range of periods for score and trend
- Check that the user metadata association is configured
- SQL job run successfully (NIA core update)
 - SQL Job prior to V12.2:
 - NESI-ETL-Batch
 - NESI-ETL-Evaluation-Batch
 - NESI-ESIDW-PerformQMCcalculations



Can't see new results in
QC SC

- Verify that the option of “Use in Portals Dashboards” is enabled in the scorecard
- Verify that the scorecard column “Included” is marked ‘Yes’
- Evaluations are done for template and template is configured with scorecard
- Evaluation should be media type not agent type evaluation
- Check for Template permission
- Evaluation score should be greater then 0
- SQLJob Run (NICE-QC-ScorecardCalculationBatch)



QC Portal

Last 7 Days

Steven Mills

VIEWING

All AGENTS

Interactions

78

7 Days

Avg. Duration

90

Seconds

Evaluation Scorecard

QC Scorecard

NO DATA

CATEGORY	TREND	PERFORMANCE	NORMALIZED SCORE	RANK	CHANGE	INTERACTIONS
Average Duration	Name: Nexidia.Evaluations.PortalDashboardsPeriod1StartDayDelta				n/a	43
Average Interaction Per Day	Value: -7				n/a	43
	Default Value: -7					
	Notes: Number of days in the past for the start of the first portal dashboard period.					

Set the portal date range

NiCE 20

The dashboard show data form the last 7 days if you want to changes the period

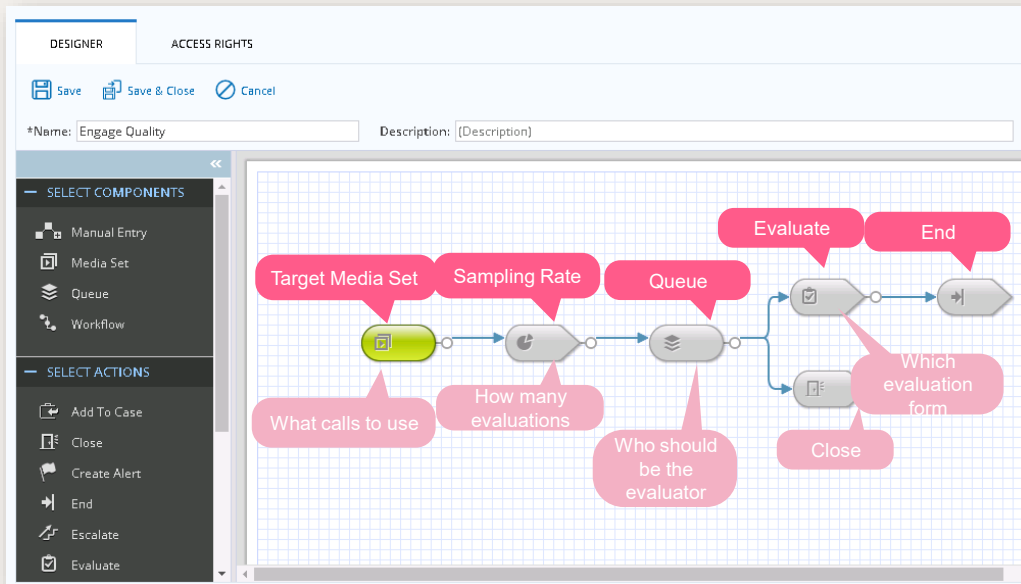
You can do it Via the management Console



WORKFLOW



Workflow



NiCE

Automate the QM process.

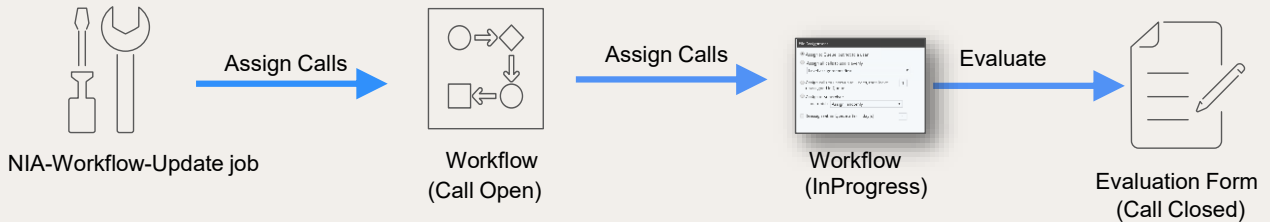
The workflow defines and controls all the relevant steps required to complete the evaluation process:

1. Identify the set of calls (population of calls) that can be used to evaluate the agents (using TMS).
2. Set the sampling rate how many calls from the population of calls are sent to the evaluators (number of evaluation per agents)
3. Assign the evaluators- which evaluators can perform the evaluation
4. Which evaluation form to use
5. Re-evaluation or calibration if needed

The End step is not configurable but is required to complete the workflow. The Close step states if Notes are required to complete the workflow or how it should be tagged. NOTE: tags are not covered as part of this course. See the QC Getting Started Guide for more details.



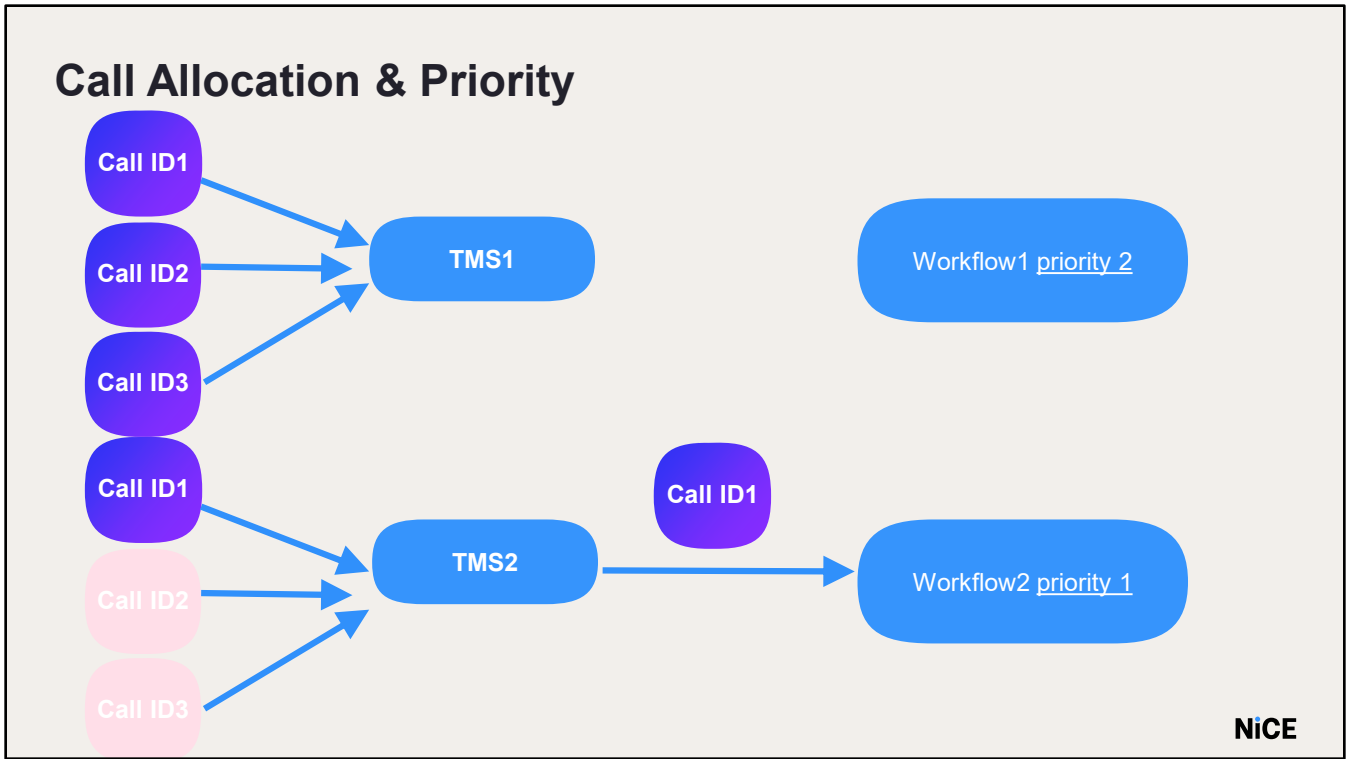
Call Allocation



Each workflow is assigned with a default priority, according to the workflow creation order. To change the workflow priority, use the “set priority” button in the workflow administration.

The NIA workflow update job insert\assign the interactions to the workflow

The NIA workflow update job insert\assign the interactions to the workflow; the calls are inserted to the workflow with status “open.” According to the workflow “file assignment.” The call is assigned to a queue or a user, and the call status changes to in-progress. Once the evaluation is done, the call status changes to closed. When the NIA workflow update job runs, it will remove “closed” calls from the workflow.



Let's look at the following example:

TMS 1 assign interactions to workflow 1, and TMS 2 assign interactions to workflow 2.

According to the TMS filters interaction ID 1 fits both TMS1 & TMS 2. The call will be assigned to the workflow with the highest priority.

In our example, it will be assigned to workflow 2.

Note: call can be assigned only to one workflow

Workflow Main Tables

```
SELECT * FROM [NexidiaESI].[Workflow].[Workflow]
```

Workflow ID

Workflow Name

	WorkflowId	WorkflowIdentifier	Name	Description	IsActive	IsDeleted	DiagramMetadata	ProcessingOrder
1	1	63C38ED5-108A-424B-BF45-B4F389836A06	Qc WF		1	0	{'surfaceOrigin':{'x':-100,'y':-102},'surfaceSiz...	1
2	2	DD1D86C8-309C-4F3E-A3C7-ADACF64A5196	Suport Team Eval		1	0	{'surfaceOrigin':{'x':-100,'y':-102},'surfaceSiz...	2

Priority



Workflow Main Tables

```
SELECT * FROM [NexidiaESI].[Workflow].[WorkflowStateLog]
```

	WorkflowId	WorkflowIdentifier	Name	Description	IsActive	IsDeleted	DiagramMetadata	ProcessingOrder
1	1	63C38ED5-108A-424B-BF45-B4F389836A06	Qc WF		1	0	{"surfaceOrigin":{"x":-100,"y":-102},"surfaceSiz...	1
2	2	DD1D86C8-309C-4F3E-A3C7-ADACF64A5196	Suport Team Eval		1	0	{"surfaceOrigin":{"x":-100,"y":-102},"surfaceSiz...	2

Call ID

Call Status

Results		Messages		WorkflowStateLogId	WorkflowStateLogEventTypeld	WorkflowId	WorkflowStepId	EntryStepId	SourceMediaId	Objectld	Activity
9	9			1	1	2	1	175	NULL	File entered workflow "Qc WF" by target media s...	
10	10			1	1	2	1	176	NULL	File entered workflow "Qc WF" by target media s...	
11	11			4	1	3	NULL	176	NULL	Assigned to queue "Evaluator"	
12	12			4	1	3	NULL	175	NULL	Assigned to queue "Evaluator"	
13	13			4	1	3	NULL	174	NULL	Assigned to queue "Evaluator"	
14	14			4	1	3	NULL	173	NULL	Assigned to queue "Evaluator"	
15	15			4	1	3	NULL	172	NULL	Assigned to queue "Evaluator"	
16	16			4	1	3	NULL	171	NULL	Assigned to queue "Evaluator"	
17	17			4	1	3	NULL	170	NULL	Assigned to queue "Evaluator"	
18	18			4	1	3	NULL	169	NULL	Assigned to queue "Evaluator"	
19	19			4	1	3	NULL	168	NULL	Assigned to queue "Evaluator"	
20	20			4	1	3	NULL	167	NULL	Assigned to queue "Evaluator"	
21	21			11	1	4	NULL	167	-2147...	Eval Published (-2147483647)	
22	22			19	1	6	NULL	167	NULL	Closed	

Present which calls associate to which workflow and the status of the call, assign to queue\evaluation published\closed, etc.



Troubleshooting Scenario:

• Peter Britt is logging into QC <http://nia/NxIA/Portal/Login.aspx> as "Agent" to view the Evaluation score card and the published evaluations in the date range **8/23/2025 - 8/29/2025**. He was able to view his Evaluation dashboard, but he could not see his individual evaluations. Identify and fix the issues so that Peter Britt can see his individual evaluations.

Agent Credentials	Evaluator Credentials
Username: pbritt	Username: dvoss
Password: nicecti	Password: nicecti



REVIEW
QUESTIONS



Review Question #1

If you want to display the performance of an agent-based on their evaluations, what needs to be created?

Session

Evaluations

Target Media Set

QC Scorecards



summary

- QC Logs & Databases
- QC Ingest Flow
- Interaction SC and QC SC
- Workflow Flow





Thank You

Course Closing

NiCE Interaction Analytics & Quality
Central Troubleshooting Fundamentals

Create a
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world ☺



Course Survey

A screenshot of the "Training satisfaction survey" form. The header shows the NiCE logo and the title "Training satisfaction survey" with a smiley face icon. The first question is "How satisfied are you with the overall training experience?". Below this is a 5-point scale with labels: "Very Dissatisfied" (1), "Dissatisfied" (2), "Neither Satisfied nor Dissatisfied" (3), "Satisfied" (4), and "Very Satisfied" (5). The second question is "On a scale of 1 to 5, how satisfied are you with the following aspects of the training course:". Below this is a table with 5 columns corresponding to the same satisfaction levels as the first question. The rows of the table are: "The instructor's knowledge of the subject matter", "The instructor's delivery, attitude and behavior", "The course content and structure (agenda, presentations, documents, etc.)", "The balance's between lectures, demos, and hands-on activities", and "The labs used for demos and exercises". Each cell in the table contains a scale from 1 to 5.

Please leave us comments on how we can improve the learning experience for this course

Scores range from "Very Dissatisfied" (left) to "Very Satisfied" (right)

You should have received a course survey in your email from CSATTeam@nice.com. Please take a few moments to take the course survey before we proceed.

If you did not receive the course survey in your Inbox, check your Junk email folder. If you did not receive the survey, please let your instructor know so that he/she can have it resent to you.



Written Exam Instructions

Examination Instructions

Welcome to your online test.

As you have only ONE ATTEMPT to pass this test, please ensure the following points are followed before you start:

1. You have covered all learning materials and feel confident to pass the test

2. You are in a quiet environment

3. Once started, this test will run for exactly 40 minutes so try to avoid any disruption

Test name: **NiCE Nexidia Analytics & QC v12.4 - Fundamentals - Theoretical Exam**

Number of questions: **20**

Allotted time: **40 minutes**

Passing score: **80**

This course and subsequent test have been designed to ensure that you have the relevant knowledge to fulfill your role with the utmost success. We therefore ask you to complete the test in the appropriate environment and answer the questions openly and honestly.

Use the assessment page to respond to or mark questions for follow-up. When finished, click the Summary button to review the answers and the questions marked for follow-up. Note that changes cannot be made once the assessment has been submitted. Upon completion, click the Submit button to save the answers.

Good Luck!

Warning:

Please do not use your Browser buttons to navigate in the Test.

Please use the navigation buttons at the bottom of each page.

Your test answers will not be recorded if you navigate using the Browser buttons.

Cancel

Continue

- Read the Written Exam instructions carefully
- Use all your resources from the course including the lab and other NiCE documents
- Complete all assigned tasks
- Email the instructor with your final score

After clicking the Launch Test link, an Examination Instructions page will appear. Make sure you read the Examination Instructions before beginning the written exam. Especially heed the warning regarding navigation. Do not use the browser forward and back buttons during the test. Always use the navigation buttons at the bottom of each test page. If you use the browser buttons, your test answers will not be recorded.





Written Exam Instructions

Complete the exam in the allotted time. Make sure to submit the exam before the 2:00 countdown

- Troubleshooting Fundamentals - Theoretical Exam

Time remaining
36:10

Use Mark for follow up to return to questions later

☐ Mark for follow up

Use the navigation at the bottom of the page and not the browser. Click Summary to review your answers before submitting them

[Save / Return Later](#) [Summary](#)

You must complete the exam in the allotted time. The clock in the upper right hand corner will specify how much time you have remaining.

NOTE: Make sure to submit your test prior to the 2:00 minute warning. If not, the test session may expire.

Do not get hung up on a single question. If you don't know the answer, make your best guess and move on. Use the Mark for follow-up box to check questions that you want to go back later and reconsider. These may be questions you have answered or not answered. You can come back later and try to answer the question again.

Use the buttons at the bottom of the test pages to navigate through the exam. The summary button will take you to a page where you can see all questions marked for follow-up and all unanswered questions.





Practical Exam

NiCE

NiCE Interaction Analytics and Quality Central Troubleshooting Fundamentals – Practical Exam

General Instructions
Students will need to resolve issues with a NiCE NIA and QC system, as reported in a case by a customer.



Please feel free to refer to any documentation, manuals, or notes to assist in completing the Practical Exam

Duration: 2 hours

You have been assigned a case for ACME Software, a call center with NIA + QC. You are provided the following information about the case:

1. Steven Mills, a Team lead was trying to create a new session named "Exam_Billing Session" import calls from the "Exam_Billing" media folder, but he is not able to find the folder.
2. Also, he was not able play the audio files from this session.
3. When tried to look upon phrases for the same "Exam_Billing Session" in discovery application no data is populated.
4. Peter Britt is logging into QC <http://nia/NxdA/Portal/Login.aspx> as "Agent" to view the Evaluation score card and the published evaluations in the previous week. He was able to view his individual evaluation, but his Evaluation dashboard is blank.

Read the Practical Exam document carefully

Complete all assigned tasks

Notify the instructor when complete

You will receive the Practical Assessment document from the instructor. Before beginning please read the entire document carefully. Complete all assigned tasks and notify the instructor when everything has been completed.





Thank You



Contact us: training@nice.com | Visit us: [NiCE Dojo](#)

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